The (Un)Fortunate Poor*

Jan Eeckhout — UPF Barcelona†

Rory McGee — Western Ontario‡

Alireza Sepahsalari — Bristol§

Peter Spittal — Bristol¶

November 6, 2026

Abstract

We document a robust V-shaped relationship between wealth and pay. Among indebted households, average pay is highest for those with the least wealth (highest debt) and falls as wealth approaches zero, before rising again at positive wealth. This pattern survives alternative measures of pay and wealth, different specifications and sample restrictions, and is present across countries with different institutions. We show that the relationship reflects endogenous selection of worker types across the wealth distribution. Higher permanent earnings types can sustain deeper debt positions following adverse shocks and accumulate larger precautionary buffers, while lower types remain concentrated closer to zero wealth. We develop an incomplete-markets model with permanent worker heterogeneity, uninsurable income risk, and borrowing constraints that increase with expected labor income. Calibrated to US data, the model reproduces the observed pay–wealth relationship through this mechanism. We then use the model to study what this selection implies for the measurement of economic need and the targeting of transfers. Current economic circumstances can be a poor guide to permanent resources, affecting both which households are identified as poor (lifetime versus resource poverty) and where transfers generate the greatest welfare gains. Our calibrated model suggests transfers should target wealth and pay jointly, with a higher weight on pay.

Keywords. Macro/Labor. Financial Frictions. The Working Poor. Wealth Inequality. Wage Inequality. Lifetime Poverty. Resource Poverty.

*We are grateful for comments and feedback from John Grigsby (discussant), Eric Hurst, Ayse Imrohoroglu, Nobu Kiyotaki, Francis Kramarz, Huw Lloyd-Ellis (discussant), Guido Menzio, Kurt Mitman, Giuseppe Moscarini, Guillermo Ordoñez, Robert Shimer, Gianluca Violante and numerous colleagues and seminar audiences. Eeckhout acknowledges support from the ERC, Advanced grant 882499, and from grant PID2022-138443NB-I00 from the Spanish Ministry of Science and Innovation.

†jan.eeckhout@upf.edu – ICREA-BSE-CREI

‡rmcgee4@uwo.ca – IFS

§alireza.sepahsalari@bristol.ac.uk

¶peter.spittal@bristol.ac.uk – IFS

1 Introduction

Do workers with the least wealth also have the lowest pay? We document that they do not. Among households with negative wealth – that is, households in debt – average pay is highest for those with the most negative wealth and declines as wealth rises toward zero. Among households with positive wealth, by contrast, average pay rises with wealth. The result is a V-shaped relationship between wealth and pay, with the lowest average pay not among the most indebted households, but among households close to zero wealth.

This pattern matters because a household's position in the wealth distribution reflects both its current circumstances and its permanent earnings capacity. We show that the pay–wealth kink arises from the endogenous selection of heterogeneous workers across the wealth distribution. Higher permanent earnings types can sustain deeper debt positions following adverse income shocks, while also accumulating larger precautionary buffers, whereas lower types remain concentrated closer to zero wealth. As a result, moving further into debt can indicate a worse current state while at the same time selecting households with higher permanent earnings capacity. This distinction has implications for how economic need is measured and how transfers are targeted.

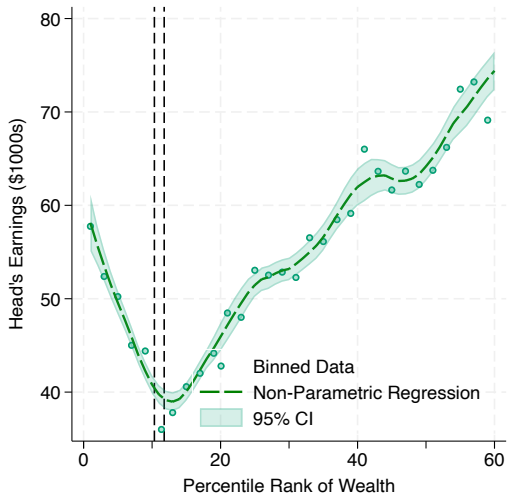


Figure 1: Non-monotone relationship between wealth and annual pay

Notes. This is the estimated kernel regression in equation (1) below, along with 95 percent confidence intervals and a binned scatter plot of the raw data where each point represents average pay in a two-percentile interval on the wealth distribution. The vertical lines delimit the range where wealth is near-zero, defined as being between −$250 and $250.

Figure 1 illustrates our main empirical finding using the Panel Study of Income Dynamics. A household head with zero net wealth earns $36,142 on average. By contrast, someone in the bottom

wealth percentile has net debt of $151,558 and earns $58,118 on average – over 60% more. Average pay at the bottom of the wealth distribution is therefore close to average pay much further up the distribution: at the 37th wealth percentile, household heads earn $58,932 on average and have wealth of $58,629.

The relationship is remarkably robust. The V-shaped pattern remains under alternative measures of labor income and wealth, across different specifications and sample restrictions, over time, and after conditioning on household demographics and other observables. It is also present across countries with very different labor-market institutions, tax-and-transfer systems and sources of household debt. The breadth of this evidence suggests that the relationship is not an artifact of a particular institutional feature of the US.

We next use the data to understand the mechanisms underlying the pay–wealth kink. We first show that the relationship is driven by persistent differences in earnings across workers, rather than transitory earnings variation or predictable changes in pay over the life cycle. An AKM-style decomposition further shows that these differences are primarily associated with workers rather than the occupations in which they work. Finally, we show that pay dispersion is itself non-monotonic in wealth, closely mirroring the pattern in average pay. Together, these findings point toward a mechanism in which workers with different permanent earnings prospects and exposure to income risk select systematically into different parts of the wealth distribution.

We formalize this mechanism in an incomplete-markets model à la Aiyagari (1994) with permanent worker heterogeneity, uninsurable labor income risk, and borrowing constraints that depend on expected labor income. Higher permanent earnings types have higher expected pay and greater pay dispersion. Their higher expected earnings give them greater access to credit, allowing them to reach deeper debt positions following adverse income realizations. At the same time, their greater exposure to income risk leads them to accumulate larger precautionary buffers. Lower permanent earnings types have less borrowing capacity and accumulate smaller buffers, leaving them concentrated closer to zero wealth. Changes in the composition of permanent worker types across the wealth distribution therefore generate the V-shaped relationship between wealth and pay.

We first establish this selection mechanism analytically in a simple two-type version of the model. When borrowing capacity scales with permanent earnings capacity, higher types occupy a wider range of the wealth distribution than lower types and extend further into both tails. In the stylized environment, within-type mean pay is constant in wealth, so the aggregate V-shape arises entirely from changes in the composition of worker types across the wealth distribution. The analytical

model therefore isolates the selection mechanism that generates the pay–wealth kink.

We then calibrate the full model to key features of the US earnings process, tax-and-transfer system, credit environment, and joint distribution of pay and wealth. The model closely reproduces the V-shaped relationship between wealth and pay through the same selection mechanism established analytically. Lower permanent earnings types remain concentrated close to zero wealth, while higher types extend further into both tails of the wealth distribution. The resulting composition of permanent worker types also generates non-monotonic aggregate relationships between wealth and both lifetime utility and consumption.

These results have implications for how economic need is measured. Building on a literature emphasizing the importance of a dynamic perspective on poverty (e.g., Poterba 1989 and Meyer and Sullivan 2012), we distinguish between *resource poverty*, defined as having very low current resources, and *lifetime poverty*, defined as having very low lifetime welfare. Because the composition of permanent worker types varies non-monotonically across the wealth distribution, these two measures identify substantially different households. Among households in the bottom 5% by current resources, only 14% are also in the bottom 5% by lifetime welfare. Resource-poor households tend to be deeply indebted and have realized pay well below their expected pay, consistent with adverse current states among workers with relatively high earnings capacity. Lifetime-poor households instead have positive wealth but much lower permanent earnings capacity. In this way, resource poor households are both ex-ante fortunate, as they tend not to be lifetime poor, but ex-post unfortunate.

This distinction has a natural counterpart in two motives for transfer policy. A transfer can provide insurance against an adverse current state, or redistribute resources toward households with low permanent resources. We formalize these motives by decomposing the marginal welfare value of a transfer into insurance within permanent worker type and redistribution across permanent worker types similarly to McGee and O'Connell (2026) in the design of social insurance (see also Lieber and Lockwood 2019 and Finkelstein, Hendren, and Luttmer 2019 in the context of Medicaid). Wealth is particularly informative about the insurance component, while current pay is more informative about redistribution. Our selection mechanism implies that wealth and pay contain different information about the source of welfare gains from transfers.

Finally, we examine the implications for targeting. Unlike Golosov, Troshkin, and Tsyvinski (2016), we do not consider the full mechanism design problem, yet our approach highlights similar lessons for the design of public policy. We consider a marginal policy that assigns an additional dollar to a fixed fraction of households and construct the optimal targeting rule when policy can condition

on current pay, current wealth, or both. Wealth performs substantially better than pay when the objective is insurance, while pay performs substantially better when the objective is redistribution across permanent types. Conditioning on both performs best when the objective incorporates both motives. The endogenous selection that generates the pay–wealth kink therefore matters not only for how economic need is measured, but also for how redistributive policy is targeted.

We contribute to two literatures that study the relationships between household balance sheets and heterogeneity in economic outcomes such as the marginal propensity to consume (e.g., Kaplan and Violante, 2022; Kaplan, Violante, and Weidner, 2014; Fagereng, Holm, and Natvik, 2021; Lewis, Melcangi, and Pilossoph, 2019; Aguiar, Bils, and Boar, 2025) or job search behavior (e.g., Eeckhout and Sepahsalari, 2024; Luo and Mongey, 2019; Krusell, Lou, and Ríos-Rull, 2019; Donaldson, Piacentino, and Thakor, 2019; Herkenhoff, Phillips, and Cohen-Cole, 2023). This work typically considers how wealth, portfolios, or access to credit affect workers' intertemporal consumption or labor-search choices. Our contribution is to focus on what the joint distribution of wealth and earnings reveals about the persistent characteristics of the households occupying different parts of the wealth distribution. We study this empirically, establishing the robustness of the non-monotonic pattern to alternative measurement choices and settings,[1] before highlighting the important implications of permanent heterogeneity in workhorse consumption-saving models.

Our work also relates to a literature studying consumption smoothing, precautionary saving, and incomplete markets. We emphasize how precautionary savings motives at the household level, common to both microeconomic models in the income fluctuation tradition (e.g., Deaton, 1991; Carroll, 1997; Zeldes, 1989) and macroeconomic models in the incomplete market tradition (e.g., Imrohoroğlu, 1989; Bewley, 1980; Huggett, 1993a; Aiyagari, 1994), deliver stark implications for the endogenous selection of permanently heterogeneous workers. Our individual agents are motivated by the same forces, in line with empirical evidence on the importance of precautionary saving including Carroll, Dynan, and Krane (2003) who finds that households facing greater earnings uncertainty hold more precautionary wealth and Jappelli and Pistaferri (2025) who show that households adjust target wealth in response to revisions in permanent income. This work emphasizes the role income

[1] Non-monotonic relationships also arise in other contexts. Kennickell (2006) similarly highlights differences between those in negative net worth close to zero and those with larger net debt holdings. Daminato and Pistaferri (2023) document a non-monotonic relationship between wealth and expected returns, while Groes, Kircher, and Manovskii (2015) document a U-shaped relationship between pay and occupational mobility. The objects and mechanisms in those papers are distinct from the pay–wealth relationship studied here. Both Brewer, Etheridge, and O'Dea (2017) and Meyer and Wu (2023) document non-monotonicity in the relationship between income and consumption for the lowest level of incomes. They find this stems from systematic measurement error as these households rely on other unmeasured sources of income. In contrast, we find that our non-monotonicity can be fully explained by how dynamic optimization generates the endogenous selection of households.

risk plays in shaping the wealth distribution and its right tail. We instead focus primarily on the left tail and complement this body of work by quantifying how a household's position in the wealth distribution reveals its permanent earnings type. Moreover, we show how a modified version of these workhorse frameworks is consistent with the facts we establish, yet has immediate consequences for fiscal policy.

The rest of the paper is organized as follows. Section 2 establishes our key fact. Section 3 describes our quantitative framework and section 4 provides analytic results for a simplified version of the model. Section 5 describes our empirical approach to calibrate the model, presents the estimates and shows how endogenous selection delivers the key fact. Section 6 draws implications for the measurement of poverty and the design of targeted transfers. Section 7 concludes.

2 Data and Empirical Results

2.1 Data

Our core data are drawn from the Panel Study of Income Dynamics (PSID), a biennial household panel survey containing detailed information on pay, occupation, wealth, and demographic characteristics for a nationally-representative sample of households in the US. We focus on the period from 1999 to 2019, corresponding to the period over which PSID has collected wealth data in every wave. Our baseline sample contains 26,383 employed observations with non-zero survey weights, comprising 6,552 households observed for an average of 4.03 waves. Appendix A describes the sample construction and variable definitions in more detail.

Our primary measure of wealth is household net worth: total assets net of mortgage and other debt, $a_{i,t}$. To reduce the influence of households with extremely high or low wealth, we use the percentile rank of wealth, $a^p_{i,t}$, in our empirical analysis. Our measure of pay is the household head's gross annual labor income, including pay from all jobs and any labor component of income from a business or farm. Net worth and labor income are expressed in 2012 prices.

Table 1 reports descriptive statistics for the baseline sample. Around 10 percent of households have negative net worth, while annual labor earnings are substantially higher among households with negative wealth than among those at zero wealth.

We also document the relationship between wealth and pay for a set of other countries. For the UK, we use the UK Household Longitudinal Study (UKHLS); for the remaining countries, we use the Luxembourg Wealth Study. Appendix B describes these data in more detail.

Table 1: Descriptive Statistics

	Full sample		Negative wealth		Zero wealth		Positive wealth	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
Annual labor earnings	79,691	121,506	49,536	32,597	36,142	28,132	83,829	128,116
Annual hours worked	2,273	626	2,166	672	2,091	742	2,288	617
Hourly wage	34.68	58.30	23.21	15.74	17.62	12.99	36.25	61.57
Net worth	299,980	778,518	-47,133	76,082	0	0	344,840	817,000
Consumption	46,180	40,994	37,513	18,188	25,470	12,956	47,482	42,934
Age	41.49	8.35	37.40	8.12	40.37	8.70	41.98	8.23
High school or less	0.33	0.47	0.34	0.47	0.65	0.48	0.33	0.47
Some college	0.24	0.43	0.28	0.45	0.17	0.38	0.23	0.42
College or more	0.43	0.49	0.38	0.48	0.17	0.38	0.44	0.50
Black	0.07	0.25	0.11	0.31	0.21	0.41	0.06	0.23
Family size	3.56	1.26	3.68	1.40	3.92	1.48	3.54	1.24
Observations	26,383	–	3,250	–	476	–	22,657	–
Households	6,552	–	1,746	–	343	–	5,942	–
Weighted sample share (%)	100.00	–	10.37	–	1.22	–	88.41	–

Notes. The table reports weighted means and standard deviations for the baseline PSID sample. Negative, zero, and positive wealth refer to household net worth. Annual labor earnings are gross annual labor income of the household head; hourly wage is wage and salary income divided by annual hours worked. Monetary variables are expressed in 2012 dollars. Education categories are defined using completed years of schooling. The final row reports each wealth group's share of the weighted sample.

2.2 The V-Shaped Relationship Between Wealth and Pay

We begin by estimating the following non-parametric relationship between pay, $y_{i,t}$, and the household's rank in the wealth distribution $a^p_{i,t}$:

$$y_{i,t} = m(a^p_{i,t}) + \epsilon_{i,t}, \tag{1}$$

where the unknown function $m(\cdot)$ defines the mean of pay conditional on wealth. We estimate this function using a local linear regression with a bandwidth selected by cross-validation.[2]

Figure 2 reports this non-parametric relationship, which reproduces the estimated function in Figure 1. In the same figure, we also plot this relation controlling for observable differences between households by estimating a semi-parametric relationship of the form

$$y_{i,t} = m(a^p_{i,t}) + X'_{i,t}\beta + \epsilon_{i,t}, \tag{2}$$

[2]We use the same bandwidth across all regressions to aid comparability and to avoid numerical issues in estimating higher order moments when the bandwidth differs.

where X includes year, a cubic in age, education, race and education-specific age trends.

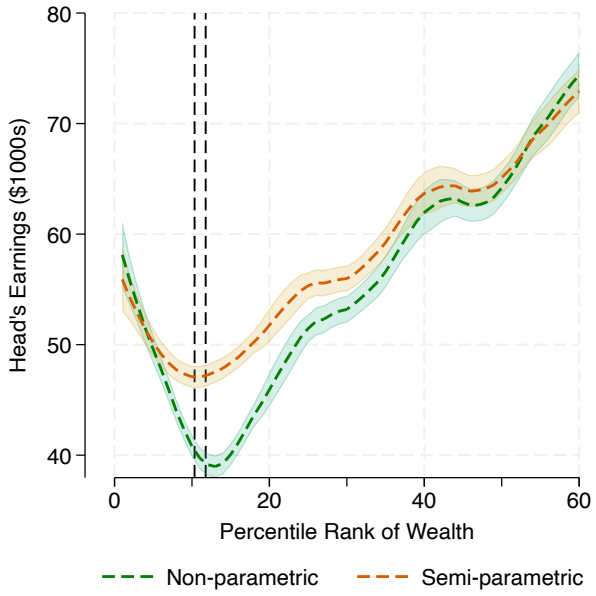


Figure 2: Relationship between wealth and annual pay

Notes. Figure plots estimated $m(\cdot)$ functions, along with 95 percent confidence intervals, from non-parametric regression (1) and semi-parametric regression (2). In the semi-parametric regression, the vector of controls X includes year, a cubic in age, education, race and education-specific age trends. The vertical lines delimit the region where wealth is near-zero, defined as lying between −$250 and $250.

Our estimates reveal a clear pattern: average pay is non-monotonic in wealth. At the bottom of the wealth distribution, average pay declines in wealth. This relationship reverses at around the 12th percentile of the wealth distribution – just above the point at which wealth equals zero – and average pay then increases in wealth. The difference is substantial: households in the bottom wealth percentile earn $58,118 on average, compared with $36,142 among households with zero net wealth. The households with the least wealth are therefore not those with the lowest average pay. Adding controls attenuates the V-shaped pattern, indicating that part of it reflects differences in observable characteristics over the wealth distribution, but a clear non-monotonicity remains even after controlling for these differences.

In Table 2, we demonstrate the robustness of the non-monotonic relationship across a range of parametric specifications. We estimate a piece-wise linear regression of the form:

$$y_{i,t} = \beta_0 + \beta_1 a^p_{i,t} + \delta_0 \mathbf{1}\left\{a_{i,t} \geq 0\right\} + \delta_1 a^p_{i,t} \mathbf{1}\left\{a_{i,t} \geq 0\right\} + X'_{i,t}\theta + \epsilon_{i,t}, \tag{3}$$

This piecewise-linear specification allows the relationship between wealth and pay to have

different slopes above and below zero wealth. In this specification we additionally control for household-level characteristics $X_{i,t}$.

Table 2 reports our estimates of the parameters of interest. Column (1) reiterates the pattern shown in Figure 2. In column (2) we add controls for age and race, and in column (3) we control also for education. While these controls attenuate the slope coefficients, the non-monotonicity remains statistically significant across all specifications. In the specification reported in column (3) an increase of one percentile rank is associated with an increase in pay of $1,274 for those with positive wealth. In contrast, for those with negative wealth an increase of one percentile rank is associated with a decrease in pay of $1,038.

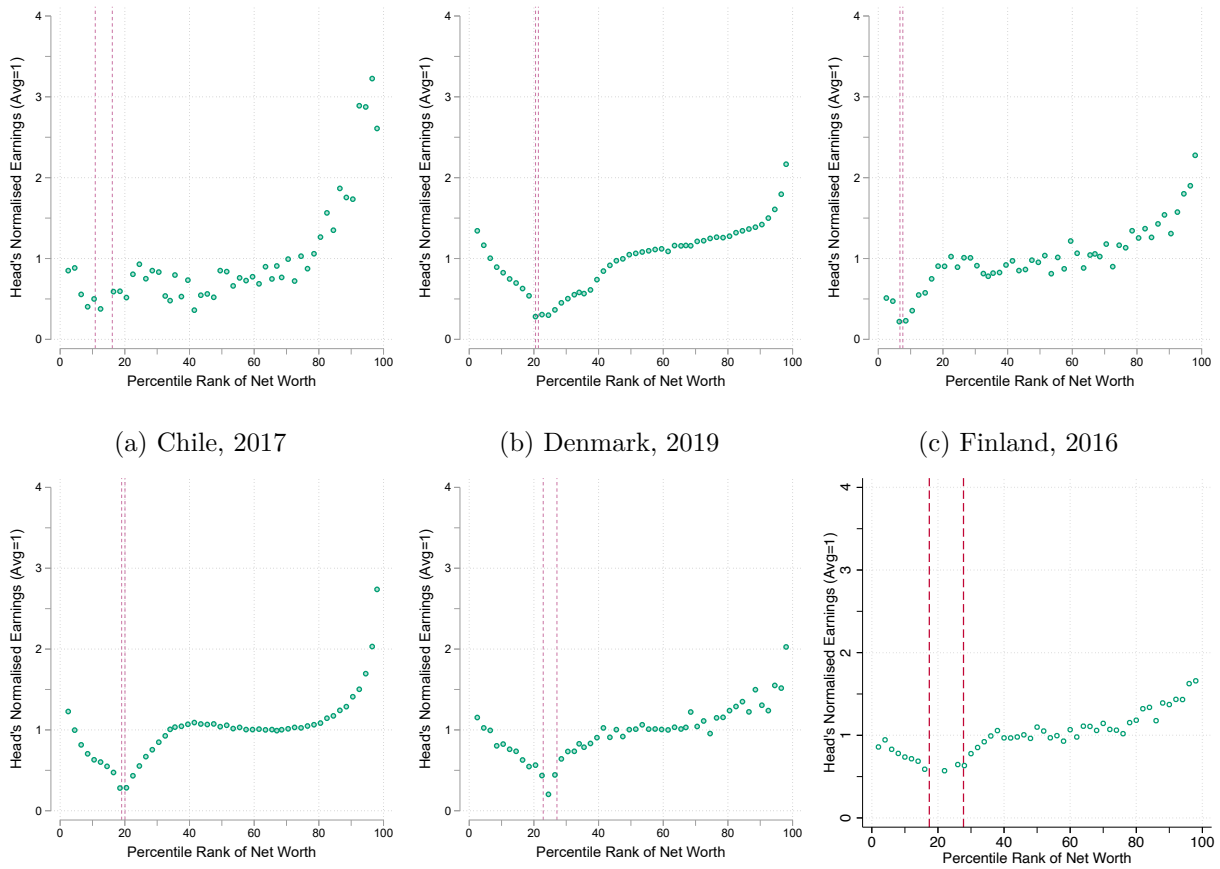
Table 2: Parametric test for non-monotonicity

	(1)	(2)	(3)
Wealth rank (for $a_{i,t} \geq 0$)	1453.1***	1420.8***	1274.2***
	(123.5)	(123.6)	(121.2)
Wealth rank (for $a_{i,t} < 0$)	-1655.3***	-1541.6***	-1038.2***
	(357.7)	(356.4)	(346.5)
State FE	✓	✓	✓
Year FE	✓	✓	✓
Age		✓	✓
Race		✓	✓
Education			✓
Observations	26,383	26,317	26,317

Notes. Coefficients report the change in annual pay associated with a one-percentile increase in wealth rank, separately for households with negative and non-negative net worth. Standard errors clustered at the household level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

2.3 Evidence Across Countries

The non-monotonic relationship between pay and wealth is also present across countries with different labor-market institutions, tax-and-transfer systems and sources of household debt. Figure 3 shows the relationship for Chile, Denmark, Finland, Norway, Sweden and the UK. We focus on the countries for which we observe sufficient mass below zero wealth to identify the downward-sloping part of the relationship: in particular, those with at least 5% with negative wealth in our data. Appendix C reports the complete set of countries for which we have constructed comparable evidence.



(a) Chile, 2017 (b) Denmark, 2019 (c) Finland, 2016

(d) Norway, 2019 (e) Sweden, 2005 (f) UK, 2012–13, 2016–17

Figure 3: Relationship between wealth and pay across selected countries

Notes. Each panel plots average pay against the percentile rank of net worth within the corresponding country. Each point represents average pay within a two-percentile interval of the wealth distribution, normalized by average pay in that country. The vertical lines delimit the percentiles within 100 local currency units of zero wealth. Data are from the Luxembourg Wealth Study (LWS) for Chile, Denmark, Finland, Norway and Sweden, and from the UK Household Longitudinal Study (UKHLS) for the UK. We select countries with at least 5% of observations below zero net worth, using the criterion described in Appendix C. Appendix B provides further details on the data and variable definitions.

As in Figure 1, each point represents average pay in a two-percentile interval on the wealth distribution. The vertical lines indicate the size and position of any mass around zero wealth measured for each country. The non-monotonic relationship between wealth and pay is clearly visible across these countries. Its presence across economies with different labor-market institutions, tax-and-transfer systems and arrangements for educational financing suggests that the pattern is not specific to a particular feature of the US institutional environment.

2.4 Understanding the Non-Monotonicity

The evidence so far establishes a robust non-monotonic relationship between wealth and pay. It survives conditioning on observable characteristics and appears across countries with very different institutional settings. We now ask: what are the economic forces that drive this relationship?

We document a series of additional empirical facts to help answer this question. First, we show that the relationship reflects persistent differences in earnings across workers, rather than differences in labor supply or transitory pay variation, and that these differences are primarily associated with workers rather than the occupations in which they work. Second, we show that pay dispersion is also non-monotonic in wealth. We finally show that the main findings are not driven by the particular way we measure wealth or pay, by educational debt, by the mass of households reporting zero wealth, or by the sample restrictions used in our baseline analysis.

Taken together, this evidence identifies the key features that an economic explanation of the non-monotonicity must account for and motivates the model we develop in Section 3. We summarize the main evidence here and report the full set of exercises in Appendices D and E .

Persistent worker heterogeneity. We begin by asking what features of workers' earnings account for the non-monotonic relationship with wealth. Annual pay could vary over the wealth distribution because hourly pay differs or because workers supply different hours. Appendix E.1 shows that annual hours conditional on working are broadly flat across the wealth distribution. We find evidence of a small non-monotonicity at the extensive margin which, if anything, would reinforce the non-monotonicity we have documented among workers. But the variation is quantitatively small relative to the kink in pay among employed household heads. This indicates that differences in labor supply – either in hours or participation – do not drive the non-monotonicity in pay, which instead reflects variation in wage rates across the wealth distribution.

We next ask whether the differences in pay across the wealth distribution reflect permanent heterogeneity across workers, or variation in workers' pay over time. Such variation could reflect either transitory earnings shocks or predictable changes in pay over the life cycle. We construct a measure of permanent earnings using workers' earnings histories: we first residualize log annual earnings with respect to a cubic in age and year fixed effects, and then average these residuals across all observations for each worker. This removes predictable life-cycle and aggregate variation in earnings, while averaging out transitory variation across years.

Figure 4 shows that the same non-monotonic pattern remains: permanent earnings are relatively

high among households at the bottom of the wealth distribution, fall as wealth approaches zero, and then rise again through the positive-wealth distribution. The pay–wealth kink therefore does not arise solely from variation in workers' earnings over time, whether due to transitory earnings shocks or predictable life-cycle changes.

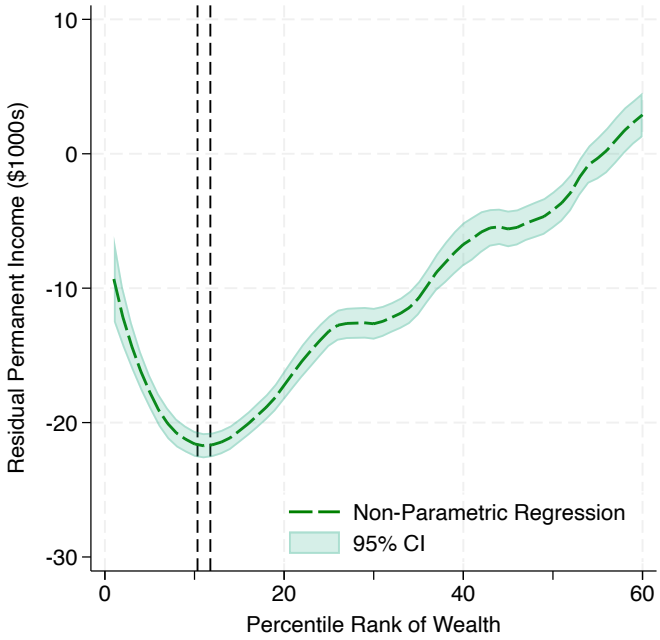


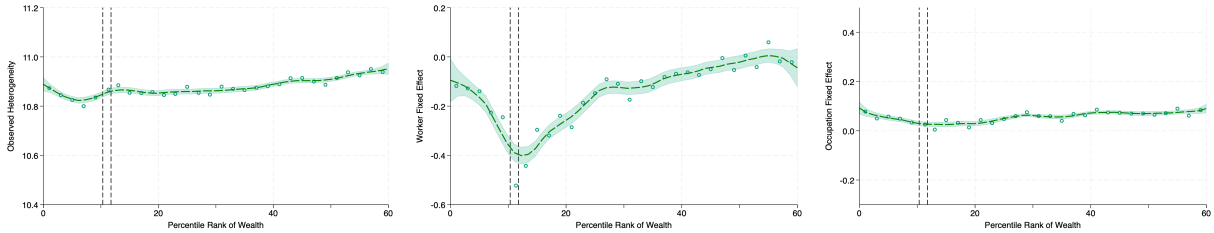
Figure 4: Permanent earnings and wealth

Notes. Figure plots the estimated relationship between permanent earnings and the percentile rank of wealth. We construct permanent earnings by first regressing log annual earnings on a cubic in age and year fixed effects. We then average the residuals across all observations for each worker.

A natural next question is what drives these persistent differences in earnings? Do they arise because workers with different characteristics are located in different parts of the wealth distribution, or because otherwise-similar workers sort into jobs with different pay? To distinguish these possibilities, we estimate an AKM-style decomposition of log earnings into an observable time-varying component (a cubic in age and year fixed effect), permanent worker heterogeneity, and permanent occupation heterogeneity. Appendix D describes the estimation and decomposition in detail.

Figure 5 plots each estimated component over the wealth distribution. The worker fixed effect varies strongly and non-monotonically across the wealth distribution, while the occupation effect exhibits a much smaller non-monotonicity. The component attributable to observable heterogeneity is flat over the wealth distribution. The evidence therefore points primarily to persistent heterogeneity

across workers as the source of the non-monotonic relationship between wealth and pay.



(a) Observed worker component (b) Worker fixed effect (c) Occupation component

Figure 5: Components of pay over the wealth distribution

Notes. Panels plot the estimated non-parametric relationship between wealth and components of log pay from the AKM decomposition described in Appendix D. Panel (a) shows observed worker heterogeneity; panel (b) shows the worker fixed effect; panel (c) shows the occupation fixed effect.

Despite being quantitatively small, the non-monotonicity in the occupation component could nevertheless reflect endogenous occupational choice if changes in wealth affect workers' occupational mobility. Appendix E.2 examines whether changes in workers' wealth percentile rank are associated with (i) the probability of changing occupation and (ii) changes in the estimated occupation pay premium, allowing these relationships to differ between workers with negative and non-negative wealth. We find no statistically significant relationship between changes in wealth and either measure of occupational mobility, nor evidence that these relationships differ across zero wealth. Taken together, these results suggest that occupations explain only a small fraction of the pay–wealth relationship, with little evidence that even this component arises from endogenous occupational choice.

Pay dispersion. We now document a second feature of the pay–wealth relationship that guides the model. Pay dispersion also follows a non-monotonic pattern over the wealth distribution. To document this fact, we construct the squared residual from regression (1), $(y_{i,t} - \hat{m}(a^p_{i,t}))^2$, and estimate

$$(y_{i,t} - \hat{m}(a^p_{i,t}))^2 = s(a^p_{i,t})^2 + u_{i,t}. \tag{4}$$

where $s(\cdot)$ defines the standard deviation of pay conditional on wealth, with $s(\cdot)^2$ the corresponding variance. We report the estimated function $\hat{s}(a^p_{i,t})$ in Figure 6.

Figure 6 shows a clear non-monotonic relationship between wealth and pay dispersion, closely mirroring the pattern for average pay. At low levels of wealth, the standard deviation of pay

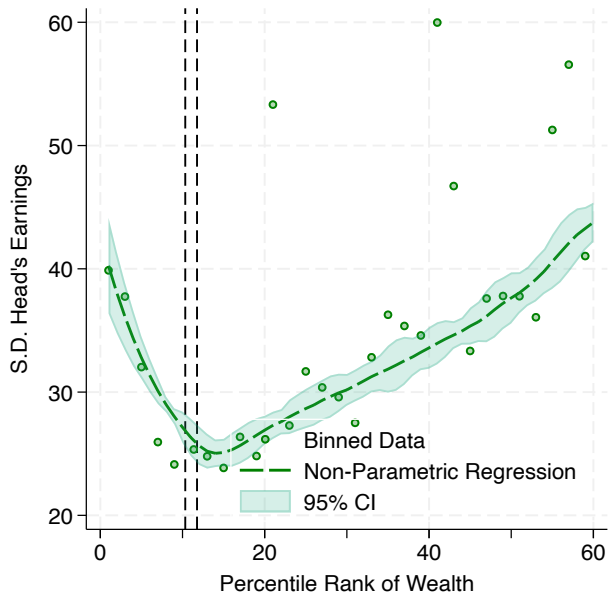


Figure 6: Non-monotonic relationship between pay dispersion and wealth

Notes. We construct our estimate of the conditional variance using residuals from the relationship estimated in Figure 1. The asymptotic distribution of this estimator is derived in Fan and Yao (1998) and we use the delta method to construct the asymptotic distribution for the conditional standard deviation.

declines as wealth increases, before rising again beyond zero. Any explanation for the non-monotonic relationship between wealth and pay must also account for this pattern of pay dispersion.

Robustness and additional results. We finally turn to a set of robustness exercises and additional results. We first examine whether the non-monotonic relationship between pay and wealth is sensitive to our variable definitions, sample restrictions, or the populations and periods considered. Appendix E.3 considers alternative definitions of wealth, including liquid wealth and wealth excluding student debt, while Appendix E.4 considers alternative measures of current and permanent pay, including hourly pay and household income after taxes and transfers. Appendix E.5 varies the age, marital-status and employment restrictions we impose on our sample; Appendix E.6 estimates the relationship by PSID wave; Appendix E.7 treats the self-employed separately; and Appendix E.8 examines the relationship throughout the pay distribution and also conditional on educational attainment. Across these exercises, we continue to find a clear non-monotonic relationship between pay and wealth.

A separate concern is that the kink may be driven by the mass of households reporting exactly zero net wealth. Across the countries we study, the turning point in the pay–wealth relationship

occurs at or near zero wealth, which is also typically a mass point in the wealth distribution. To assess whether these observations drive our findings, Appendix E.9 re-estimates the piecewise-linear specification after progressively excluding observations around zero wealth. The negative slope below zero and positive slope above zero persist even when increasingly wide intervals are excluded, indicating that the kink is not an artifact of the mass point at zero.

Finally, Appendix E.10 extends the analysis to household consumption, documenting a similar V-shaped relationship with wealth.

Summary. Taken together, these results suggest that the pay–wealth kink is driven primarily by persistent heterogeneity across workers, rather than by differences in labor supply, transitory earnings variation, predictable life-cycle earnings profiles, or occupational sorting. At the same time, pay dispersion follows a similar non-monotonic pattern over the wealth distribution. These relationships are robust to alternative definitions of wealth and pay, different sample restrictions, and the mass of households reporting zero wealth.

These facts directly guide the model we build next, in which workers with permanent earnings heterogeneity, facing uninsurable wage risk, select endogenously into different parts of the wealth distribution.

3 The Model

We develop an incomplete-markets model in which households differ in permanent earnings capacity and face uninsurable income risk. Households smooth consumption through saving and borrowing, subject to borrowing limits that depend on expected earnings. Workers with higher permanent earnings capacity can take on larger debt positions, while also accumulating greater precautionary savings. As a result, higher-earning worker types extend further into both tails of the wealth distribution, while lower-earning types remain concentrated closer to zero wealth. This endogenous selection of worker types across the wealth distribution generates a non-monotonic relationship between wealth and pay.

Setup. Time is discrete and indexed by t, and the horizon is infinite. The economy is populated by a continuum of households of measure one. Households supply labor inelastically, receive stochastic labor income, consume, and trade a risk-free asset. Financial markets are incomplete: households

can save and borrow to smooth consumption, but cannot insure directly against idiosyncratic income risk.

Preferences. Households have preferences over consumption c_t represented by a von Neumann–Morgenstern utility function $u(c)$ that is twice differentiable, increasing, and strictly concave ($u' > 0$, $u'' < 0$). For the quantitative exercise below, we assume constant relative risk aversion, $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$, with risk aversion given by γ. Agents discount future utility at rate $\beta < 1$.

Labor Income. We represent persistent heterogeneity through an exogenously given permanent worker type α, drawn from a finite set.

In each period, a household supplies one unit of labor at wage w_t, which is stochastic. The wage depends on the household's permanent type, as well as the sum of a persistent component, z_t, which follows an AR(1), and a transitory component, ν_t, which is modeled as serially uncorrelated:

$$\log w_t = \mu + \alpha + z_t + \nu_t \tag{5}$$

$$z_t = \rho z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad \nu_t \sim \mathcal{N}(0, \sigma_\nu^2), \quad \alpha \sim \mathcal{N}(0, \sigma_\alpha^2), \tag{6}$$

where $\rho \in (0, 1)$ is the persistence of the process, μ governs the unconditional mean, and σ_ε is the standard deviation of the innovation. The transitory component is MA(0), a choice governed by our biennial data, with standard deviation σ_ν. Log pay thus follows a normal distribution with mean $\mu + \alpha$ and variance $\frac{\sigma_\varepsilon^2 + (1-\rho^2)\sigma_\nu^2}{1-\rho^2}$. Pay in levels is therefore lognormally distributed. Higher-type workers have both higher expected pay and greater dispersion of pay in levels.

Financial Frictions. There are no insurance markets to hedge labor income risk. Households instead self-insure through saving and borrowing. They can trade a risk-free bond, but face a wedge between borrowing and lending rates:

$$r(a) = \begin{cases} r_L & \text{if } a \geq 0, \\ r_B & \text{otherwise,} \end{cases} \tag{7}$$

where $r_B > r_L$ and the spread reflects the cost of default.[3] We interpret the spread between borrowing and lending rates as a reduced-form cost of unsecured borrowing. As shown by Achdou et al. (2022), such a wedge also helps match the wealth distribution by generating a concentration of households around zero wealth, as observed in the data.

Household borrowing is also subject to a type-specific lower bound on wealth, $\underline{a}(\alpha) \leq 0$, which depends on expected labor income. The corresponding maximum amount of debt for a household of type α is therefore $-\underline{a}(\alpha)$.

We impose a loan-to-income constraint and a common ceiling on debt. The loan-to-income constraint allows borrowing up to κ times expected labor income. We denote by $\underline{a}_{\max} < 0$ the common minimum permitted wealth position, corresponding to maximum debt of $-\underline{a}_{\max} > 0$. The effective lower bound on wealth is the more restrictive (less negative) of the two limits:

$$\underline{a}(\alpha) = \max\left\{-\kappa \mathbb{E}(w \mid \alpha), \underline{a}_{\max}\right\}, \tag{8}$$

where $\mathbb{E}(w \mid \alpha)$ denotes expected labor income conditional on permanent worker type α, integrating over the stationary distributions of the persistent and transitory income components. The parameter κ governs the loan-to-income constraint, while $\underline{a}_{\max}$ limits the maximum amount of debt irrespective of expected income. This reduced-form specification captures the idea that lenders extend larger credit lines to borrowers with higher expected income, subject to an overall debt ceiling (e.g., Stiglitz and Weiss, 1981; Chatterjee et al., 2007; Livshits, MacGee, and Tertilt, 2007).

Because borrowing limits are finite, households cannot fully insure against adverse income realizations. Instead, they smooth consumption using two imperfect instruments: precautionary saving and debt accumulation. This determines how different permanent worker types select into different parts of the wealth distribution.

Taxes and Transfers. Disposable income differs from market income in two ways. First, households are guaranteed a minimum level of pre-tax income, $\underline{Y}$. This captures both formal social-insurance programs and informal sources of support that are not modeled explicitly.

Second, households face a progressive tax-and-transfer schedule. We adopt an isoelastic functional form following Feldstein (1969); Bénabou (2002) and Heathcote, Storesletten, and Violante (2017),

[3] We do not model default explicitly. The differential between borrowing and lending rates does not drive the V-shaped pattern but plays a role in matching the data quantitatively.

where the effective average tax rate on taxable income w is

$$\mathcal{T}(w) = 1 - \lambda\,(w)^{-\tau}\,, \tag{9}$$

where $\lambda > 0$ governs the overall level of taxation and $\tau \geq 0$ controls progressivity. When $\tau = 0$ the tax schedule is proportional; larger values of τ imply greater redistribution. After-tax income, incorporating the income floor, is therefore

$$\mathcal{W}(w) = \lambda\,(\max\{\underline{Y}, w\})^{1-\tau}\,. \tag{10}$$

Timing. A period is one year. A household enters period t with wealth a_t and permanent type α. The persistent and transitory earnings components, z_t and ν_t, are then realized, determining current labor income w_t. After observing current income, the household chooses consumption c_t and next-period wealth a_{t+1}.

Actions and value functions. The problem of a household with permanent worker type α, current wealth a, and wage w is

$$\begin{aligned} V(a, z, \nu; \alpha) = \max_{c,\, a'} & \left\{ u(c) + \beta\, \mathbb{E}_{z',\nu'} \Big[V(a', z', \nu'; \alpha) \;\Big|\; z \Big] \right\} \\ \text{s.t.} \quad & (1 + r(a))\, a + \mathcal{W}(w) - c \geq a', \\ & a' \geq \underline{a}(\alpha). \end{aligned} \tag{11}$$

Steady state. Given the interest rates r_L and r_B, a steady state consists of household policy functions for consumption and saving and a stationary distribution of households over permanent worker types, earnings states, and wealth. Household policy functions solve the problem in equation (11), and the stationary distribution is invariant under these policies and the earnings process. The interest rates are disciplined by the wealth distribution in the quantitative exercise in Section 5.

4 A Simple Example to Illustrate the Mechanism

Before turning to our full quantitative exercise, we first build intuition for how self-insurance decisions can lead to endogenous selection of workers with different permanent earnings types across the wealth distribution, generating the V-shaped relationship between wealth and pay. To do so, we

consider a deliberately stylized version of our model that is analytically tractable.

We focus on the special case where there are two worker types, $\alpha \in \{\alpha_L, \alpha_H\}$. Consistent with our empirical evidence, we use $m(a) = \mathbb{E}[w_t | a_t = a]$ to denote expected pay at a given level of wealth, measured at the start of the period, and $m_j(a) = \mathbb{E}[w_t | a_t = a, \alpha = \alpha_j]$ for type $j \in \{L, H\}$.

Let p_j denote the population share of type j and let $p_j(a) = \Pr(\alpha = j \mid a)$ denote the share of households at wealth level a who are type j. Average pay across types then decomposes as

$$m(a) = \underbrace{m_H(a)}_{\text{within-type mean}} + \underbrace{p_L(a)\left(m_L(a) - m_H(a)\right)}_{\text{composition effect}}. \tag{12}$$

Even if the within-type means $m_j(a)$ are flat or monotone in wealth, non-monotonic composition—driven by the saving and borrowing choices of heterogeneous households—can generate the V-shape we document above.

To isolate this composition effect, we deliberately focus on a case in which pay is IID across periods and takes a finite number of values: a worker of type j earns $w_t = \theta_j z_t$, where $z_t \in \{z_L, z_H\}$, with $z_H > z_L > 0$, is drawn independently each period, taking the value z_H with probability π, and $\theta_H > \theta_L$.[4] In this stylized setting, since current wealth is chosen before the income shock is realized, mean pay within each type, $\overline{w}_j = \theta_j \, \mathbb{E}[z_t] = \theta_j \left(\pi z_H + (1-\pi) z_L\right)$, does not vary with wealth, so the only source of non-monotonicity in the pay–wealth relationship is changing compo...sition:

$$m(a) = \overline{w}_H + \underbrace{p_L(a)\left(\overline{w}_L - \overline{w}_H\right)}_{\text{composition effect}}. \tag{13}$$

Our stylized version of the model is analytically tractable and eliminates other, quantitatively important, features of the model that are not central to characterizing the channel. We therefore assume preferences are CRRA, that taxes are proportional ($\tau = 0$) with no binding income floor, and that there is no wedge between borrowing and lending rates, so that $r(a) = r$. We additionally assume that the common debt ceiling does not bind for either type, so that borrowing limits are proportional to expected pay, $\underline{a}(\alpha_j) = -\kappa \, \overline{w}_j$. Under these assumptions, the high type's problem is an exact scaled-up copy of the low type's: in every period, both income and borrowing capacity are $\eta = \theta_H / \theta_L > 1$ times larger. Full details of the environment are in Appendix F. This delivers the following analytic characterization of our key mechanism.

[4] In Appendix F.1, we show that the IID assumption can be relaxed and extended to arbitrarily many income realizations.

Proposition 1 (V-shaped pay–wealth profile) *In the two-type IID environment of Appendix F, suppose $\beta R < 1$, where $R = 1 + r$. Let $\underline{a}_L$ and $\overline{a}_L$ denote the lowest and highest wealth levels held by low types in the stationary distribution, and suppose low types both borrow and save, $\underline{a}_L < 0 < \overline{a}_L$. Then $\overline{a}_L$ is finite and:*

i. The lowest and highest wealth levels held by high types are $\eta\,\underline{a}_L$ and $\eta\,\overline{a}_L$, so the high-type wealth distribution extends strictly beyond the low-type distribution in both tails.

ii. Only high types hold wealth below $\underline{a}_L$ or above $\overline{a}_L$, and in these tails $m(a) = \overline{w}_H$.

iii. Wherever low types are present, $m(a) < \overline{w}_H$.

Hence expected pay is high in both tails of the wealth distribution and lower wherever low types are present: a non-monotone relationship between wealth and pay. Furthermore, when the stationary distributions for each type are single peaked, there is a single interior minimum and so the non-monotone relationship is a V-shape.

The endogenous mechanism delivering this composition channel is as follows. Because utility is homothetic and both income and borrowing limits scale with type, high-type policy functions are simply η-scaled versions of low-type policies. Facing the same sequence of low income shocks, the high type's greater borrowing capacity allows them to take on more debt to self-insure. Bounded patience, $\beta R < 1$, ensures even the most fortunate households who experience infinite sequences of positive income realizations choose to accumulate a finite buffer stock and consume excess resources today. The result is that high types *exclusively* occupy both tails of the wealth distribution. Since shocks are IID, within-type mean income is independent of wealth, so composition – not selection on income realizations – drives the V-shape entirely through equation (13). This is demonstrated in Figure 7.

In the quantitative model, we reintroduce persistent and transitory earnings shocks, taxes and transfers, the borrowing–lending rate wedge, and the richer borrowing constraint described above. We show that the same underlying selection mechanism continues to generate the non-monotonic relationship between wealth and pay in this richer environment. Appendix F.1 shows that the example itself extends to persistent income shocks, provided differences in pay across types are large relative to the persistence of income.

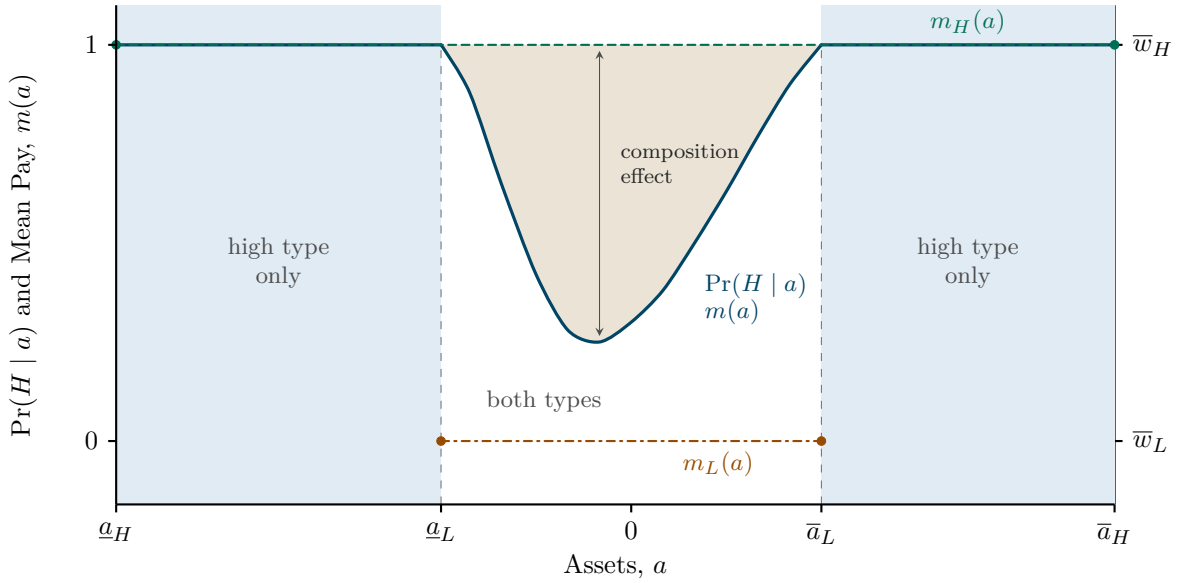


Figure 7: Pay–wealth profile in the stylized model

Notes. This figure illustrates the mechanism of Proposition 1 in the two-type IID environment of Appendix F. The left axis plots the conditional probability of the high type, $\Pr(H \mid a)$, over the wealth distribution. Shaded regions indicate wealth levels occupied exclusively by high types: the left tail below $\underline{a}_L$ (due to wider borrowing capacity) and the right tail above $\overline{a}_L$ (due to larger buffer stocks). The right axis plots within-type mean pay $m_H(a)$ and $m_L(a)$, which are flat under IID shocks, alongside the aggregate profile $m(a)$. The shaded wedge shows the composition effect $p_L(a)\big(m_H(a) - m_L(a)\big)$ from equation (12), which pulls $m(a)$ below $\overline{w}_H$ in the overlap region and generates the V-shaped pay–wealth profile.

5 Quantitative Exercise

We now quantify the model and assess its ability to account for the empirical patterns documented in Section 2. We first describe the calibration strategy and then assess the fit of the model to the joint distribution of pay and wealth. We finally use the calibrated model to quantify how workers with different permanent earnings types select into different parts of the wealth distribution.

5.1 Calibration

We solve the model numerically and compute its stationary distribution, with the computational procedure described in Appendix G.1. We adopt a two-step strategy to calibrate the model in the spirit of Gourinchas and Parker (2002).

External calibration. In the first step, we estimate or calibrate those parameters that, given our assumptions, can be cleanly identified outside our model. These include the discount rate, parameters of the income process, including the distribution of permanent heterogeneity and income

risk, parameters of the tax system, and a subset of the parameters disciplining financial markets. We summarize these in Table 3.

We fix the annual discount factor, β, at 0.96, a parameterization consistent with common values in the heterogeneous-agent literature (Krusell, Lou, and Ríos-Rull, 2019; Aiyagari, 1994; Huggett, 1993b).

To estimate the stochastic process for pay, we assume observed pay is given by

$$\log w_{i,t}^{\text{observed}} = X_{i,t}\beta + \log w_{i,t} + \text{meas. error}_{i,t}, \tag{14}$$

where $\log w_{i,t}$ follows equations (5) and (6) while also allowing for measurement error (Meghir and Pistaferri, 2011). The vector $X_{i,t}$ includes a cubic polynomial in age and calendar year fixed effects. All other (time-invariant) worker heterogeneity is captured by the persistent type. We demean these controls before estimation, so that the regression intercept captures average log pay; we set μ equal to this intercept. We then remove the predictable life-cycle and aggregate components of earnings and recover the remaining parameters of the earnings process using a standard minimum-distance procedure. The implementation of this estimation, together with the numerical solution and simulation of the model, is described in Appendix G.1.

Our data do not allow us to separately identify measurement error and transitory earnings shocks. We therefore assume that measurement error accounts for 25% of transitory variation, following Pischke (1995)'s validation study of the PSID. The resulting estimates imply that permanent heterogeneity accounts for approximately 65% of the cross-sectional variation in residual earnings, while the persistent component contributes a further 22%. The remaining 13% reflects a combination of transitory shocks and measurement error. We use these to calibrate the variances of the normally distributed components in equation (6).

Our estimate of the persistence parameter, ρ, is somewhat lower than is typically reported in the literature. This reflects the fact that our specification includes an additional source of permanent heterogeneity, α, which absorbs part of the persistence that is usually captured by the autoregressive component.[5]

We set parameters of the tax system to match the US tax system using values in Eeckhout and Guner (2026). Their estimates imply an average tax rate of 14.4% for a worker with average

[5]Indeed, when we estimate a version of the model without permanent heterogeneity α, the estimated persistence parameter rises to 0.96 in a specification that continues to account for measurement error, very close to conventional estimates in the literature.

earnings.[6] The income floor is set to $\underline{Y} = \$10,000$ which is approximately two thirds of the federal poverty limit for a two person household in 2012 and around 80% of the 2012 SSI federal maximum for a couple.[7] The value is smaller than the real value of broad means-tested consumption floors calibrated by Hubbard, Skinner, and Zeldes (1995) or Scholz, Seshadri, and Khitatrakun (2006) and is similar, after equivalizing, to the value in Guvenen, Ozkan, and Madera (2024). It is, however, more generous than estimated Medicaid consumption floors (e.g., De Nardi et al. 2025) or values for non-medical transfers reported in Guner, Rauh, and Ventura (2024). This is consistent with other unmodeled sources of self-insurance as well as broad working-age social insurance floors.

The real return on saving is $r_L = 3\%$ per year and, in the baseline, we assume all workers have access to credit markets.

We set $\kappa = 3$, implying that borrowing capacity can reach three times expected annual labor income. This constraint is best interpreted as a reduced-form borrowing capacity across all available credit lines and borrowing instruments. However, it is broadly consistent with conventional loan-to-income ratios in various settings (Díaz and Luengo-Prado, 2008; Rosen, 2011; Peydró et al., 2024; Catherine, Ebrahimian, and Yannelis, 2026).

Internal calibration. In the second step, we calibrate the remaining parameters of the model: the coefficient of relative risk aversion, γ, and the borrowing rate, r_B. The maximum borrowing limit is not calibrated separately, but is pinned down by $\underline{a}_{\max} = -\frac{Y}{r_B}$ which ensures that debt repayment remains feasible for all workers and that borrowing limits remain below the natural borrowing limit implied by progressive taxation. Taking the externally calibrated parameters as given, we calibrate the remaining parameters using a minimum-distance procedure. Specifically, we choose parameter values so that moments generated by the simulated economy match their empirical counterparts as closely as possible, as measured by a GMM criterion function. Appendix G.1 describes the numerical implementation of this procedure.

Our objective is to explain the non-monotonic relationship between pay and wealth in the bottom 60 percent of the wealth distribution. We therefore target moments that characterize the joint distribution of pay and wealth in the lower tail of the wealth distribution. Let $\mathcal{P} = \{1, 5, 10, 15, \ldots, 60\}$ denote the set of target wealth percentiles. Then, the moments we target in

[6] We translate the parameter values defined for average earnings via $\lambda = \lambda^{EG} \frac{\mathbb{E}(w)}{\left(\mathbb{E}(w)\right)^{1-\tau}}$, which accounts for differences in the numeraire while maintaining comparability with the literature.

[7] U.S. Department of Health and Human Services (2012) and Social Security Administration (2012). Both De Nardi, Fella, and Paz-Pardo (2025) and Guvenen, Ozkan, and Madera (2024) emphasize the importance of income floors for quantifying consumption risk.

Table 3: External Calibration

Parameter	Definition	Value	Source/Target
Preferences			
β	Discount factor	0.96	Krusell, Lou, and Ríos-Rull (2019)
Income			
μ	Average (Log-)Pay	10.725	PSID
σ_α	Permanent Heterogeneity	0.680	PSID
ρ	Persistence of AR(1)	0.757	PSID
σ_ε	Innovations to AR(1)	0.219	PSID
σ_ν	Transitory Shock	0.281	PSID; Pischke (1995)
λ	Average Tax Rate	0.856	Eeckhout and Guner (2026)
τ	Tax Progressivity	0.127	Eeckhout and Guner (2026)
$\underline{Y}$	Income Floor	\$10,000	See text
Financial Frictions			
r_L	Return on saving (lending rate)	3%	
κ	Loan-to-Income Constraint	3.0	

our minimum distance procedure are:

1. Mean gross pay at each percentile in $\mathcal{P}$ of the wealth distribution.
2. The quantile function of the wealth distribution (i.e. the right cutpoint of the CDF), at each percentile in $\mathcal{P}$ of the wealth distribution.
3. The quantile function of the wealth distribution at the 80th percentile of the wealth distribution.
4. The share of households with negative net wealth (which is 10.4% in the PSID).

Our pay targets correspond to the non-parametric estimates reported in Figure 2. These targets help separately identify households' precautionary saving motives, and the wedge between borrowing and saving. We additionally target wealth holdings at the 80th percentile for two reasons: first, it provides additional identifying variation, as we discuss below; and second, it provides additional discipline on the upper part of the wealth distribution, which is important for the subsequent welfare and policy analysis.

We use a diagonal weighting matrix (Pischke, 1995) where each diagonal element corresponds to the inverse of the variance of the underlying moment condition in the data. This weighting scheme

Table 4: Internal Calibration

Parameter	Definition	Value	Source/Target
γ	Relative risk aversion	2.438	Jointly calibrated
r_B	Borrowing rate	7.194%	Jointly calibrated
$\underline{a}_{\max}$	Maximum borrowing capacity	-\$139,005	Implied by $\underline{a}_{\max} = -\underline{Y}/r_B$

accounts for both differences in scale across moments and differences in their statistical precision. Although less efficient asymptotically than the optimal weighting matrix, it is less susceptible to the small-sample problems discussed by Altonji and Segal (1996).

Calibration results. Table 4 reports the jointly calibrated parameters. The calibrated coefficient of relative risk aversion is $\gamma = 2.438$, close to a common benchmark value of $\gamma = 2$. This value is conservative relative to many structural estimates that identify risk aversion from observed precautionary wealth accumulation. For example, Cagetti (2003) estimates coefficients above 3 by matching median wealth-age profiles in the PSID, Gourinchas and Parker (2002) estimate $\gamma = 4.1$ in a life-cycle setting, and De Nardi et al. (2025) estimate $\gamma = 3.7$ by fitting the asset holdings of retired couples and singles facing medical-expense and longevity risk. Consequently, our calibration does not rely on an unusually strong curvature of utility. Instead, precautionary saving arises from the interaction of income risk, incomplete markets, and income-contingent borrowing constraints.

The borrowing rate is calibrated at $r_B = 7.194\%$, implying a spread of 4.194 percentage points over the lending rate. This value is broadly consistent with unsecured consumer credit rates in the United States over our sample period and with values used in related quantitative work. For example, Kaplan, Moll, and Violante (2018) calibrate a borrowing wedge of 6.5 percentage points in their HANK model, while Chatterjee et al. (2007) calibrate effective borrowing costs between 7 and 12 percent in models with endogenous default. Observed consumer credit spreads in the Federal Reserve G.19 data vary considerably across borrowing instruments, and our calibrated value should be interpreted as reflecting the average use of these instruments.

The maximum borrowing limit, $\underline{a}_{\max}$, is not a free parameter: it is pinned down by the cost of borrowing and the income floor through $\underline{a}_{\max} = -\underline{Y}/r_B = -\$139,005$. This allows high type workers to access large lines of credit, which is important to match observed utilization. Even with this generous borrowing limit, however, the model does not fully reproduce the extreme left tail of the wealth distribution. Matching the multi-million-dollar debt positions observed in the data would require a borrowing limit that implies almost no interest on debt or an unrealistically high

income floor.

We show the implications of these parameters for the borrowing capacity of different workers in Figure 8. Our calibrated parameters suggest the LTI constraint is binding for around 36% of workers, while the ceiling on borrowing capacity binds for the remaining 64%.

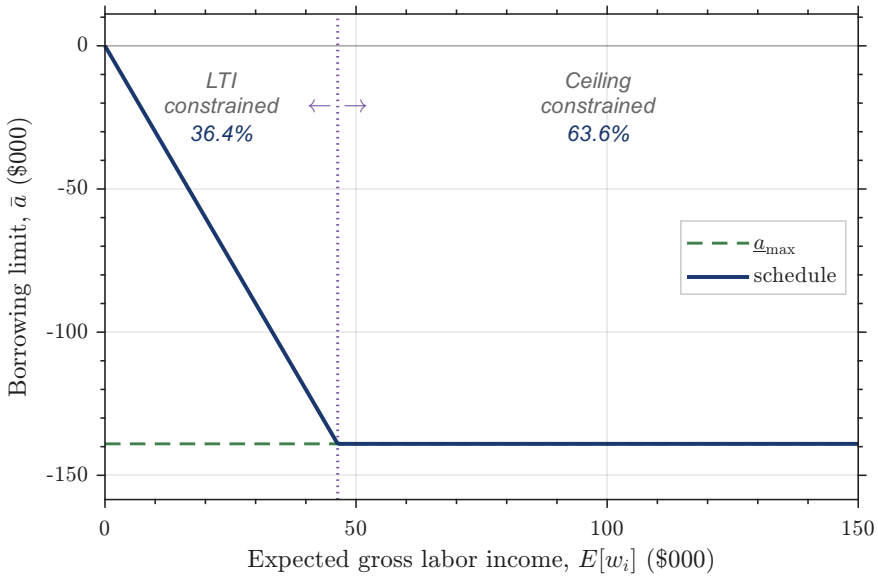


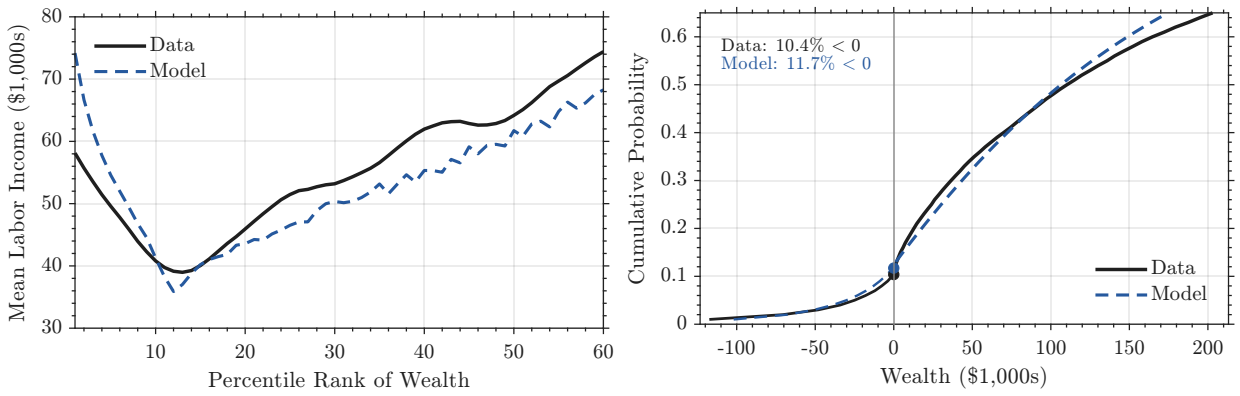
Figure 8: Heterogeneous Borrowing Capacity by Expected Income

Notes. This shows the calibrated worker-type-specific borrowing constraints as a function of their expected earnings. The functional form is piecewise linear as described in equation 8 and shown at the calibrated values reported in Table 3 and described in the main text. We display values in 1,000s of dollars.

Model fit. Figure 9 compares the empirical moments we target in estimation with their simulated analogs at the calibrated parameter values. The left panel plots average gross pay by wealth percentile in the data (black line) and in the simulated model (blue line). The model reproduces the central empirical pattern: pay is non-monotonic over the bottom 60 percent of the wealth distribution, falling as wealth rises from the left tail toward zero and then increasing again over the rest of the lower-wealth region.

The model generates this pattern through endogenous selection of permanent worker types across the wealth distribution. Higher-type workers face greater dispersion in pay and have access to more credit, allowing them to reach deeper debt positions following adverse income realizations, while also accumulating larger precautionary buffers. Lower-type workers are instead concentrated closer to zero wealth.

Panel (b) of Figure 9 compares the wealth distribution in the data and the model. The model



(a) Wealth and Annual Pay

(b) Distribution of Wealth

Figure 9: Model Fit: Wealth-Pay relationship and the Wealth Distribution

Notes. The left panel plots average gross annual pay against the percentile rank of wealth for households in the bottom 60 percent of the wealth distribution. The right panel plots the empirical and model cumulative distribution functions of wealth. The black line denotes empirical moments computed from the PSID baseline sample described in Section 2.1, and the blue line shows the corresponding moments simulated from the stationary distribution of the calibrated model at the parameter values reported in Tables 3 and 4. Externally set parameters are taken from the first-step calibration; the remaining parameters are calibrated by the method of simulated moments, minimizing a GMM criterion function over the moments described in the text.

captures the key features of the lower tail of the wealth distribution: a sizable mass of households with negative net worth, a sharp transition around zero wealth, and a gradual increase in holdings through the lower-middle percentiles. These patterns are important because they reflect the extent of debt utilization, the effective cost of borrowing, and the strength of precautionary saving in the calibrated economy.

The model abstracts from several sources of heterogeneity that influence households' position in the wealth distribution, including housing, student loans, medical debt, household composition, and differences in expected income growth. It therefore does not reproduce all local features of the empirical pay–wealth relationship: the relationship for those with negative wealth is slightly steeper in the model than the data, while the reverse is true for those with positive wealth. Nevertheless, the model closely reproduces the broad shape of the relationship and, importantly, generates the V-shaped pattern without relying on preference heterogeneity or behavioral assumptions beyond standard precautionary saving in the presence of incomplete markets and income-contingent borrowing constraints.

5.2 Endogenous Selection Across the Wealth Distribution

We now examine how the selection mechanism characterized in Section 4 operates quantitatively in the calibrated model. Households differ in their permanent earnings types, exposure to income risk, and borrowing capacity. Through their saving and borrowing decisions, workers of different permanent types select systematically into different parts of the wealth distribution.

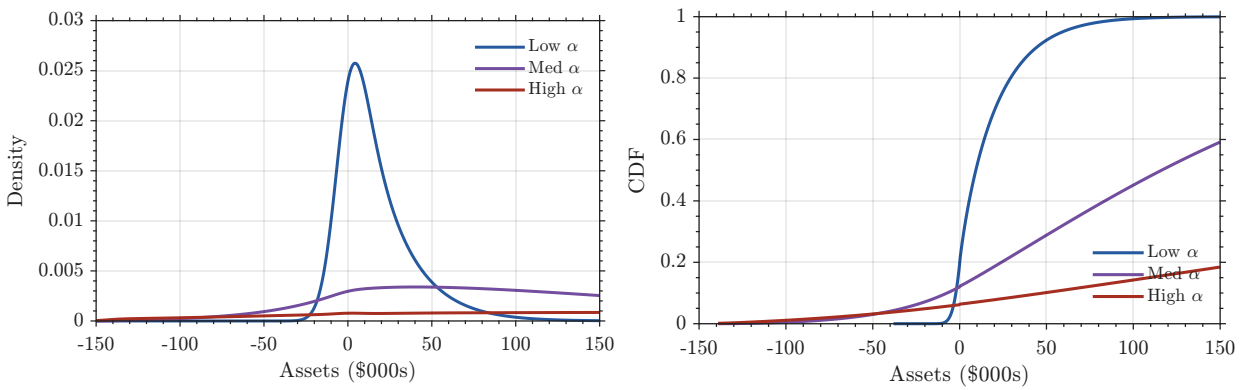
Figure 10 illustrates this selection by decomposing the stationary wealth distribution for three representative permanent earnings types, corresponding to the low (1st percentile), median, and high (99th percentile) points of the α distribution. Low types are concentrated around zero wealth. Because the income floor and progressive tax-and-transfer system provide relatively strong insurance against adverse income realizations, these households have limited incentives to accumulate precautionary wealth. At the same time, their low expected earnings restrict access to credit. As a result, they typically hold only modest amounts of debt and rarely accumulate substantial financial wealth.

Medium types (who are a larger share of the population than low types) are distributed much more widely across the wealth distribution. More than 70 percent accumulate over \$50,000 in savings, and their wealth distribution exhibits a substantially longer right tail. Although the fraction of households holding debt is similar to that among low worker types, debt utilization is considerably higher conditional on borrowing. These households therefore appear throughout a much larger portion of the wealth distribution.

Finally, higher types extend further into both tails of the wealth distribution. Because the income floor provides relatively little insurance against fluctuations in their earnings, these households face stronger precautionary-saving motives and accumulate large buffer stocks of wealth. At the same time, their high expected future earnings allow them to sustain substantial debt positions following severe adverse shocks. Consequently, high types appear disproportionately in both tails of the wealth distribution: they are over-represented among wealthier households, but also among those with the deepest debt positions.

Taken together, these patterns provide a quantitative counterpart to the selection mechanism in Section 4. Lower permanent earnings types are concentrated close to zero wealth, while higher types extend progressively further into both tails of the wealth distribution. The composition of permanent earnings types therefore varies non-monotonically across the wealth distribution.

Figure 11 shows how this selection generates the non-monotonic relationship between wealth



(a) Density by Type

(b) CDF by Type

Figure 10: The Composition of Worker Types in the Wealth Distribution

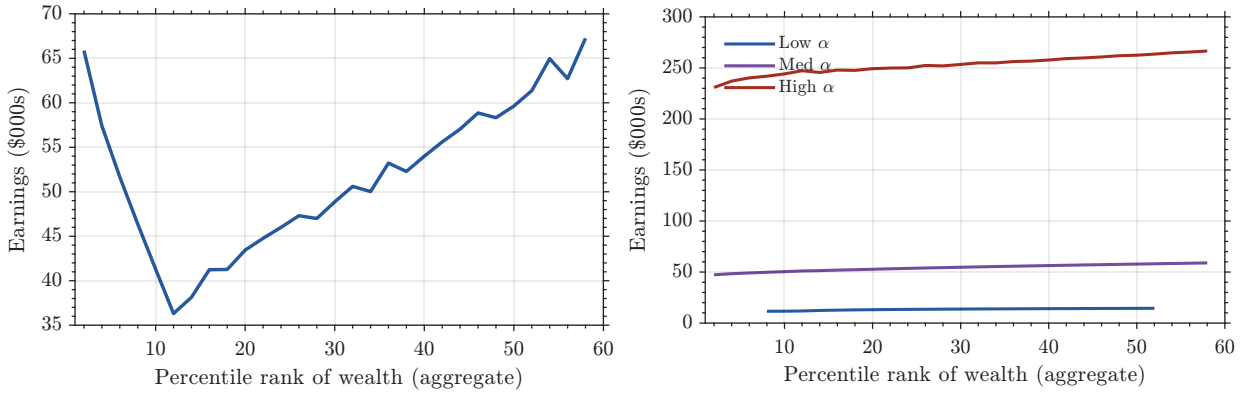
Notes. This figure decomposes the stationary wealth distribution from the calibrated model into contributions from three worker-type groups: low, medium and high. The model is evaluated at the parameter values reported in Tables 3 and 4.

and pay. The left panel reproduces the aggregate relationship in the simulated model from Figure 9, while the right panel shows the same relationship separately for the three worker types considered in Figure 10.

Within each type, pay is monotonically increasing in wealth, and at any given level of wealth higher types earn more on average than lower types. The aggregate non-monotonicity therefore does not arise within worker types. Instead, it reflects changes in composition across the wealth distribution: households close to zero wealth are disproportionately drawn from lower permanent earnings types, while households in both tails are increasingly drawn from higher types.

6 Welfare, Poverty and Redistributive Policy

Having shown that the non-monotonic relationship between wealth and pay arises from endogenous selection through precautionary saving and debt utilization, we now consider what this implies for welfare and redistributive policy. A central problem for such policy is identifying which households should receive support. While current income and wealth provide natural measures of current economic circumstances, they may not provide a good guide to economic need more broadly: households with few resources today may have relatively high expected future resources, while those with more current resources may face persistently low earnings prospects. The selection mechanism developed above makes this distinction especially relevant, since the most indebted households include workers with relatively high permanent earnings capacity following adverse



(a) Aggregate Pay

(b) Pay by Type

Figure 11: Wealth-Pay Relationship by Worker Type

Notes. The left panel plots average gross annual pay against the percentile rank of wealth in the simulated stationary distribution of the calibrated model, replicating the model series from Figure 9. The right panel plots the same relationship separately for low, medium and high worker types. In the right panel, pay for each group is plotted against percentiles of the population wealth distribution. The model is evaluated at the parameter values reported in Tables 3 and 4.

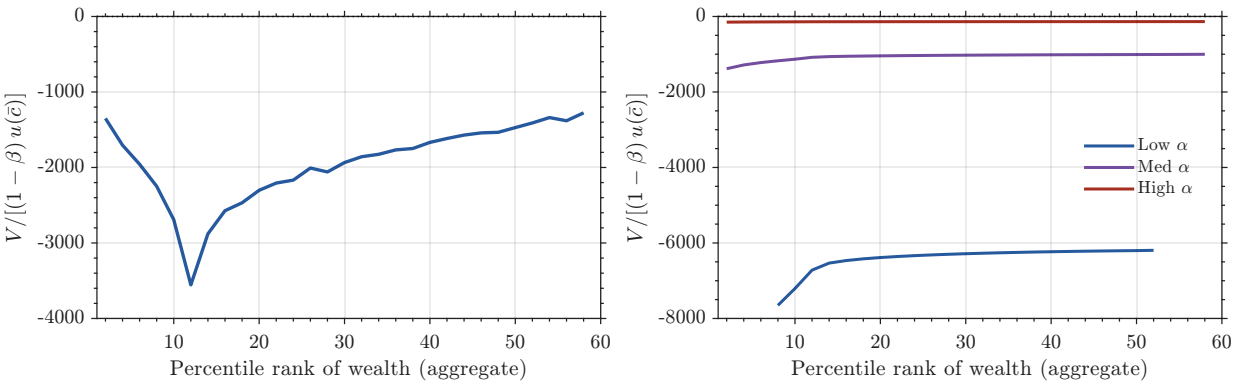
income realizations.

We use the calibrated model to examine these implications for welfare and policy. We first show how wealth maps into lifetime welfare, and compare poverty defined by current resources with poverty defined by lifetime welfare. We then decompose the marginal welfare value of a transfer into insurance against adverse current states and redistribution across permanent worker types. This distinction reveals that current pay and wealth may be informative about different sources of welfare gains from transfers. Finally, we examine how these differences affect the targeting of marginal transfers.

6.1 Wealth and Lifetime Welfare

The selection mechanism established above has a direct implication for welfare. Because higher worker types are disproportionately represented in both tails of the wealth distribution, while lower types are concentrated closer to zero wealth, wealth need not provide a monotone ranking of households by lifetime welfare.

Figure 12 plots expected lifetime utility against wealth, both in aggregate and separately by household type. Within each type, average lifetime utility is increasing in wealth because households with high wealth are better insured against future income risk and, on average, also have higher income draws. However, the composition of households changes systematically across the wealth



(a) Aggregate Value Function

(b) Value Function by Type

Figure 12: Lifetime Welfare Across the Wealth Distribution

Notes. The left panel plots average lifetime utility, measured by the value function $V(a, z, \nu; \alpha)$, against the percentile rank of wealth in the simulated stationary distribution. The right panel plots the same relationship separately for each of three types—low, medium and high. In both panels, the average at each wealth percentile integrates over the distribution of persistent labor income realizations conditional on wealth. The model is evaluated at the parameter values reported in Tables 3 and 4.

distribution. As a result, average lifetime utility is itself non-monotonic in wealth, closely mirroring the pattern we document for pay.

A similar pattern appears in consumption. Within each worker type consumption is increasing in wealth, while aggregate consumption is non-monotonic because the composition of worker types changes across the wealth distribution. Appendix G, Figure G.2 reports this pattern in the model, which corresponds closely to the empirical relationship shown in Appendix E, Figure E.9.

These results reveal a disconnect between wealth rank and welfare rank. Households near the bottom of the wealth distribution are not uniformly the worst off: those holding the most debt are disproportionately high-type workers borrowing against high expected future earnings, while lower-type households cluster near zero wealth with lower lifetime utility despite holding less debt.[8]

This disconnect motivates a distinction between two notions of poverty. A household may have few resources available today, or it may have low expected lifetime welfare. Because the composition of permanent worker types varies non-monotonically with wealth, these two notions need not identify the same households. We consider this distinction next.

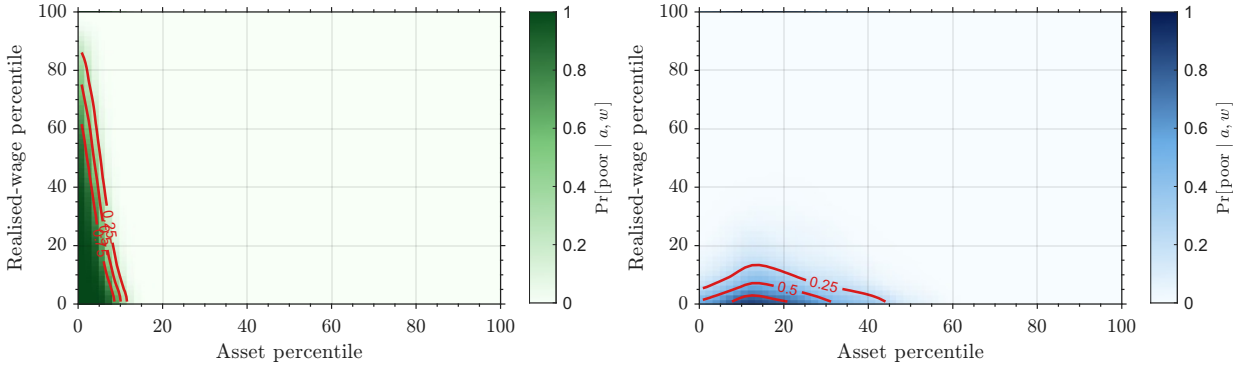
[8]This echoes, in a household context, the broader insight that the distribution of heterogeneous agents across states is itself an equilibrium object with first-order consequences for aggregate efficiency and the evaluation of policy (e.g., Hsieh and Klenow, 2009; Restuccia and Rogerson, 2008).

6.2 Resource and Lifetime Poverty

The non-monotonicity in lifetime utility shows that low current resources and low lifetime welfare need not coincide. To make this distinction explicit, we consider two measures of poverty in the stationary distribution of the calibrated economy. The first provides a natural model-agnostic benchmark: we classify a household as *resource poor* if its cash on hand, $x = (1 + r(a))a + \mathcal{W}(w)$, lies in the bottom q percent of the population distribution. This ranks households according to the resources currently available to them, without taking a stand on their future income prospects. We classify a household as *lifetime poor* if its lifetime utility, $V(a, z, \nu; \alpha)$, lies in the bottom q of its corresponding distribution. This second measure instead incorporates both the household's current state and its expected future resources.

To characterize how the two notions of poverty vary over the joint distribution of wealth and pay, we compute the probability that a household is poor conditional on its current wealth and pay. For each pair (a, w), this probability integrates over the distribution of permanent types and persistent income states consistent with that combination:

$$\Pr[\text{poor} \mid a, w] = \sum_{\alpha} \int \mathbf{1}\left[\text{poor}(a, w, z; \alpha)\right] \mathrm{d}\Pr(\alpha, z \mid a, w). \tag{15}$$



(a) Resource Poverty (b) Lifetime Poverty

Figure 13: Alternative Poverty Metrics

Notes. Each panel reports $\Pr[\text{poor} \mid a, w]$ from equation (15) within cells defined by percentile ranks of wealth and realized pay in the stationary distribution of the calibrated model, with $q = 0.05$; red lines are iso-poverty contours. Each percentile cell contains a distribution of wealth and pay, which explains the non-binary classification. Panel (a) uses cash on hand, Panel (b) lifetime utility. Appendix G.3 reports $q = 0.10$ and $q = 0.20$. Parameters as in Tables 3 and 4.

Figure 13 reports these two poverty measures for the bottom 5% of their respective distributions. Panel (a) reports resource poverty, or those belonging to the bottom 5% of households by available

cash on hand. As expected, this is concentrated in the bottom-left corner, where households have both low wealth and low realized pay. The frontier slopes downward and is steeper in wealth-rank space.

This measure, however, does not distinguish between households with similar current resources but different expected future resources. Panel (b) instead shows the bottom 5% of households by lifetime utility. The lifetime poor are concentrated around zero wealth, with the iso-poverty lines exhibiting an inverse V-shape that flips the kink pattern we document in the data. This reflects the importance of permanent type heterogeneity in determining lifetime poverty.

The contrast is particularly stark over the debt region. Between the fifteenth wealth percentile and the bottom of the distribution, resource poverty rises while lifetime poverty falls. Moving further into debt therefore makes a household increasingly likely to be poor in terms of its current resources, but increasingly unlikely to be poor in terms of lifetime welfare. This follows from the same selection mechanism that generates the pay–wealth kink. The deepest debt positions are disproportionately occupied by higher permanent-income types experiencing adverse income realizations, while lower types are concentrated closer to zero wealth. The two measures therefore capture different forms of disadvantage: resource poverty identifies households with few resources today, while lifetime poverty identifies households with low expected lifetime welfare.

More generally, the two measures place different weight on wealth and pay. Because wealth is more dispersed than incomes, resource poverty is driven primarily by wealth. Lifetime poverty instead depends on permanent type, for which realized pay is more informative. The same qualitative pattern holds at higher poverty thresholds (Appendix G.3).

Table 5 shows that resource and lifetime poverty identify substantially different groups of households. At $q = 0.05$, only 13.5% of resource-poor households are also lifetime poor. The overlap increases with the poverty threshold, but remains only 54.0% at $q = 0.20$.

The characteristics of the two groups are also different. At $q = 0.05$, resource-poor households hold substantial debt and have realized pay well below their expected pay, at \$36.2 thousand compared with \$55 thousand. Lifetime-poor households, by contrast, hold positive wealth and have much lower expected pay, of only \$14.4 thousand. They therefore have considerably more cash on hand than resource-poor households, despite having much lower permanent earnings capacity. Resource poverty therefore tends to identify households in adverse current states, whereas lifetime poverty identifies households with persistently low resources who may not appear poor based on their current resources alone.

Table 5: Resource and Lifetime Poverty

	$q = 0.05$	$q = 0.10$	$q = 0.20$
Panel A. Overlap between poverty measures			
Share of resource-poor households also lifetime poor (%)	13.5	38.8	54.0
Panel B. Resource-poor households			
Wealth ($1,000s)	-49.1	-28.5	-9.6
realized annual pay ($1,000s)	36.2	31.0	32.6
Expected annual pay, $\mathbb{E}[w \mid \alpha]$ ($1,000s)	55.0	44.9	43.5
Panel C. Lifetime-poor households			
Wealth ($1,000s)	13.8	24.3	39.5
realized annual pay ($1,000s)	12.9	16.3	21.3
Expected annual pay, $\mathbb{E}[w \mid \alpha]$ ($1,000s)	14.4	17.9	23.1

Notes: Resource-poor households are those in the bottom q of the cash-on-hand distribution, while lifetime-poor households are those in the bottom q of the lifetime-welfare distribution. Panel A reports Pr(lifetime poor | resource poor), the share of either poverty group that is also classified as poor under the other definition. Because both poverty groups contain share q of the population, this is also equal to the share of lifetime-poor households who are resource poor. Panels B and C report means of household characteristics within each poverty group. Expected annual pay is the household's expected labor income conditional on permanent worker type, $\mathbb{E}[w \mid \alpha]$.

Taken together, Figure 13 and Table 5 show that low current resources and low lifetime welfare capture distinct dimensions of economic need. The same selection mechanism that generates the pay–wealth kink separates households experiencing adverse current states from those with persistently low resources.

6.3 Insurance and Redistribution

This distinction has a natural counterpart in the welfare value of a transfer. A transfer can provide insurance against an adverse current state, or redistribute resources toward households with low permanent resources. We formalize these two motives by decomposing the marginal welfare value of a transfer into an insurance component and a redistribution component.

Consider the marginal welfare value of assigning an additional dollar to a household in state $(a, z, \nu; \alpha)$. Let $M_\alpha \equiv \mathbb{E}_{a,z,\nu}[u'(c(a, z, \nu; \alpha)) \mid \alpha]$ denote average marginal utility within permanent type α, and let $M \equiv \mathbb{E}_{\alpha,a,z,\nu}[u'(c(a, z, \nu; \alpha))]$ denote average marginal utility in the population.[9] Normalizing by M, the marginal welfare value of a dollar can be decomposed as

[9] By the envelope theorem, the marginal effect of an additional unit of current resources on lifetime utility equals current marginal utility, with the effect on future utility operating through the household's optimal saving response.

$$\frac{u'(c(a,z,\nu;\alpha))}{M} = (1 + \text{Ins}(a,z,\nu;\alpha))\,(1 + \text{Redist}(\alpha))\,. \tag{16}$$

where

$$\text{Ins}(a,z,\nu;\alpha) = \frac{u'(c(a,z,\nu;\alpha)) - M_\alpha}{M_\alpha} \qquad \text{and} \qquad \text{Redist}(\alpha) = \frac{M_\alpha - M}{M}.$$

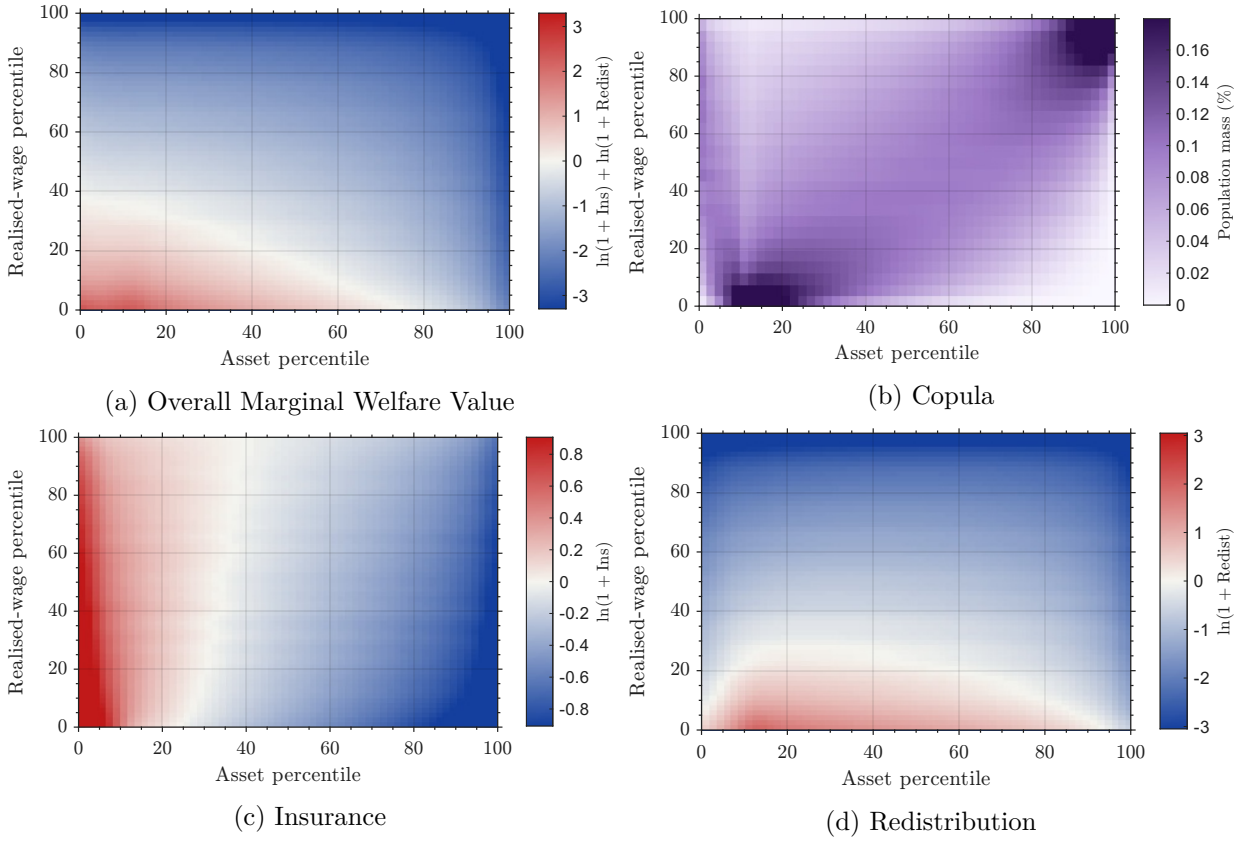
The first component measures the value of insurance across states within a permanent worker type. It is positive when a household has high marginal utility relative to other states experienced by households of the same type. The second component measures redistribution across permanent worker types. It is positive for types whose average marginal utility is high relative to the population average. Since both gross components are strictly positive, the decomposition can equivalently be written in logs:

$$\log\left(\frac{u'(c(a,z,\nu;\alpha))}{M}\right) = \log\left(1 + \text{Ins}(a,z,\nu;\alpha)\right) + \log\left(1 + \text{Redist}(\alpha)\right). \tag{17}$$

This gives zero a natural interpretation: $\log(1 + \text{Ins}(a,z,\nu;\alpha)) = 0$ when marginal utility equals the average within the household's type, while $\log(1 + \text{Redist}(\alpha)) = 0$ when average marginal utility for the type equals the population average. The log marginal welfare value of a transfer can therefore be read as the sum of an insurance component, capturing the value of transferring resources across states within type, and a redistribution component, capturing the value of transferring resources across permanent worker types.

Figure 14 maps the conditional expected marginal welfare value of a transfer and its two components over the joint distribution of wealth and realized pay. Panel (a) plots the overall marginal welfare value, while Panels (c) and (d) separately plot its insurance and redistribution components. Panel (b) shows the corresponding joint population distribution of wealth and realized pay. The figure therefore shows where an additional dollar has greatest marginal welfare value as a function of current wealth and current pay.

Panel (a) shows that the overall marginal welfare value of a transfer varies substantially over the joint distribution of wealth and realized pay. Neither low wealth nor low current pay alone identifies all households for whom an additional dollar is most valuable. Panel (b) reports the corresponding population distribution, showing which regions of the welfare maps contain substantial numbers of households.



(a) Overall Marginal Welfare Value

(b) Copula

(c) Insurance

(d) Redistribution

Figure 14: Insurance and Redistribution Across Wealth and Pay

Notes. Panels plot the marginal welfare value of a transfer and its components across pairs of wealth and realized-wage percentile ranks in the stationary distribution of the calibrated model. Panel (a) plots the conditional expectation of overall log marginal welfare value, $\log\left(u'(c(a,z,\nu;\alpha))/M\right) = \log\left(1+\text{Ins}(a,z,\nu;\alpha)\right) + \log\left(1+\text{Redist}(\alpha)\right)$. Panel (b) plots the joint stationary population mass over wealth and realized-wage percentile cells. Panel (c) plots $\log(1+\text{Ins})$, the within-type insurance component, and Panel (d) plots $\log(1+\text{Redist})$, the between-type redistribution component. Parameters are as reported in Tables 3 and 4.

The insurance component in panel (c) varies primarily with wealth. Conditional on type, low wealth identifies adverse states in which marginal utility is high relative to the type average. This is closely related to the notion of resource poverty in the previous subsection. In addition, the left tail of the wealth distribution is disproportionately populated by higher types, who face greater earnings risk and, through their type-specific borrowing constraints, are able to reach deeper debt positions following adverse shocks. The value of insurance therefore rises sharply toward the bottom of the wealth distribution.

Redistribution has a markedly different pattern. Panel (d) shows that its value varies primarily with realized pay. Although the redistribution component is defined over permanent worker types, current pay is informative about type: households with low realized pay are more likely to

have low permanent resources and therefore high average marginal utility across states. In this sense, the pattern mirrors our measure of lifetime poverty in the previous subsection. Wealth nevertheless provides additional information because the composition of permanent worker types varies systematically across the wealth distribution. Conditional on similar realized pay, households further into the left tail are more likely to be higher-type workers experiencing adverse income realizations, and therefore have lower redistributive value than households closer to zero wealth. This is the same selection mechanism that generates the pay–wealth kink: the households with the least wealth are not necessarily those with the greatest redistributive value.

Taken together, Figure 14 provides a marginal counterpart to the distinction between resource and lifetime poverty. Resource poverty is closely related to the insurance motive: low current resources tend to identify adverse states in which the marginal value of a transfer is high relative to others of the same permanent type. Lifetime poverty is more closely related to redistribution across permanent types, since households with low permanent resources have high marginal utility on average across states. Wealth is therefore particularly informative about insurance, while realized pay is particularly informative about redistribution. Neither wealth nor current pay alone captures both sources of welfare gains from transfers.

6.4 Targeting Transfers

The distinction between insurance and redistribution has direct implications for targeting transfers. Policies directed toward households with the fewest current resources will tend to place greater weight on insurance against adverse current states. Redistribution across permanent economic circumstances instead should target households with more current resources but lower permanent earnings capacity. Since wealth and current pay contain different information about these two dimensions, neither alone need identify the households with the highest overall marginal welfare value of a transfer. This raises a practical question for redistributive policy: how effectively can transfers be targeted using these observable measures of economic circumstances?

We consider a marginal policy that assigns an additional dollar to a fraction q of households, meaning the total fiscal cost is held fixed across targeting rules. We compare three information sets: current pay, current wealth, and pay and wealth jointly. For each of the three welfare objectives, we construct the optimal targeting rule available under each information set. For insurance, the relevant value is $1 + \text{Ins} = u'(c)/M_\alpha$; for redistribution, it is $1 + \text{Redist} = M_\alpha/M$; and for overall welfare, it is $u'(c)/M$. For a given objective and information set, households are ranked by the

conditional expectation of the relevant welfare value and transfers are assigned to the fraction q with the highest predicted value. The optimal pay- or wealth-based rule therefore need not simply target the bottom q of the corresponding distribution.

Table 6: Targeting Marginal Transfers

	Insurance	Redistribution	Overall Welfare
Random targeting	1.000	1.000	1.000
Pay only	1.254	4.265	4.797
Wealth only	2.222	2.807	3.448
Pay and wealth	2.242	4.409	5.197

Notes: The table reports the performance of alternative rules for targeting an additional dollar to 10% of households. For each information set, households are ranked by the conditional expectation of the relevant welfare object, and transfers are assigned to the 10% with the highest predicted value. The insurance column reports the average value of $1 + \text{Ins} = u'(c)/M_\alpha$ among recipients, the redistribution column reports the average value of $1 + \text{Redist} = M_\alpha/M$, and the overall welfare column reports the average value of $u'(c)/M$. Each welfare object has population mean one, so random targeting yields a value of one. Higher values indicate that the targeting rule directs transfers toward households with higher marginal value under the corresponding objective.

Table 6 reports the welfare value achieved by these optimal targeting rules. Because each welfare object has population mean one, the entries can be interpreted relative to random targeting. For example, a value of 2.222 means that the optimal wealth-based targeting rule delivers 2.222 times the average welfare value of random targeting. Or, equivalently, that a \$1 targeted transfer delivers the same welfare gain among recipients as a \$2.22 randomly targeted transfer.

The table shows that wealth and pay are informative about different welfare objectives, consistent with the patterns in Figure 14. Wealth is much more effective than pay at targeting insurance: the wealth-only rule achieves a value of 2.222, compared with 1.254 using pay alone. Adding pay to wealth improves this only slightly, to 2.242. For redistribution, the pattern is reversed. Pay-only targeting achieves a value of 4.265, compared with 2.807 using wealth alone, while adding wealth raises this further to 4.409. These results show that wealth is particularly informative about adverse current states, while pay is more informative about permanent economic circumstances.

Conditioning on both observables allows policy to capture both sources of welfare gains from transfers. Joint targeting achieves a value of 5.197, 8.3% higher than using pay alone and 50.7% higher than using wealth alone. When jointly targeting pay and wealth, the biggest welfare effect derives from redistribution, which is best achieved by targeting pay because current pay is most informative about the worker type and hence about the lifetime earnings potential. While the best

policy is to target pay and wealth jointly, the primary concern of policy makers is to put most weight on the pay and less on wealth.

7 Conclusion

The working poor are not all the same. We find that households close to zero wealth have the lowest average pay, while those with more debt have higher pay. This gives rise to a V-shaped relationship between wealth and pay, with the kink around zero wealth. The relationship is robust to different measures of pay and wealth, and appears across countries. We also find a similar non-monotonic relationship between wealth and the dispersion of pay. Persistent worker heterogeneity accounts for substantially more of these patterns than either time-varying differences across workers or occupation effects.

We analyze an Aiyagari (1994) model with permanent worker heterogeneity and a borrowing constraint that depends on expected labor earnings. Higher permanent earnings types have both higher expected pay and greater dispersion in pay. Their greater borrowing capacity allows them to sustain deeper debt positions following adverse shocks, while precautionary saving leads them to accumulate more wealth. Lower types remain concentrated closer to zero wealth. The calibrated model reproduces the aggregate non-monotonic relationship between wealth and pay. It also generates non-monotonic aggregate relationships between wealth and lifetime utility and consumption, again through changes in the composition of permanent worker types across the wealth distribution. Current wealth therefore need not provide a good guide to permanent resources or lifetime welfare. Households deep in debt may have relatively high earnings capacity but be experiencing adverse current income realizations, while households with lower permanent resources can hold more wealth.

This selection of households across the wealth distribution has implications for how we identify economic need and target transfers. A measure of poverty based on current cash on hand identifies a substantially different set of households from a measure based on lifetime welfare. We show that the marginal welfare value of a transfer can similarly be decomposed into insurance against adverse current states and redistribution across permanent worker types. Wealth is particularly informative about the insurance component, while current pay is more informative about redistribution. As a result, neither pay nor wealth alone identifies all households with high marginal welfare value, and targeting on both improves performance when the objective incorporates both motives. That said, for our calibrated model, targeting based on pay alone has a substantially higher welfare value than

conditioning on wealth alone, and achieves over 90% of the value of targeting on both measures. This is because differences in lifetime earnings potential have the biggest impact on welfare, and pay is the most accurate measure of those permanent income differences.

More generally, measures of current economic circumstances contain both adverse current states and persistently low resources. The selection of heterogeneous households across wealth and pay therefore matters both for how economic need is measured and for how redistributive policy is targeted.

References

Achdou, Yves, Jiequn Han, Jean-Michel Lasry, Pierre-Louis Lions, and Benjamin Moll. 2022. "Income and Wealth Distribution in Macroeconomics: A Continuous-Time Approach." *The Review of Economic Studies* 89 (1):45–86.

Aguiar, Mark, Mark Bils, and Corina Boar. 2025. "Who Are the Hand-to-Mouth?" *The Review of Economic Studies* 92 (3):1293–1340.

Aiyagari, S. Rao. 1994. "Uninsured Idiosyncratic Risk and Aggregate Saving." *The Quarterly Journal of Economics* Vol. 109, No. 3.:659–684.

Altonji, Joseph G and Lewis M Segal. 1996. "Small-sample bias in GMM estimation of covariance structures." *Journal of Business & Economic Statistics* 14 (3):353–366.

Bénabou, Roland. 2002. "Tax and Education Policy in a Heterogeneous-Agent Economy: What Levels of Redistribution Maximize Growth and Efficiency?" *Econometrica* 70 (2):481–517.

Bewley, T F. 1980. "Interest Bearing Money and the Equilibrium Stock of Capital." Mimeo.

Brewer, Mike, Ben Etheridge, and Cormac O'Dea. 2017. "Why Are Households That Report the Lowest Incomes So Well-off?" *The Economic Journal* 127 (605):F24–F49.

Cagetti, Marco. 2003. "Wealth accumulation over the life cycle and precautionary savings." *Journal of Business & Economic Statistics* 21 (3):339–353.

Carroll, Christopher D. 1997. "Buffer-Stock Saving and the Life Cycle/Permanent Income Hypothesis." *The Quarterly Journal of Economics* 112 (1):1–55.

Carroll, Christopher D., Karen E. Dynan, and Spencer D. Krane. 2003. "Unemployment Risk and Precautionary Wealth: Evidence from Households' Balance Sheets." *The Review of Economics and Statistics* 85 (3):586–604. URL http://ideas.repec.org/a/tpr/restat/v85y2003i3p586-604.html.

Catherine, Sylvain, Mehran Ebrahimian, and Constantine Yannelis. 2026. "How do income-driven repayment plans benefit student debt borrowers?" *Journal of Financial Economics* 178:104253.

Chatterjee, Satyajit, Dean Corbae, Makoto Nakajima, and José-Víctor Ríos-Rull. 2007. "A quantitative theory of unsecured consumer credit with risk of default." *Econometrica* 75 (6):1525–1589.

Daminato, Claudio and Luigi Pistaferri. 2023. "Heterogeneity in returns to wealth and consumption inequality." Stanford mimeo.

De Nardi, Mariacristina, Giulio Fella, and Gonzalo Paz-Pardo. 2025. "Wage risk and government and spousal insurance." *Review of Economic Studies* 92 (2):954–980.

De Nardi, Mariacristina, Eric French, John Bailey Jones, and Rory McGee. 2025. "Why do couples and singles save during retirement? household heterogeneity and its aggregate implications." *Journal of Political Economy* 133 (3):750–792.

Deaton, Angus. 1991. "Saving and Liquidity Constraints." *Econometrica* 59 (5):1221–1248.

Díaz, Antonia and María José Luengo-Prado. 2008. "On the User Cost and Homeownership." *Review of Economic Dynamics* 11 (3):584–613.

Donaldson, Jason Roderick, Giorgia Piacentino, and Anjan Thakor. 2019. "Household debt overhang and unemployment." *The Journal of Finance* 74 (3):1473–1502.

Eeckhout, Jan and Nezih Guner. 2026. "Optimal Spatial Taxation: Are Big Cities Too Small?" SSRN. URL https://ssrn.com/abstract=6555395.

Eeckhout, Jan and Alireza Sepahsalari. 2024. "The effect of wealth on worker productivity." *Review of Economic Studies* 91 (3):1584–1633.

Fagereng, Andreas, Martin B Holm, and Gisle J Natvik. 2021. "MPC heterogeneity and household balance sheets." *American Economic Journal: Macroeconomics* 13 (4):1–54.

Fan, Jianqing and Qiwei Yao. 1998. "Efficient estimation of conditional variance functions in stochastic regression." *Biometrika* 85 (3):645–660.

Feldstein, Martin S. 1969. "The effects of taxation on risk taking." *Journal of Political Economy* 77 (5):755–764.

Finkelstein, Amy, Nathaniel Hendren, and Erzo FP Luttmer. 2019. "The value of medicaid: Interpreting results from the oregon health insurance experiment." *Journal of Political Economy* 127 (6):2836–2874.

Golosov, Mikhail, Maxim Troshkin, and Aleh Tsyvinski. 2016. "Redistribution and Social Insurance." *American Economic Review* 106 (2):359–386.

Gourinchas, Pierre-Olivier and Jonathan A Parker. 2002. "Consumption over the life cycle." *Econometrica* 70 (1):47–89.

Groes, Fane, Philipp Kircher, and Iourii Manovskii. 2015. "The U-shapes of occupational mobility." *The Review of Economic Studies* 82 (2):659–692.

Guner, Nezih, Christopher Rauh, and Gustavo Ventura. 2024. "Means-tested transfers in the US: facts and parametric estimates." Tech. rep., IZA Discussion Papers.

Guvenen, Fatih, Serdar Ozkan, and Rocio Madera. 2024. "Consumption dynamics and welfare under non-gaussian earnings risk." *Journal of Economic Dynamics and Control* 169:104945.

Heathcote, Jonathan, Kjetil Storesletten, and Giovanni L. Violante. 2017. "Optimal Tax Progressivity: An Analytical Framework." *Quarterly Journal of Economics* 132 (4):1693–1754.

Herkenhoff, Kyle, Gordon Phillips, and Ethan Cohen-Cole. 2023. "How credit constraints impact job finding rates, sorting, and aggregate output." *Review of Economic Studies* :rdad104.

Hsieh, Chang-Tai and Peter J Klenow. 2009. "Misallocation and manufacturing TFP in China and India." *The Quarterly journal of economics* 124 (4):1403–1448.

Hubbard, R Glenn, Jonathan Skinner, and Stephen P Zeldes. 1995. "Precautionary saving and social insurance." *Journal of political Economy* 103 (2):360–399.

Huggett, Mark. 1993a. "The risk-free rate in heterogeneous-agent incomplete-insurance economies." *Journal of Economic Dynamics and Control* 17:953–969.

———. 1993b. "The risk-free rate in heterogeneous-agent incomplete-insurance economies." *Journal of economic Dynamics and Control* 17 (5-6):953–969.

Imrohoroğlu, Ayşe. 1989. "Cost of business cycles with indivisibilities and liquidity constraints." *Journal of Political economy* 97 (6):1364–1383.

Jappelli, Tullio and Luigi Pistaferri. 2025. "Permanent Income Shocks, Target Wealth, and the Wealth Gap." *American Economic Journal: Macroeconomics* 17 (1):102–125.

Kaplan, Greg, Benjamin Moll, and Giovanni L Violante. 2018. "Monetary policy according to HANK." *American Economic Review* 108 (3):697–743.

Kaplan, Greg and Giovanni L. Violante. 2022. "A Model of the Consumption Response to Fiscal Stimulus Payments." *Annual Review of Economics* 14.

Kaplan, Greg, Giovanni L Violante, and Justin Weidner. 2014. "The wealthy hand-to-mouth." Tech. rep., National Bureau of Economic Research.

Kennickell, Arthur B. 2006. "A Rolling Tide: Changes in the Distribution of Wealth in the US, 1989–2001." In *International perspectives on household wealth*. Edward Elgar Publishing.

Krusell, Per, Jinfeng Lou, and Victor Jose Ríos-Rull. 2019. "Wealth, Wages, and Employment." *Working Paper* .

Lewis, Daniel J, Davide Melcangi, and Laura Pilossoph. 2019. "Latent heterogeneity in the marginal propensity to consume." *FRB of New York Staff Report* (902).

Lieber, Ethan MJ and Lee M Lockwood. 2019. "Targeting with in-kind transfers: Evidence from Medicaid home care." *American Economic Review* 109 (4):1461–1485.

Livshits, Igor, James MacGee, and Michele Tertilt. 2007. "Consumer bankruptcy: A fresh start." *American Economic Review* 97 (1):402–418.

Luo, Mi and Simon Mongey. 2019. "Assets and job choice: Student debt, wages and amenities." Tech. rep., National Bureau of Economic Research.

McGee, Rory and Martin O'Connell. 2026. "Risk Protection and Redistribution in the Design of Social Insurance." Tech. rep., Centre for Economic Policy Research.

Meghir, Costas and Luigi Pistaferri. 2011. "Earnings, consumption and life cycle choices." In *Handbook of labor economics*, vol. 4. Elsevier, 773–854.

Meyer, Bruce D. and James X. Sullivan. 2012. "Identifying the Disadvantaged: Official Poverty, Consumption Poverty, and the New Supplemental Poverty Measure." *Journal of Economic Perspectives* 26 (3):111–136.

Meyer, Bruce D and Derek Wu. 2023. "Poverty in the United States." *University of Chicago, Working Paper* .

Peydró, José-Luis, Francesc Rodriguez-Tous, Jagdish Tripathy, and Arzu Uluc. 2024. "Macroprudential Policy, Mortgage Cycles, and Distributional Effects: Evidence from the United Kingdom." *The Review of Financial Studies* 37 (3):727–760.

Pischke, Jörn-Steffen. 1995. "Measurement error and earnings dynamics: Some estimates from the PSID validation study." *Journal of Business & Economic Statistics* 13 (3):305–314.

Poterba, James M. 1989. "Lifetime Incidence and the Distributional Burden of Excise Taxes." *American Economic Review* 79 (2):325–330.

Restuccia, Diego and Richard Rogerson. 2008. "Policy distortions and aggregate productivity with heterogeneous establishments." *Review of Economic dynamics* 11 (4):707–720.

Rosen, Richard J. 2011. "Competition in Mortgage Markets: The Effect of Lender Type on Loan Characteristics." *Economic Perspectives* 35 (1):2–21.

Scholz, John Karl, Ananth Seshadri, and Surachai Khitatrakun. 2006. "Are Americans saving "optimally" for retirement?" *Journal of political economy* 114 (4):607–643.

Social Security Administration. 2012. "Supplemental Security Income Program Description and Legislative History." Annual Statistical Supplement, 2012. URL https://www.ssa.gov/policy/docs/statcomps/supplement/2012/ssi.html.

Stiglitz, Joseph E and Andrew Weiss. 1981. "Credit rationing in markets with imperfect information." *The American economic review* 71 (3):393–410.

U.S. Department of Health and Human Services. 2012. "2012 HHS Poverty Guidelines." URL https://aspe.hhs.gov/2012-hhs-poverty-guidelines. Federal Register, Vol. 77, No. 17, January 26, 2012, pp. 4034–4035.

Zeldes, Stephen P. 1989. "Consumption and liquidity constraints: an empirical investigation." *Journal of political economy* 97 (2):305–346.

Supplementary Appendix

Appendices A and B describe the data and sample construction. Appendix C reports the complete cross-country evidence, Appendix D reports the worker–occupation decomposition, and Appendix E collects empirical robustness and alternative explanations. Appendix F contains the proofs for the stylized model, and Appendix G contains computational details and additional quantitative results.

A Description of US Data

Our core data are drawn from the Panel Study of Income Dynamics (PSID), a biennial household panel survey containing detailed information on pay, wealth and demographic characteristics for a nationally-representative sample of households in the US. We focus on the period from 1999 to 2019, corresponding to the period over which PSID has collected asset data in every wave.

A.1 Sample Selection

To construct our baseline sample, we begin with an unbalanced panel of all PSID individuals in the sample period. We restrict attention to households in which the head is employed, defining employment as annual earnings at least equivalent to 20 hours per week at the federal minimum wage over one quarter.[10] We then focus only on married households, and restrict further to those with a head aged between 25 and 55 to abstract from education or retirement decisions. Following Blundell, Pistaferri, and Saporta-Eksten (2016), whenever there is a change in marital composition we drop the household in the year of the change and treat it as a new family in subsequent observations. Finally, to reduce the influence of measurement error, we drop 33 observations with extremely high asset values ($20 million or more). Our final sample contains a total of 26,383 employed observations with non-zero survey weights, comprised of 6,552 households observed for an average of 4.03 waves.

A.2 Wealth

Our primary measure of wealth is intended to capture households' net worth. It is defined as the household's total assets—the sum of cash, bonds, stocks, the value of any business, pension funds,

[10] This follows Guvenen et al. (2021), who study income dynamics in US administrative social security records, and recode incomes below this threshold to zero.

housing, and other real estate, and vehicles—net of any mortgage or other debt. We adjust our measure of wealth to 2012 prices using the CPI.

We also consider a more restrictive measure of wealth, "liquid" wealth, which is intended to capture resources that are easier for households to access or borrow against on short notice. Liquid wealth excludes the value of home equity, the value of vehicles, and the value of pension funds.

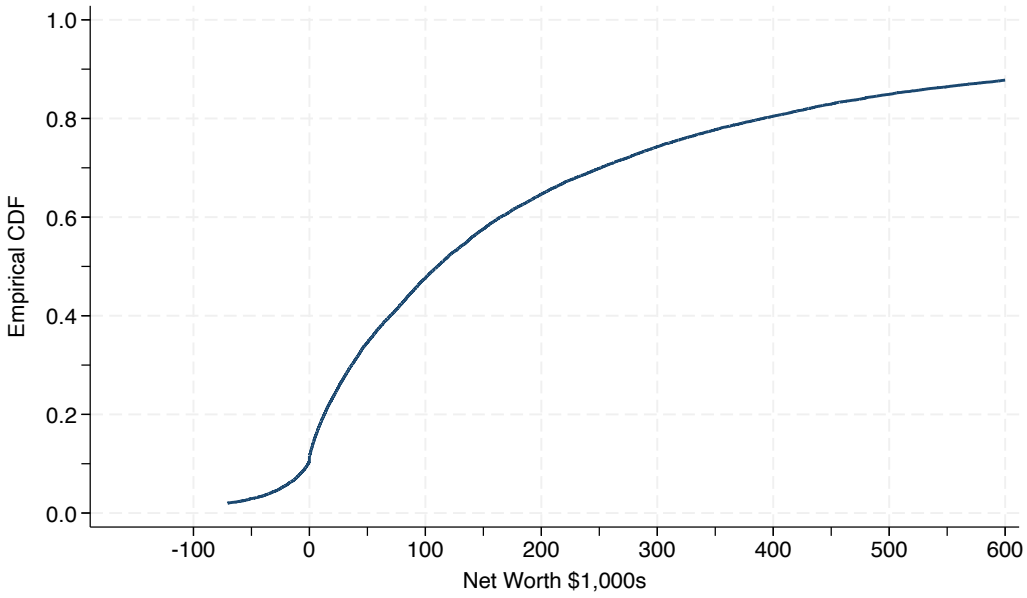
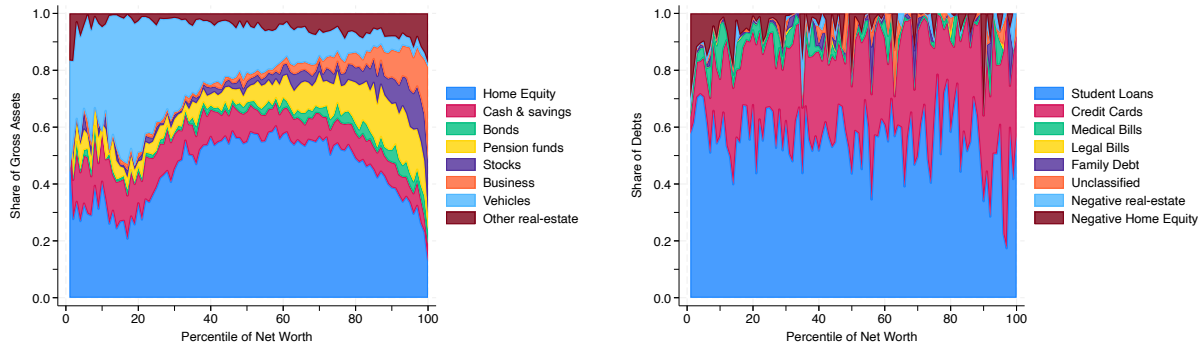


Figure A.1: Empirical Distribution of Wealth

The empirical CDF in Figure A.1 shows that 10.4 percent of households have negative wealth, and a small mass of 1.2 percent report precisely zero wealth. Above this, the distribution has a long tail: half of households report wealth above around $110,000, while around 10% report wealth in excess of $700,000. To reduce the influence of households with extremely high or low wealth on our results, we use the percentile rank of wealth rather than its level throughout our analysis.

Figure A.2 describes how the composition of household wealth portfolios varies across the wealth distribution. Panel (a) shows the share of total assets accounted for by different asset classes, while Panel (b) reports the composition of liabilities. At low levels of wealth, wealth holdings are dominated by liquid assets and vehicles, with relatively little exposure to housing, pensions, or financial wealth. As wealth increases, portfolios shift toward housing and pension assets, which together account for the majority of wealth in the upper part of the distribution. On the liability side, there is surprisingly little variation in the composition of debt across the wealth distribution: in particular, student loans make up a majority of household debt for households across the entire

distribution. However, negative home equity is a larger share of debt for households at the bottom of the wealth distribution than for those further up, while the share of credit card debt increases in wealth.



(a) Assets (b) Debts

Figure A.2: Portfolio Composition

A.3 Pay

Our measure of pay is the household head's gross annual labor income. This captures pay from all jobs, plus any labor component of income from a business or farm, inflated to 2012 prices using CPI. Pay is thus annual labor income, and not the hourly or weekly wage rate.

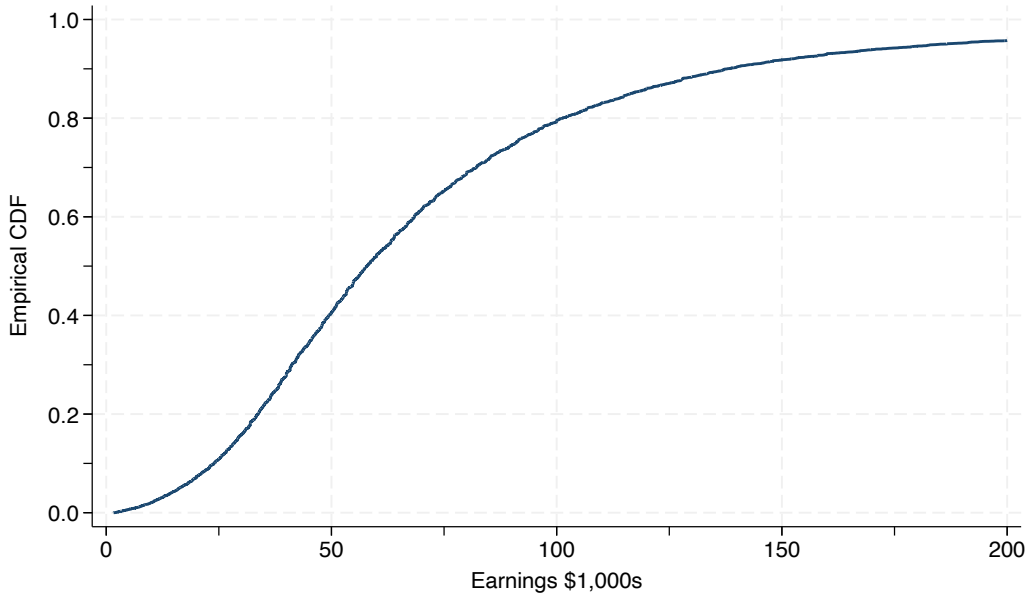


Figure A.3: Empirical Distribution of Pay

B Description of Non-US Data

This appendix describes the two sources used for the non-US evidence. We use the UK Household Longitudinal Study (UKHLS) and the Luxembourg Wealth Study (LWS).

B.1 UK Household Longitudinal Study

The UKHLS is an annual panel survey of UK households with information on pay, wealth and demographic characteristics for a representative sample of individuals in the UK. In our main analysis we focus on information from waves 4 and 8 of the survey, collected throughout 2012/13 and 2016/17 respectively, as these waves contain a set of additional detailed questions on wealth and debts. Our sample contains 33,973 person-wave observations of employed or self-employed individuals aged 16–65 with positive labor income and non-missing net worth.

We construct net liquid wealth as the value of savings and financial investments less credit-card and other non-housing debt. Our main measure, net total wealth, additionally includes the value of the main residence less outstanding mortgage debt. We use respondents' reported asset values and allocate jointly held assets and debts across their holders.[11] Housing values are based on respondents' estimates; where necessary, outstanding mortgage debt is estimated from the available mortgage information. All amounts are expressed in January 2022 pounds using the Consumer Price Index.

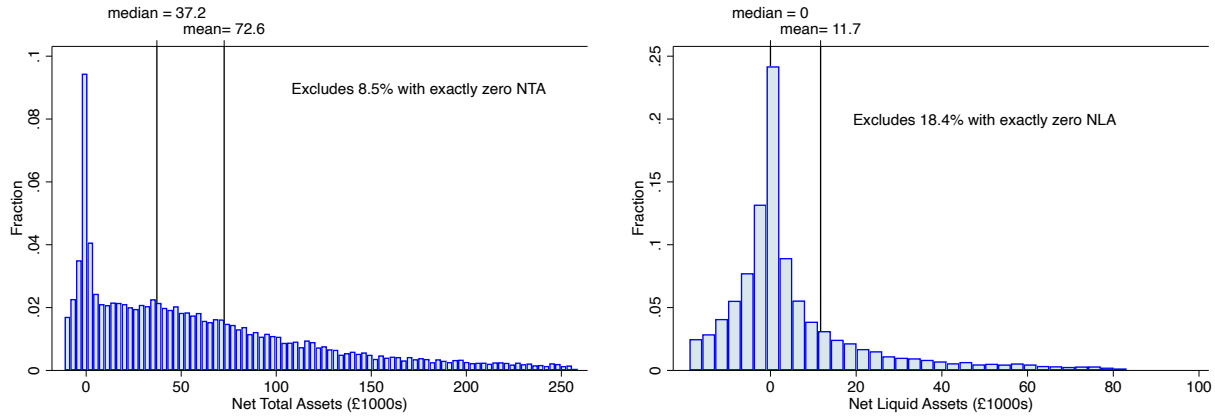
Our measure of pay is the individual's gross labor income in the month before interview, including earnings from main and secondary jobs and income from self-employment. We use percentile ranks of net worth to reduce the influence of extreme observations.

B.2 Luxembourg Wealth Study

The Luxembourg Wealth Study (LWS) contains harmonized data on income and wealth for a selection of countries around the world. We use the most recent available dataset with wealth and income available jointly that also overlaps with the sample period for our US data.

We define household wealth as non-financial assets plus financial assets less total liabilities. Non-financial assets include real estate and other non-financial holdings, while financial assets include deposits and investments. Our measure of pay is total household labor income, comprising income from paid employment and self-employment across all household members.

[11] In cases where a respondent is unable to provide an estimate of the value of an asset or debt, they are asked a sequence of branching questions to bound its value. In these cases, we use the midpoint of reported value brackets as an estimate of the amount.

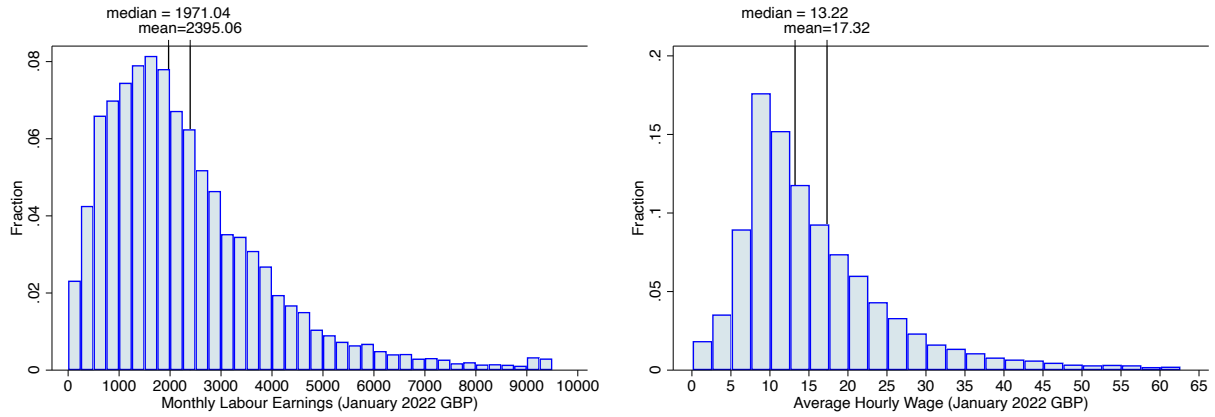


(a) Net total wealth

(b) Net liquid wealth

Figure B.1: Distribution of wealth in the UKHLS (January 2022 GBP)

Notes. For readability, the histograms exclude observations at exactly zero wealth and the top and bottom five percent of the distributions.



(a) Monthly pay

(b) Hourly pay

Figure B.2: Distribution of pay in the UKHLS (January 2022 GBP)

Notes. The histograms omit the top one percent of earners for readability.

Within each country-year, we rank households by net worth and calculate mean labor income within wealth-percentile groups using the household survey weight. For presentation, the percentile means are grouped into approximately two-percentile intervals and normalized by the country-specific mean of the underlying percentile-cell means. This allows the profiles to be compared without converting incomes or assets to a common currency. To avoid compressing the income distribution in all countries due to a long right tail in a single country, we truncate earnings at four times the national average.

C Cross-Country Evidence

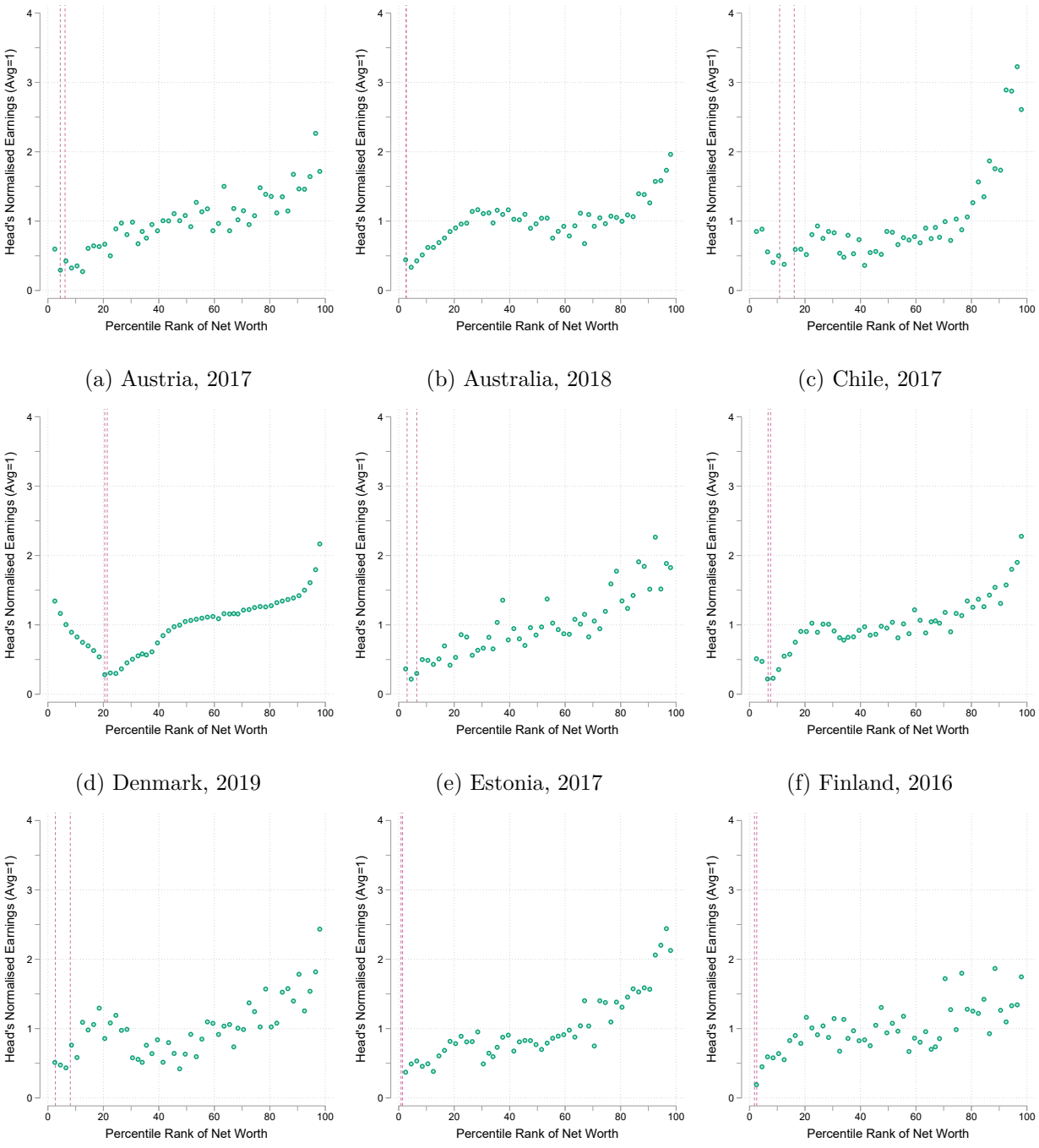
This appendix reports the cross-country evidence underlying Section 2.3. We use the harmonized household files from the Luxembourg Wealth Study (LWS), taking one country-year per country. We define household net worth as non-financial assets plus financial assets less liabilities. We rank households by net worth within each country-year and calculate average household labor income within wealth-percentile groups, using household survey weights for income means. The same procedure is applied across LWS countries.

Table C.1 summarizes the position of zero wealth in each country. Several LWS datasets contain very few recorded households with negative net worth. We do not interpret such figures as necessarily indicating that household indebtedness is rare in these countries: differences in survey coverage, measurement and the availability of particular assets or liabilities may affect the negative tail, even in a harmonized dataset. The UK illustrates this limitation. In the LWS UK file, which is based on the UK's Wealth and Assets Survey, the estimated negative-wealth share is only 1.1%, whereas our separate UKHLS analysis suggests a substantially larger share (approximately one-fifth). As a result, we interpret the very low negative wealth shares for some countries as representing measurement error, at least in part.

For the main-text cross-country comparison we use a minimum threshold of *5%* of observations estimated to have negative net worth, to ensure that the negative-wealth portion of the relationship has sufficient support. The country panels below show the estimated relationships for all countries.

Table C.1: Estimated prevalence of negative net worth across countries

Country (year)	Data source	Negative net worth (%)
Australia (2018)	LWS	2.7
Austria (2017)	LWS	4.5
Chile (2017)	LWS	10.9
Denmark (2019)	LWS	20.5
Estonia (2017)	LWS	3.0
Finland (2016)	LWS	6.7
Greece (2018)	LWS	2.7
Italy (2016)	LWS	0.8
Luxembourg (2018)	LWS	1.8
Norway (2019)	LWS	19.0
Slovakia (2017)	LWS	2.3
Slovenia (2017)	LWS	2.8
South Africa (2017)	LWS	2.0
South Korea (2019)	LWS	3.0
Spain (2017)	LWS	3.6
Sweden (2005)	LWS	22.9
United Kingdom (2019)	LWS	1.1
United Kingdom (2012–13, 2016–17)	UKHLS	17.1

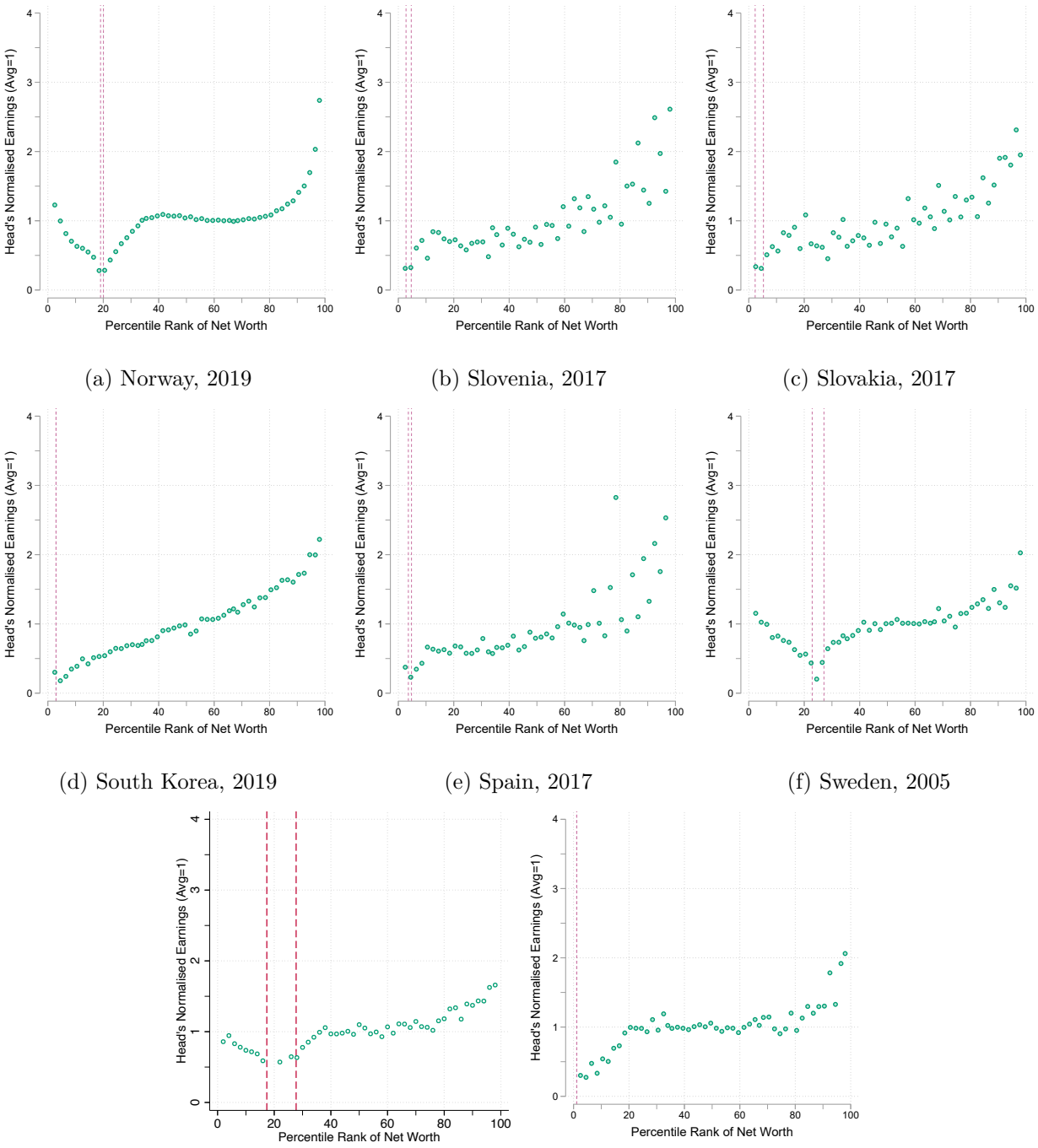


(a) Austria, 2017 (b) Australia, 2018 (c) Chile, 2017

(d) Denmark, 2019 (e) Estonia, 2017 (f) Finland, 2016

(g) Greece, 2018 (h) Italy, 2016 (i) Luxembourg, 2018

Figure C.1: Relationship between wealth and pay across countries (I)



(a) Norway, 2019 (b) Slovenia, 2017 (c) Slovakia, 2017

(d) South Korea, 2019 (e) Spain, 2017 (f) Sweden, 2005

(g) UK, UKHLS, 2012–13, 2016–17 (h) UK, LWS, 2019

Figure C.2: Relationship between wealth and pay across countries (II)

D Worker and Occupation Decomposition

This appendix reports the worker–occupation decomposition summarized in Section 2.4. The objective of this analysis is to quantify how much of the kink is due to heterogeneity in worker characteristics, versus characteristics of the jobs they are located in. Our overall finding that the kink is driven primarily by permanent worker-side heterogeneity motivates the source of heterogeneity in our model.

To quantify the contribution of these channels in the overall kink, we estimate a two-way fixed effects regression in the style of Abowd, Kramarz, and Margolis (1999) (AKM). Unlike AKM and much of the subsequent literature, in our specification we estimate fixed effects for workers and for occupations (rather than firms). That is, we specify that

$$\ln(y_{i,t}) = X'_{i,t}\beta + \alpha_i + \phi_{j(i,t)} + \epsilon_{i,t}, \tag{D.1}$$

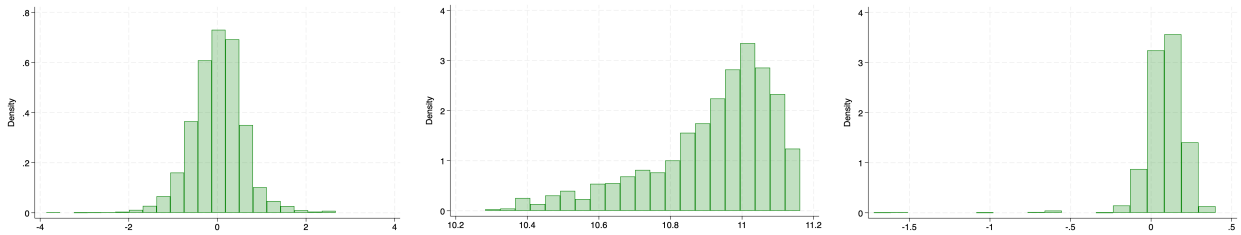
where $y_{i,t}$ is worker i's annual labor earnings at time t; $X_{i,t}$ contains a cubic polynomial in age and indicators for year; α_i is a worker fixed effect; $\phi_{j(i,t)}$ is an occupation fixed effect; and $\epsilon_{i,t}$ is the residual component of log pay. The worker fixed effect $\hat{\alpha}_i$ includes all time-invariant differences between workers, including those related to observed characteristics such as education and race. We therefore decompose log pay into the contributions of three components: observed age and time effects; time-invariant worker heterogeneity; and occupational heterogeneity.

Estimating specification (D.1) requires a definition of occupations. Our PSID data categorizes workers' occupation using census occupation codes, and the period that we study contains codes across three classification regimes. We use a crosswalk to map these into five-digit Standard Occupational Classification (SOC) codes. But there remains a challenge, since identification of the occupation fixed effects relies on the network of occupations being connected by worker flows: attempting to estimate a large number of occupation fixed effects relative to the sample size results in many occupations (or groups of occupations) which are disconnected from the rest of the network.

We therefore create 35 "pseudo"-occupations by aggregating five-digit SOC codes into groups with similar pay distributions using a k-means procedure, in the spirit of Bonhomme et al. (2023). In particular, we first construct residualized pay by regressing log pay on a cubic polynomial in age and year fixed effects, and then we cluster on the 10th, 25th, 50th, 75th and 90th percentiles of the SOC-level residual pay distribution. We verify that all 35 pseudo-occupations are connected by worker flows – i.e., the largest connected component of the network of occupations contains all 35

occupations – and therefore all occupation fixed effects are identified.

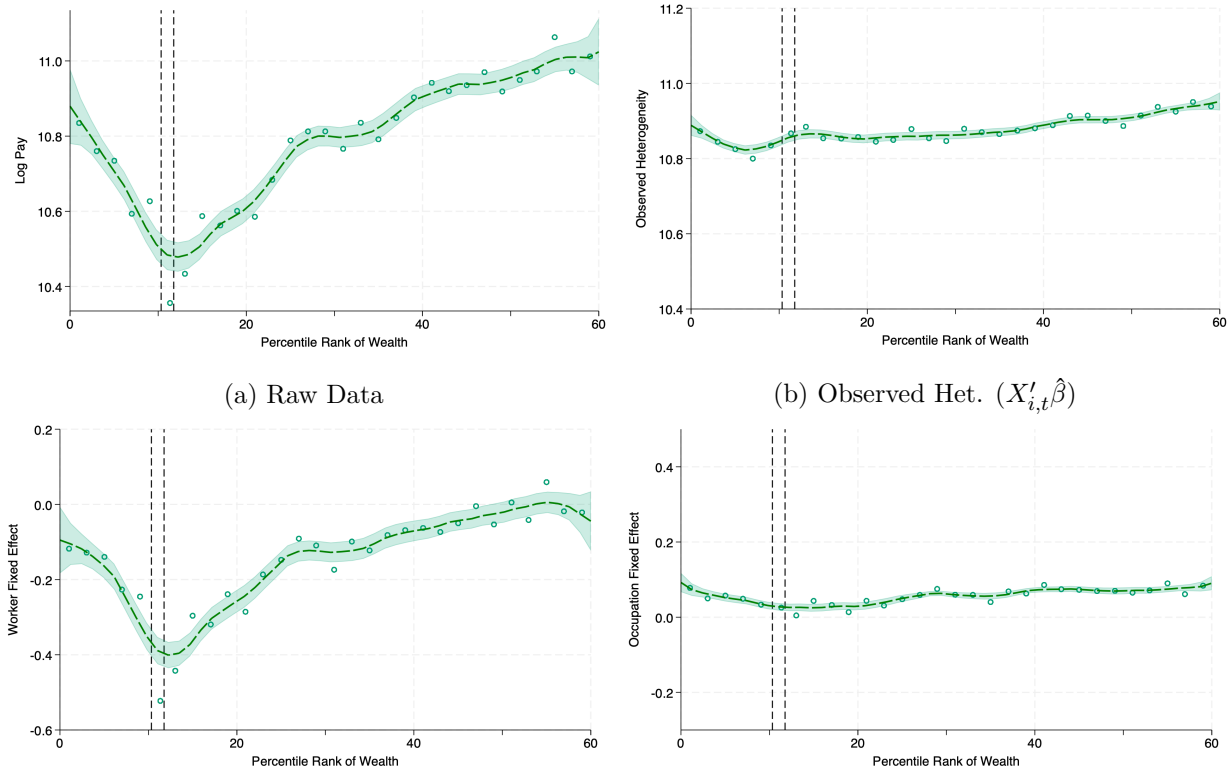
Figure D.1 shows the distributions of each of the estimated AKM components.



(a) Worker FE ($\hat{\alpha}_i$) (b) Observed Het. ($X'_{i,t}\hat{\beta}$) (c) Occupation FE (ϕ_j)

Figure D.1: Distributions of AKM components

Notes. Figure shows distributions of the estimated AKM components of log pay from equation (D.1).



(a) Raw Data (b) Observed Het. ($X'_{i,t}\hat{\beta}$)

(c) Worker FE ($\hat{\alpha}_i$) (d) Occupation FE (ϕ_j)

Figure D.2: AKM components over the wealth distribution

Notes. Panels contain the estimated regression in equation (1) for each estimated AKM component, along with 95 percent confidence intervals and a binned scatter plot of the estimated components. Panel (a) shows variation in overall log pay over the wealth distribution. Panel (b) shows the variation in $X'_{i,t}\hat{\beta}$ over the wealth distribution—that is, the variation in log pay attributable to age and year. Panel (c) shows variation in the average worker fixed effects $\hat{\alpha}_i$ over the wealth distribution. Panel (d) shows variation in the average fixed effect for occupations matched with workers at different points in the wealth distribution, $\hat{\phi}_{j(i,t)}$.

Figure D.2 shows the relationship between each of the three estimated components from specification (D.1) and wealth percentile. Each panel shows the estimated non-parametric regression which is similar to equation (1), except we redefine the dependent variable to be the estimated contribution of each AKM component to worker i's log pay in time t. There is a clear and statistically significant non-monotonicity in the relationship between each of the three components and wealth, although the size of the kink is quantitatively much larger for the worker-side heterogeneities than for occupations.

Panel (a) shows the relationship between overall log pay and wealth percentile. The pattern closely mirrors Figure 1 which documents the relationship between the level of pay and wealth. To benchmark the contribution of each component from the AKM regression to the overall variation in log pay, note that the conditional mean estimated from the kernel regression falls by around 37 log points between the first percentile of the wealth distribution and its minimum around the 12th percentile.

Panel (b) shows the contribution of observable age and time effects to the non-monotone relationship between wealth percentile and pay. This component is much flatter than the others: the estimated conditional mean falls by around 1 log point between the first and 12th percentiles. Observable age and time effects therefore account for around 3% of the overall decline in pay over this range.

Panel (c) shows the contribution of time-invariant worker heterogeneity. The pattern is strongly non-monotonic, with the worker effect declining to a clear minimum around the 12th percentile and then rising steadily across the positive wealth range. The conditional mean of the worker fixed effect falls by around 28 log points between the first and 12th percentiles, accounting for around 75% of the overall decline in log pay.

Finally, panel (d) shows the relationship between the occupation fixed effect and wealth percentile. In contrast to the worker fixed effect, the pattern here is much flatter and the overall variation more limited. The conditional mean of the occupation fixed effect falls by around 7 log points between the first and 12th percentiles, accounting for around 19% of the overall decline. This indicates that differences in occupational allocation contribute only modestly to the non-monotonic relationship between wealth and pay, relative to the role of worker heterogeneity.

The three channels account for around 3% (observed age and time effects), 75% (worker fixed effect), and 19% (occupation fixed effect) of the decline in log pay between the first and 12th wealth percentiles. The remaining 3% reflects the residual.

E Empirical Robustness and Additional Results

This appendix organizes additional empirical evidence and robustness exercises for the main results documented in Section 2. We first examine labor supply responses, then alternative definitions of wealth and pay, sample construction, distributional and education subgroups, the mass point at zero wealth, and consumption.

E.1 Labor Supply

In Figure E.1, we use semi-parametric regressions to ask whether the non-monotonic relationship between pay and wealth can be accounted for by labor supply behavior. We plot both annual hours worked (conditional on employment) and employment rates across the wealth distribution. The intensive margin shows little systematic variation: conditional on working, hours are broadly flat across wealth percentiles, with no evidence of a V-shaped pattern. Since our main analysis focuses on employed individuals, this indicates that the kink is not driven by variation in hours within our estimation sample.

On the extensive margin, employment rates decline modestly at the bottom of the wealth distribution, but the effect is small—around 5 percentage points (which is also around 5%) between the lowest wealth levels and zero wealth. This is much smaller than the nearly 20% decline in pay conditional on working shown in the equivalent semi-parametric regression in Figure 2. If anything, this extensive-margin pattern would reinforce the non-monotonicity observed among workers, but it is quantitatively minor and does not feature in our main analysis. Taken together, these results indicate that differences in labor supply – either in hours or participation – do not drive the non-monotonicity in pay, which instead reflects variation in wage rates across the wealth distribution.

E.2 Occupational allocation and switching

The AKM decomposition indicates that occupation pay premia account for only a small part of the non-monotonic relationship between wealth and pay. Nevertheless, workers might move between occupations endogenously as a function of their wealth. We investigate this possibility by asking two questions: whether changes in wealth are associated with the probability of changing occupation, and whether they are associated with moves into occupations with higher or lower pay premia.

We use observations of the same worker in consecutive two-year PSID waves, restricting the

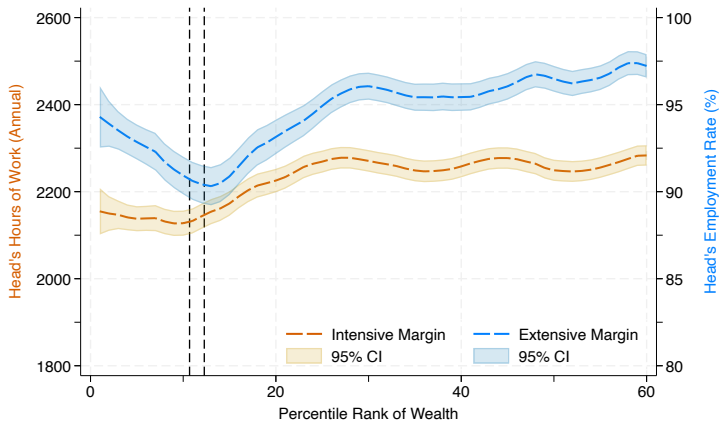


Figure E.1: Semi-parametric relationship between wealth and labor supply

Notes. Figure plots estimated $m(\cdot)$ functions, along with 95 percent confidence intervals, from semi-parametric regressions of annual hours worked (conditional on working) and the probability of working on the percentile rank of wealth. Regressions take the form $y_{i,t} = m(a^p_{i,t}) + X'_{i,t}\beta + \epsilon_{i,t}$ for each labor supply measure $y_{i,t}$, where X includes year fixed effects and a cubic polynomial in age, education and race indicators, education-specific linear age trends, and family size. The vertical lines delimit the range where wealth is near-zero, defined as being between −$250 and $250.

sample to pairs in which wealth is in the first 60 percentiles of the pooled distribution in both waves. Occupations are defined by the 35 pseudo-occupation groups used in our AKM decomposition (Appendix D). For each outcome, we relate occupational movement between waves to the change in the worker's wealth percentile rank. We allow different coefficients on changes in wealth rank for negative and non-negative wealth, and include the change in an indicator for negative wealth to account for movements across zero. This is the first-difference analog of the piecewise wealth-rank specification in our main analysis. The tables also report tests of the null hypothesis that the two wealth-rank coefficients are equal.

The specifications add controls sequentially: column (1) includes the wealth-rank terms alone; column (2) adds wave fixed effects; column (3) adds changes in the quadratic and cubic age terms and education indicators; and column (4) additionally controls for the estimated occupation fixed effect in the previous wave. All columns use the same worker-wave pairs, current-wave PSID survey weights, and standard errors clustered by worker.

Table E.1 considers the extensive margin of occupational mobility. The dependent variable indicates whether the worker changes pseudo-occupation between waves, irrespective of the direction of the move in terms of pay. Only 248 of the 4,694 worker-wave pairs involve a switch (5.28%). In every specification, the estimated wealth-rank coefficients are small and statistically insignificant. We also cannot reject equality of the negative- and positive-wealth coefficients: the p-values range from

0.892 to 0.994. The results therefore provide little evidence that changes in wealth are associated with a greater propensity to switch occupations, or that this association differs across zero wealth.

Table E.1: Occupation Switching and Changes in Wealth

	(1)	(2)	(3)	(4)
Change in wealth rank (debt)	0.000483	0.000063	0.000252	0.000260
	(0.0024)	(0.0024)	(0.0024)	(0.0024)
Change in wealth rank (non-debt)	0.000144	0.000044	-0.000002	-0.000001
	(0.000285)	(0.000284)	(0.000286)	(0.000285)
Change in debt indicator	0.0040	0.0047	0.0035	0.0033
	(0.0190)	(0.0186)	(0.0187)	(0.0187)
Lagged occupation fixed effect				-0.0144
				(0.0250)
Constant	0.0509***	0.0646***	0.1890*	0.1902*
	(0.0041)	(0.0158)	(0.1121)	(0.1123)
Wave fixed effects	No	Yes	Yes	Yes
Changes in demographic controls	No	No	Yes	Yes
Lagged occupation effect	No	No	No	Yes
Sample occupation switch rate (%)	5.28	5.28	5.28	5.28
Equality of wealth-rank slopes (p-value)	0.892	0.994	0.918	0.917
Observations	4,694	4,694	4,694	4,694

Notes: Standard errors clustered by worker (in parentheses). Demographic controls are an age-polynomial and education. Coefficients multiplied by 100 give percentage-point effects. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table E.2 examines the direction of occupational mobility. Here the dependent variable is the change in the estimated AKM occupation fixed effect between the two waves. This measure is zero for workers who remain in the same pseudo-occupation, positive for moves to occupations with higher estimated pay premia, and negative for moves to occupations with lower premia. It therefore tests whether wealth changes are associated with systematic movement toward better- or worse-paying occupations, rather than switching per se.

The estimated wealth-rank coefficients are again small and statistically insignificant in every specification. The tests for equality of the negative- and positive-wealth coefficients give p-values between 0.752 and 0.799. Hence we find little evidence that changes in wealth predict moves toward occupations with higher or lower pay premia, or that the relationship differs on either side of zero wealth.

Table E.2: Changes in Occupation Pay Premia and Changes in Wealth

	(1)	(2)	(3)	(4)
Change in wealth rank (debt)	-0.000266	-0.000222	-0.000229	-0.000211
	(0.000845)	(0.000833)	(0.000839)	(0.000837)
Change in wealth rank (non-debt)	-0.000004	-0.000009	-0.000005	-0.000001
	(0.000048)	(0.000049)	(0.000048)	(0.000048)
Change in debt indicator	0.0080	0.0076	0.0076	0.0073
	(0.0097)	(0.0096)	(0.0097)	(0.0097)
Lagged occupation fixed effect				-0.0341***
				(0.0103)
Constant	-0.0016	0.0027	0.0149	0.0177
	(0.000989)	(0.0019)	(0.0203)	(0.0205)
Wave fixed effects	No	Yes	Yes	Yes
Changes in demographic controls	No	No	Yes	Yes
Lagged occupation effect	No	No	No	Yes
Equality of wealth-rank slopes (p-value)	0.752	0.794	0.786	0.799
Observations	4,694	4,694	4,694	4,694

Notes: Standard errors clustered by worker (in parentheses). Demographic controls are an age-polynomial and education. * p < 0.10, ** p < 0.05, *** p < 0.01.

Taken together, the two exercises provide little evidence that endogenous occupational mobility explains the non-monotonic relationship between wealth and pay.

E.3 Alternative Definitions of Wealth

Liquid wealth. The measure of wealth we use throughout our analysis is *total* wealth—that is, the sum of cash, bonds, stocks, business wealth, pension funds, housing, other real estate and vehicles, net of mortgage and other debt. In this section, we consider a more restrictive measure—*liquid* wealth—which is intended to capture financial resources that are more readily accessible to households. We define liquid wealth as the sum of cash, bonds and stocks, net of non-mortgage debt. Relative to total wealth, this excludes home equity, other real estate, business wealth, pension funds and vehicles.

Figure E.2 shows estimates of the non-parametric relationship from equation (1) between the percentile rank of liquid wealth and annual pay. Note that, because the mass point of households reporting zero liquid wealth is much larger than for total wealth, we estimate two separate regressions

either side of the mass point, and indicate average earnings among those with zero liquid wealth as a separate point. The non-monotonicity remains apparent using this more restrictive measure of wealth, indicating that the relationship is not driven by the inclusion of less-liquid assets in household wealth.

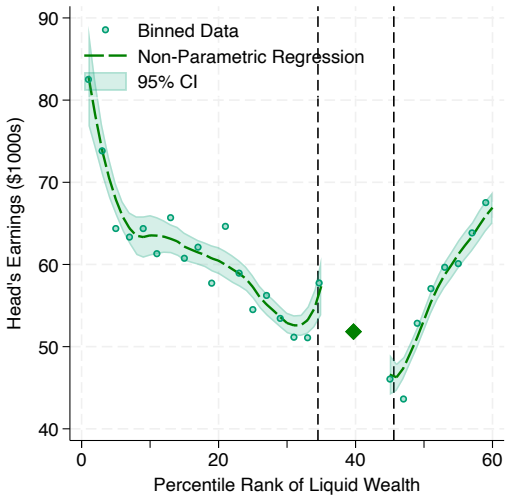


Figure E.2: Non-monotonic Relationship Between Liquid Wealth and Annual Pay

Notes. The figure plots the estimated non-parametric relationship from equation (1), with 95 percent confidence intervals and a binned scatter plot in which each point represents average pay within a two-percentile interval of the liquid wealth distribution. Liquid wealth is defined as cash, bonds and stocks, net of non-mortgage debt. The vertical lines indicate the range of wealth percentiles corresponding to liquid wealth between -$250 and $250.

Excluding student debt. Student debt could contribute to the high earnings observed among households with negative wealth, since some indebted households may have relatively high current or expected earnings. To assess its importance, we exclude student debt and recalculate wealth ranks. Figure E.3 shows that the non-monotonic relationship persists in both the non-parametric and semi-parametric specifications. The kink is therefore not explained by the inclusion of student loans in household debt.

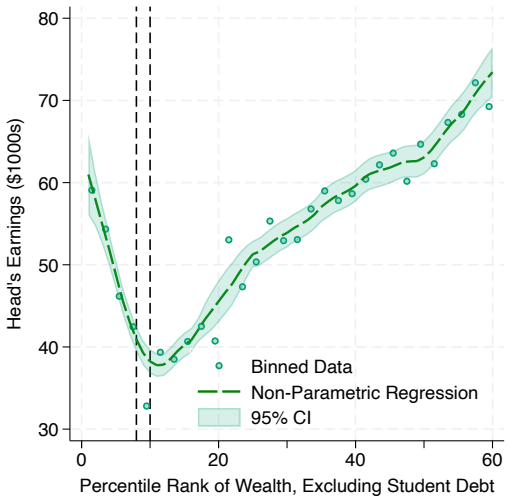


Figure E.3: Excluding Student Debt

Notes. This is the estimated kernel regression in equation (1), along with 95 percent confidence intervals and a binned scatter plot of the raw data where each point represents average pay in a two-percentile interval on the wealth distribution. The measure of wealth is as in our main analysis, but excludes student debt.

E.4 Alternative Definitions of Pay

Our baseline measure of pay is the household head's gross annual labor earnings. In this subsection we consider alternative outcomes to assess whether the non-monotonic relationship with wealth reflects particular features of that measure, including the influence of extreme earnings observations, persistent differences between workers, variation in hours worked, or the distinction between individual earnings and household resources after taxes and transfers. Figure E.4 presents these comparisons. Panel (a) reproduces the semi-parametric relationship for annual pay, providing a benchmark for the other panels.

We first consider permanent differences in pay across workers. For panels (b) and (d), we estimate the following earnings equation with worker fixed effects:

$$y_{i,t} = X'_{i,t}\beta + \alpha_i + \epsilon_{i,t}, \tag{E.1}$$

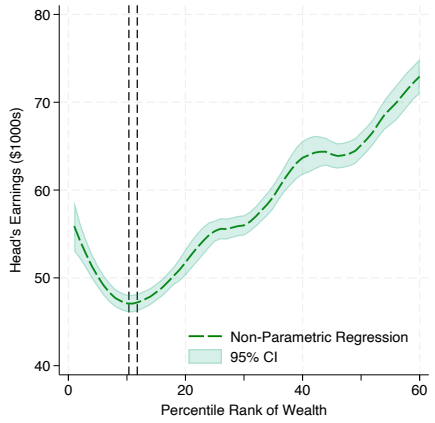
where $y_{i,t}$ is the household head's annual labor earnings in levels, $X_{i,t}$ contains a cubic polynomial in age and calendar year fixed effects, and α_i is a worker fixed effect. We use the estimated worker effect $\widehat{\alpha}_i$ to summarize persistent differences in annual earnings after removing predictable age and aggregate-year variation. Panel (b) reproduces Figure 4 from the main text, relating this estimated worker effect, measured in dollars, to wealth rank using the semi-parametric specification. Its

non-monotonic profile indicates that the relationship is not confined to temporary movements in current earnings.

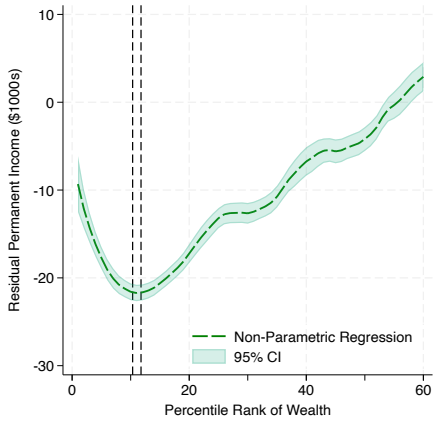
Panels (c) and (d) instead express pay as a percentile rank. Panel (c) uses the rank of current annual earnings; panel (d) uses the rank of the worker fixed effect $\widehat{\alpha}_i$. Ranks limit the influence of exceptionally large earnings observations and allow the relationship to be assessed independently of the dollar scale of the outcome. Both rank-rank profiles display the non-monotonicity. Panel (c) uses the broader sample of household heads rather than the baseline prime-age married sample, while panel (d) uses the baseline sample.

Finally, panels (e) and (f) consider different measures of pay. Panel (e) uses annual household income after taxes and transfers, rather than the head's gross labor earnings, to assess whether the pattern persists in a measure more closely related to disposable household resources. Panel (f) uses the head's hourly wage, defined as annual earnings divided by annual hours worked. This helps distinguish variation in wage rates from variation in annual labor supply. The non-monotonic relationship remains under both alternatives.

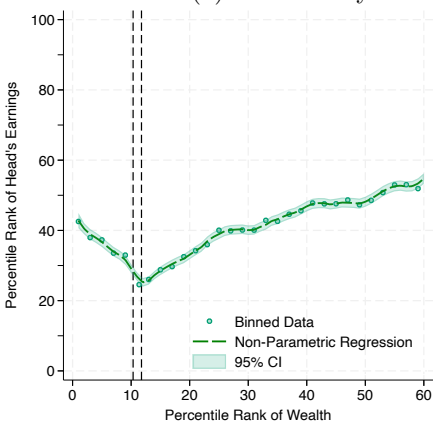
Taken together, the panels show that the relationship between wealth and pay is robust to changes in the scale and definition of the earnings measure. In particular, it appears in measures of persistent worker earnings and hourly wages, as well as in household income after taxes and transfers.



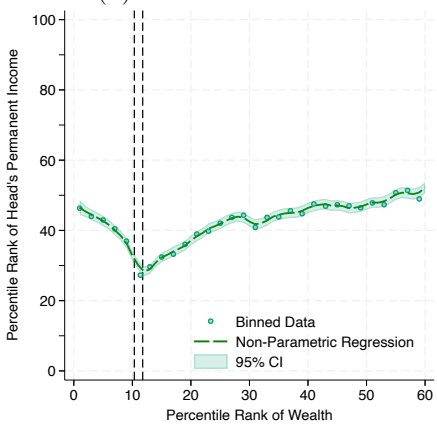
(a) Annual Pay



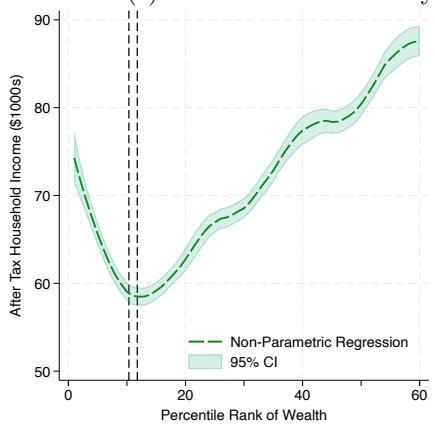
(b) Residual Permanent Income



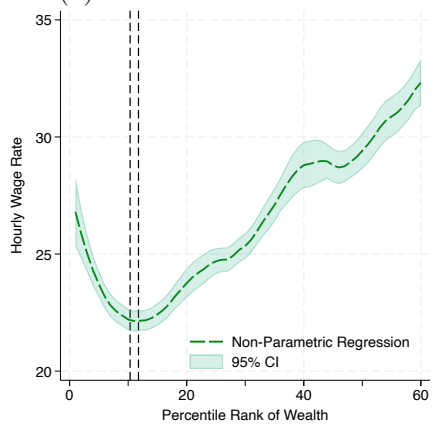
(c) Percentile Rank of Pay



(d) Percentile Rank of Permanent Pay



(e) Annual Post-Tax Household Income



(f) Hourly Pay

Figure E.4: Alternative Measures of Pay

Notes. The horizontal axis gives the percentile rank of household net worth. Panels (a), (b), (e) and (f) plot semi-parametric relationships controlling for a cubic polynomial in age, calendar year effects, education, race, education-specific age trends and family size; panels (c) and (d) plot non-parametric rank-rank relationships. Panel (b) uses the worker fixed effect from equation (E.1), and panel (d) ranks that effect. Panel (c) is estimated on the broader all-heads sample; the other panels use the baseline sample. Panel (e) measures income after taxes and transfers at the household level, and panel (f) uses wage and salary income divided by annual hours. Shaded regions show 95 percent confidence intervals. Vertical reference lines mark the region around zero wealth.

E.5 Alternative Sample Restrictions

Sample restrictions. Our baseline sample consists of married male household heads aged 25–55 with positive labor earnings. We now examine whether the non-monotonic relationship between wealth and pay depends on these restrictions. Table E.3 reports the piecewise-linear estimates from equation (3) across five samples, retaining controls for age, race, education, state and year. Column (1) reproduces the baseline specification from column (3) of Table 2. Column (2) includes non-workers and allows annual labor earnings to equal zero. Column (3) includes all prime-age male heads, regardless of marital status; column (4) extends the prime-age sample to female heads; and column (5) includes heads of all ages. Wealth percentiles are reconstructed for each sample, so the coefficients refer to relative wealth positions within the corresponding population.

Across all five columns, the estimated slope is negative for households with negative wealth and positive for those with non-negative wealth, and both slopes remain statistically significant. The magnitude of the left-hand slope becomes somewhat smaller as the sample broadens, but the qualitative pattern is stable. Including workers with zero earnings also leaves the relationship largely unchanged.

Table E.3: Robustness to Sample Composition

	(1) Baseline	(2) Include 0s	(3) Include Single Men	(4) Include Single Women	(5) All Ages
Wealth rank ($fora_{i,t} \geq 0$)	1274.231***	1272.661***	1273.957***	1166.857***	1111.099***
	(121.196)	(117.867)	(96.695)	(78.510)	(70.646)
Wealth rank ($fora_{i,t} < 0$)	-1038.230***	-977.500***	-933.525***	-875.493***	-763.101***
	(346.488)	(332.808)	(227.930)	(148.003)	(134.577)
Controls					
Age	Yes	Yes	Yes	Yes	Yes
Race	Yes	Yes	Yes	Yes	Yes
Education	Yes	Yes	Yes	Yes	Yes
N	26,317	27,699	35,544	48,411	64,313

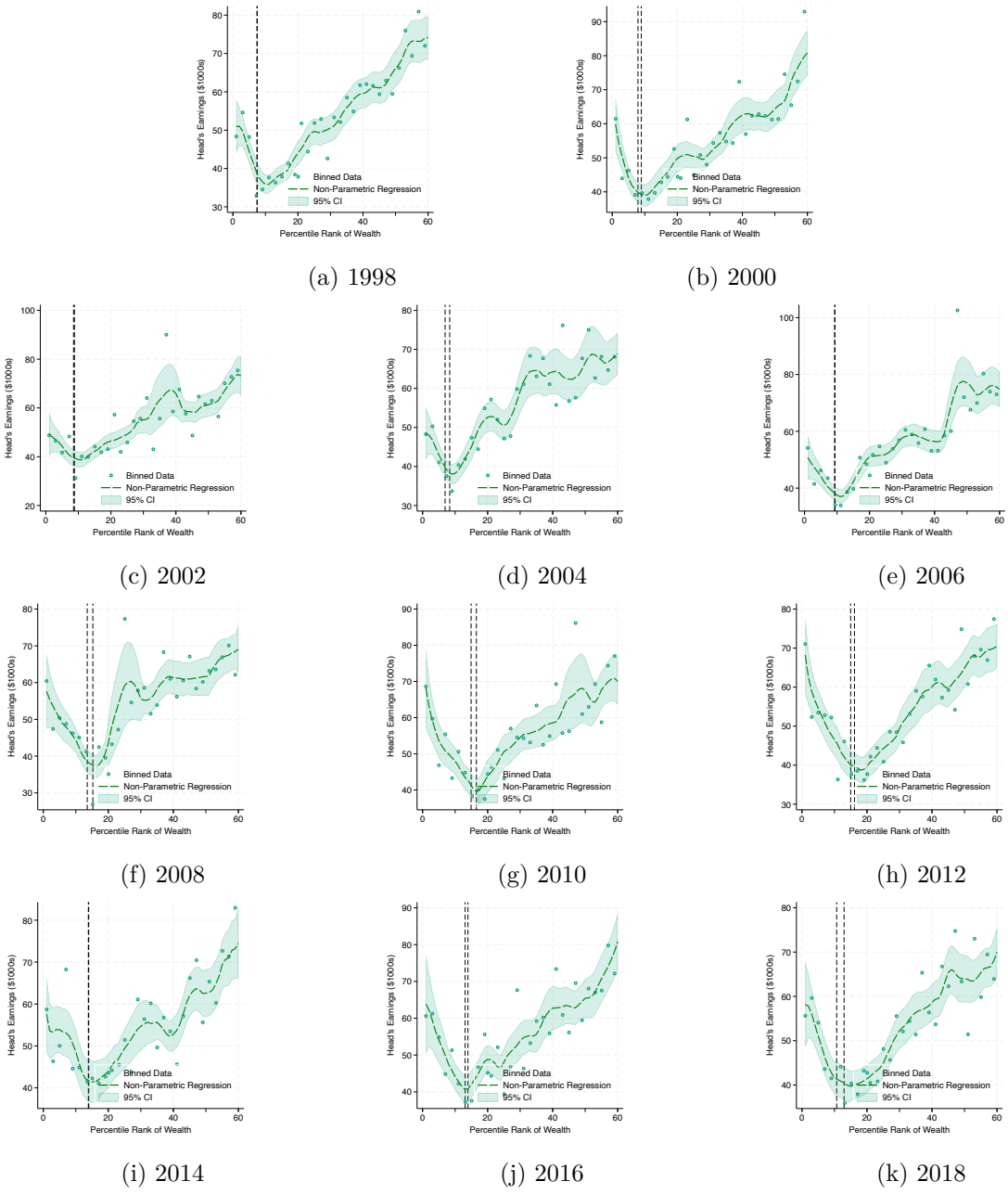
Notes. The table reports survey-weighted piecewise-linear regressions of annual labor earnings on wealth percentile rank, allowing distinct slopes below and above zero net worth. All specifications control for age, race, education, and state and year fixed effects; standard errors are clustered by person (in parentheses). Column (1) uses the baseline employed sample; column (2) adds non-workers with zero earnings; columns (3)–(5) successively relax marital-status, sex and age restrictions. Wealth percentiles are calculated within each sample. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E.6 Estimates by PSID wave

As a further check, Figure E.5 estimates the non-parametric relationship between annual pay and wealth separately in each biennial PSID wave. For this exercise, wealth percentile ranks are calculated within each wave rather than pooled across waves. Although the estimates are less precise in the smaller wave-specific samples, the non-monotonic relationship is visible throughout the sample period. This suggests that the aggregate relationship is not an artifact of combining observations from different years.

E.7 Self-employment

Finally, business income and self-employment may influence the observed relationship because business assets enter household wealth and the head's labor earnings can contain income from a business or farm. Figure E.6 examines this possibility in several ways. Panel (a) uses wage and salary earnings excluding farm and business income, while panel (b) excludes self-employed workers. These comparisons assess whether the relationship depends on the treatment of business income or on the composition of employment. The non-monotonicity is robust to these exclusions, indicating that it is not driven by self-employment.



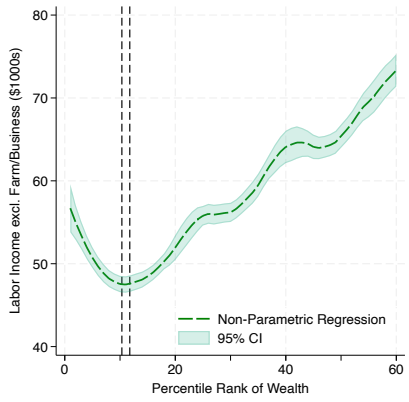
(a) 1998 (b) 2000

(c) 2002 (d) 2004 (e) 2006

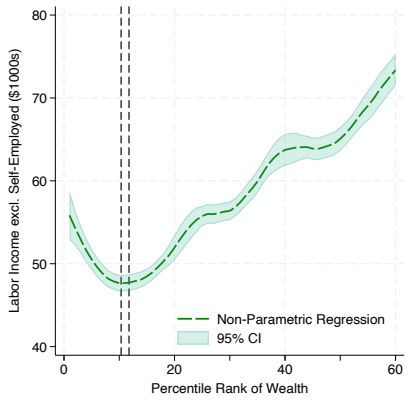
(f) 2008 (g) 2010 (h) 2012

(i) 2014 (j) 2016 (k) 2018

Figure E.5: Non-monotonicity estimated separately by year



(a) No Farm/Bus. Inc



(b) No Self Employed

Figure E.6: Self-Employment and the Wealth–Pay Relationship

Notes. All panels plot semi-parametric relationships with the percentile rank of household net worth, controlling for observable characteristics. Panel (a) uses wage and salary earnings rather than total labor earnings. Panel (b) excludes self-employed heads.

E.8 Distributional and Education Subgroups

Distribution. Our baseline analysis focuses on mean pay conditional on wealth. We now examine whether the non-monotonicity is also present at different points of the conditional earnings distribution. This is useful because a relationship in the mean alone could be influenced by a relatively small number of very high earners. Figure E.7 plots the 25th, 50th and 75th percentiles of the distribution of pay ranks conditional on wealth rank. Both variables are measured as percentiles of their respective distributions.

The figure shows that the non-monotonic pattern is not limited to average earnings: it is apparent at the lower quartile, median and upper quartile of the conditional pay-rank distribution. Households with very low net worth therefore tend to be drawn from a higher part of the pay distribution than those closer to zero wealth, with pay ranks rising again among households with positive net worth. This evidence indicates that the relationship is a relatively broad feature of the earnings distribution, rather than one generated exclusively by the upper tail.

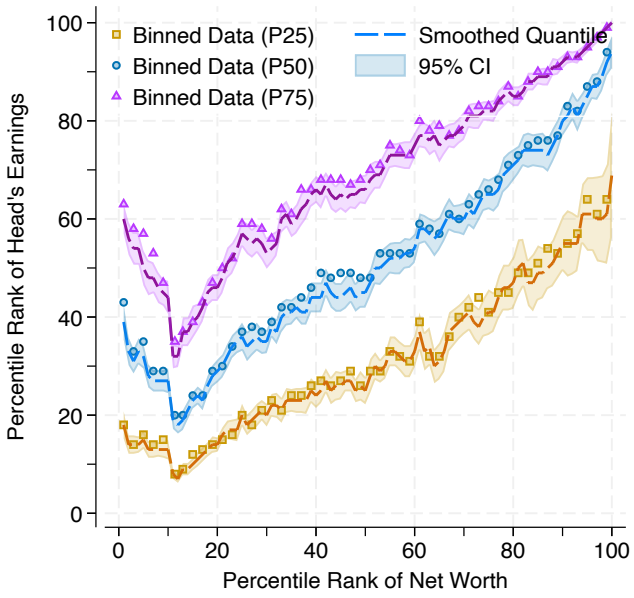


Figure E.7: The V-shape over the Distribution: p75, p50 and p25

Notes. The horizontal axis is the percentile rank of household net worth; the vertical axis is the percentile rank of the household head's annual labor earnings. The curves show estimates of the 25th, 50th and 75th conditional percentiles of pay rank, with 95 percent confidence intervals. Markers show the corresponding pay-rank quantiles within two-percentile wealth bins.

Education groups. Differences in educational attainment are a potential source of heterogeneity in pay and wealth. In particular, if households with different levels of education occupy different parts of the wealth distribution, the aggregate relationship could partly reflect differences in their average earnings rather than variation within educational groups. We therefore examine the relationship separately by educational attainment.

Figure E.8 plots mean annual pay against wealth percentile rank for household heads with fewer than 12 grades of schooling, those who completed high school, and those with post-high-school education. The estimates use the same baseline employed population, split into these three groups. Wealth ranks are defined relative to the full baseline sample, rather than recalculated within each education group, so that a given point on the horizontal axis represents the same position in the overall wealth distribution across groups.

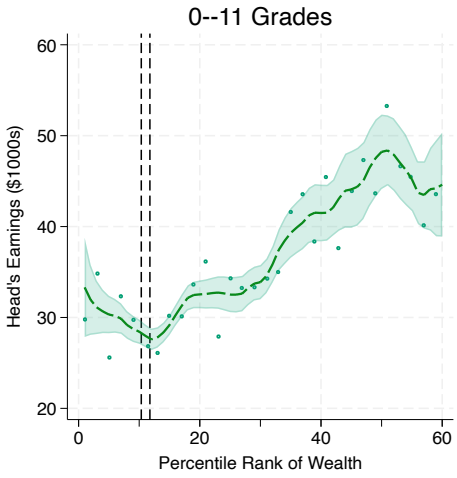
The non-monotonic relationship is visible in all three education groups: average earnings fall toward the wealth ranks associated with zero net worth and rise again over the positive-wealth range. The decline on the negative-wealth side is clearest among heads with post-high-school education, while the profile is shallower for those with fewer than 12 grades. The estimates therefore indicate that the aggregate pattern is not solely a consequence of differences in the educational composition of households across wealth ranks.

E.9 The Mass Point at Zero Wealth

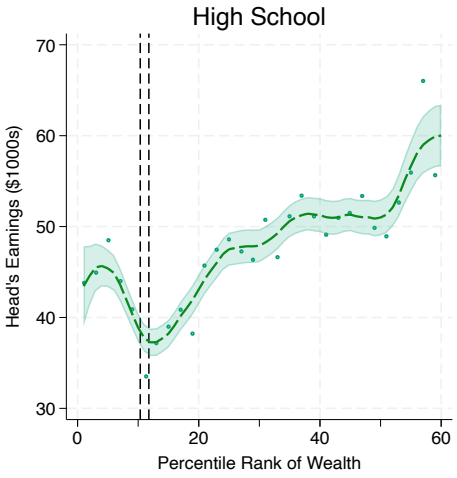
A potential concern is that the non-monotonic relationship between wealth and pay could be driven by the mass of observations at exactly zero wealth. Households reporting exactly zero wealth may differ from those on either side of zero. This mass of households could have a disproportionate influence on the estimated pattern near the kink.

To assess this possibility, we re-estimate the piecewise-linear specification in equation (3) after excluding observations in progressively wider intervals around zero wealth. Table E.4 reports the baseline estimates in column (1) — replicating the estimates from column (3) of Table 2 — and then excludes households with wealth between −$100 and $100, between −$250 and $250, and so on, up to −$10,000 and $10,000 in column (8). The exercise tests whether the slopes on either side of zero persist when estimation is based on households farther from the mass point.

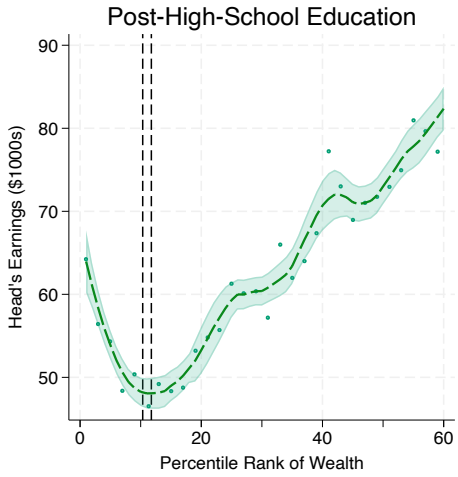
The estimated relationship is robust to these exclusions. Across all columns, the slope is negative among households with negative wealth and positive among those with non-negative wealth. The magnitudes remain broadly similar even after excluding households within $10,000 of zero: in



(a) Fewer than 12 grades



(b) High school



(c) Post-high-school education

Figure E.8: Relationship Between Wealth and Annual Pay by Education Group

Notes. The panels plot local-linear estimates of average annual labor earnings against the percentile rank of household net worth for heads with (a) 0–11 grades of schooling, (b) exactly 12 grades (high school), and (c) 13–17 grades (post-high-school education). Wealth ranks are calculated over the full baseline employed sample, rather than within education groups. Shaded bands are 95 percent confidence intervals and dots show binned average earnings. Vertical dashed lines indicate wealth within $250 of zero.

Table E.4: Donut Hole Estimator

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Wealth rank (for $a_{i,t} \geq 0$)	1274.2***	1290.5***	1289.7***	1291.9***	1297.1***	1317.5***	1350.6***	1403.6***
	(121.2)	(123.7)	(123.7)	(124.2)	(125.1)	(128.9)	(134.6)	(144.5)
Wealth rank (for $a_{i,t} < 0$)	-1038.2***	-1014.1***	-999.7***	-939.0***	-968.9***	-923.7**	-1099.7***	-978.5*
	(346.5)	(349.2)	(351.0)	(330.8)	(341.2)	(371.9)	(419.9)	(526.5)
Excluded Region	n/a	[-100,100]	[-250,250]	[-500,500]	[-1k,1k]	[-2.5k,2.5k]	[-5k,5k]	[-10k,10k]
Observations	26,317	25,816	25,772	25,688	25,502	24,930	24,068	22,692

Notes. The table reports estimates of the piecewise-linear regression in equation (3), allowing separate relationships between annual pay and wealth percentile rank for households with negative and non-negative net worth. Column (1) reports the baseline specification. Columns (2)–(8) progressively exclude households with net worth within the indicated intervals around zero. Wealth percentile ranks are calculated using the full baseline sample and are not recalculated following these exclusions. All specifications control for age, age squared, education, interactions of education with age and age squared, race, and state and year fixed effects. Regressions are weighted using PSID survey weights. Standard errors clustered at the household-head level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

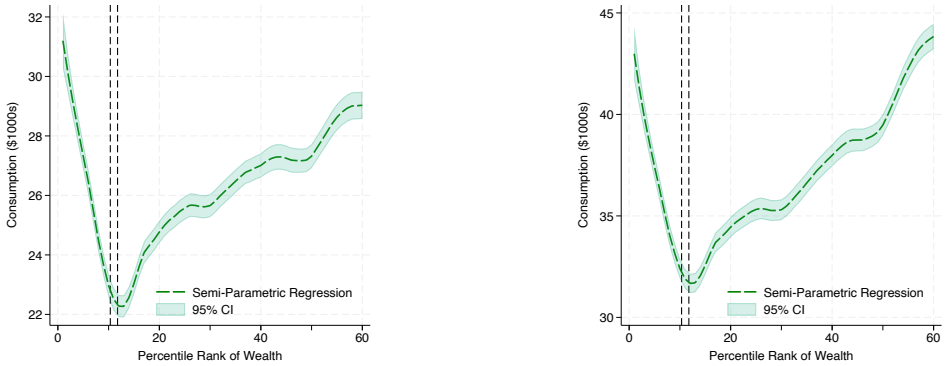
the widest-window specification, the estimated slopes are −$979 and $1,404 per wealth percentile, respectively, compared with −$1,038 and $1,274 in the baseline. The estimates become less precise as the excluded region expands, but both slopes remain statistically significant at the 10% level in every specification. The non-monotonicity is therefore not attributable solely to observations at or immediately around zero wealth.

E.10 Consumption

We next examine whether the non-monotonic relationship between wealth and pay is reflected in measures of household consumption. Figure E.9 plots average consumption against the percentile rank of wealth, using two alternative measures: non-durable and services expenditures, and a broader measure that additionally includes imputed rental services from owner-occupied housing.

Across both measures, we find a clear non-monotonic relationship between consumption and wealth that closely mirrors the pattern observed for pay. Consumption is relatively high among households at the bottom of the wealth distribution, declines as wealth rises toward zero, and then increases steadily across the positive wealth range. The minimum again occurs in the vicinity of zero net worth.

This finding is consistent with the selection mechanism in our model, in which higher permanent-earnings types can occupy both deeply indebted and high-wealth positions, while lower types are more concentrated around zero wealth.



(a) Non-durable and services expenditures (b) Including rental equivalent housing

Figure E.9: Non-monotone relationship between wealth and consumption

Notes. The panels plot estimates of the non-parametric component $m(\cdot)$ in the semi-parametric specification $c_{i,t} = m(a^p_{i,t}) + X'_{i,t}\beta + \epsilon_{i,t}$, where $a^p_{i,t}$ is the percentile rank of household net worth. Controls include calendar year effects, a cubic in age, education, race, education-specific age trends and household size. Each consumption measure is capped at its survey-weighted 95th percentile before estimation. Consumption is expressed in thousands of 2012 dollars, and shaded regions show 95 percent confidence intervals. The vertical reference lines indicate the region around zero net worth (observations with wealth within $250 of zero). Panel (a) excludes imputed rental services from owner-occupied housing; panel (b) includes them.

F Proofs for the Stylized Model

This appendix describes the stylized environment of Section 4 and proves Proposition 1. Appendix F.1 then shows that the result extends to persistent income. Two properties are key: homogeneity of degree $1-\gamma$ and impatience keeps wealth bounded.

Environment. There are two worker types, $j \in \{L, H\}$. A worker of type j earns $w_t = \theta_j z_t$, where z_t takes one of the values $0 < z_1 < \cdots < z_N$ and is drawn independently each period with probabilities $\pi_1, \ldots, \pi_N$.[12] High types are $\eta > 1$ times more productive, $\theta_H = \eta\,\theta_L$, so mean pay $\overline{w}_j = \theta_j \sum_n \pi_n z_n$ satisfies $\overline{w}_H = \eta\,\overline{w}_L$. Relative to Section 3, we make four simplifications:

(i) utility is CRRA with risk aversion γ;

(ii) taxes are proportional ($\tau = 0$) and the income floor does not bind, so disposable income is proportional to pay and, without loss of generality, equal to it;

(iii) there is a single interest rate $r > 0$, with $R = 1 + r$ and $\beta R < 1$; and

(iv) the maximum borrowing constraint is slack for both types, so borrowing limits are proportional to mean pay, $\underline{a}(\alpha_j) = -\kappa\,\overline{w}_j$. In terms of equation (8), this requires $\underline{a}_{\max} < -\kappa\,\overline{w}_H$, so the loan-

[12] We prove it in the N case extension to avoid repetition.

to-income constraint determines the wealth floor for both types. We also require $\kappa\,\overline{w}_j < \theta_j z_1 / r$, so that a worker at the borrowing limit who draws the lowest pay can still cover the interest on their debt.

A worker of type j therefore solves

$$V_j(a,z) = \max_{a' \geq \underline{a}(\alpha_j)} \Big\{ u\Big(Ra + \theta_j z - a'\Big) + \beta\, E\Big[V_j(a',z') \mid z\Big] \Big\}.$$

We refer to the long-run distribution of wealth for each type as its stationary distribution.[13] We begin by constructing two useful lemmas before proving the main result.

Lemma 1 (Scaling) *A high type with wealth ηa behaves exactly like a low type with wealth a, scaled up by η: they save and consume η times as much. Consequently, the high-type wealth distribution is the low-type distribution stretched by the factor η: the share of high types with wealth below ηa equals the share of low types with wealth below a.*

Proof. Because the high type's pay and borrowing limit are η times the low type's in every state, any savings plan available to a low type starting from wealth a is available to a high type starting from ηa once scaled up by η, and it finances η times the consumption in every period. With CRRA utility, scaling consumption by η multiplies lifetime utility by $\eta^{1-\gamma}$ (or adds a constant when $\gamma = 1$), which leaves the ranking of plans unchanged. The two types therefore choose the same plans up to the factor η. Facing the same shocks, a high type's wealth path is η times the corresponding low type's path, so the high-type distribution is the low-type distribution stretched by η. ■

Lemma 2 (Bounded wealth) *If $\beta R < 1$, there is a finite level of wealth that no worker of either type exceeds in the stationary distribution.*

Proof. Let $x_t = Ra_t + w_t - \underline{a}(\alpha_j)$ denote the most a worker could consume this period. It evolves as $x_{t+1} = R(x_t - c_t) + y_{t+1}$, where $y_{t+1} = w_{t+1} + r\,\underline{a}(\alpha_j)$ is pay net of interest on the maximum debt. By simplification (iv), y_{t+1} is positive, and it never exceeds some upper bound $\overline{y}$. This is the standard income fluctuation problem with no borrowing. With CRRA utility and a constant interest rate, Ma and Toda (2022) show that the marginal propensity to consume out of x_t converges to $1 - (\beta R)^{1/\gamma}/R$ as x_t grows. Rich households therefore save, after interest, less than a fraction

[13] As is standard, we take this distribution to be unique (e.g., Huggett, 1993; Ma, Stachurski, and Toda, 2020). Wealth is measured at the start of the period, before current pay is drawn.

$\rho < 1$ of x_t, for any ρ between $(\beta R)^{1/\gamma}$ and one: $x_{t+1} \leq \rho\, x_t + \overline{y}$ once x_t is large enough. Wealth therefore falls whenever it is large, so every wealth path eventually drops below, and never again exceeds, a finite level.[14] ■

Proof of Proposition 1. Lemma 2 shows that $\overline{a}_L$ is finite. By Lemma 1, the lowest and highest wealth levels held by high types are $\eta\, \underline{a}_L$ and $\eta\, \overline{a}_L$. Since $\eta > 1$, $\underline{a}_L < 0$ and $\overline{a}_L > 0$, the first lies strictly below $\underline{a}_L$ and the second strictly above $\overline{a}_L$. This is part (i).

By definition, no low type has wealth below $\underline{a}_L$ or above $\overline{a}_L$, so only high types are present in these tails. Because pay is IID and wealth is chosen before current pay is drawn, pay is independent of wealth within each type, so $m_j(a) = \overline{w}_j$. In the tails, $m(a) = \overline{w}_H$, which is part (ii).

Wherever low types are present, expected pay is the average $m(a) = p_L(a)\, \overline{w}_L + \Big(1 - p_L(a)\Big)\overline{w}_H$ with $p_L(a) > 0$, which lies strictly below $\overline{w}_H$ because $\overline{w}_L = \overline{w}_H / \eta$. Since low types are found arbitrarily close to $\underline{a}_L$ and $\overline{a}_L$, this is part (iii). ■

Proposition 1 pins down the tails of the pay–wealth profile rather than its full shape: within the overlap region, $m(a)$ inherits whatever variation there is in the low-type share $p_L(a)$. The width of each high-type-only tail grows with $\eta - 1$, so greater skill heterogeneity produces a more pronounced V-shape.

F.1 Extension to Markov Income Dynamics

In our main analytic exercise we focus on the IID case as it highlights the role of the composition effect in our more general model. We now show that the IID structure is a useful, but not essential, simplification.

Assumption 1 (Markov income) *The productivity state z_t follows a finite-state Markov chain that is irreducible and aperiodic. All other features of the environment are unchanged.*

Neither Lemma 1 nor Lemma 2 uses the IID assumption: the scaling argument holds shock by shock, and the bound on wealth applies equally to Markov income. What changes is that current pay now carries information about past shocks, and hence about wealth, so mean pay within each type, $m_j(a)$, varies with wealth. The restriction we need is that this within-type variation is small relative to the pay gap between types. Let $\underline{z}^P$ and $\overline{z}^P$ denote the lowest and highest values of expected productivity next period, $E[z_{t+1} \mid z_t]$, across current states z_t.

[14] Explicitly, if the bound applies for $x_t \geq X$, then x_t never again exceeds the larger of $RX + \overline{y}$ and $2\overline{y}/(1-\rho)$ once it falls below that level, and it does so in finite time from any starting point. For IID income, this result dates back to at least Schechtman and Escudero (1977); see also Aiyagari (1994).

Proposition 2 (Pay–wealth profile under persistent income) *Maintain Assumption 1, suppose $\beta R < 1$ and $\underline{a}_L < 0 < \overline{a}_L$, and suppose $\eta > \overline{z}^P / \underline{z}^P$. Then:*

i. Only high types have wealth below $\underline{a}_L$ or above $\overline{a}_L$, and in these tails $m(a) = m_H(a)$.

ii. Wherever both types are present, $m_H(a) > m_L(a)$, so $m(a) < m_H(a)$.

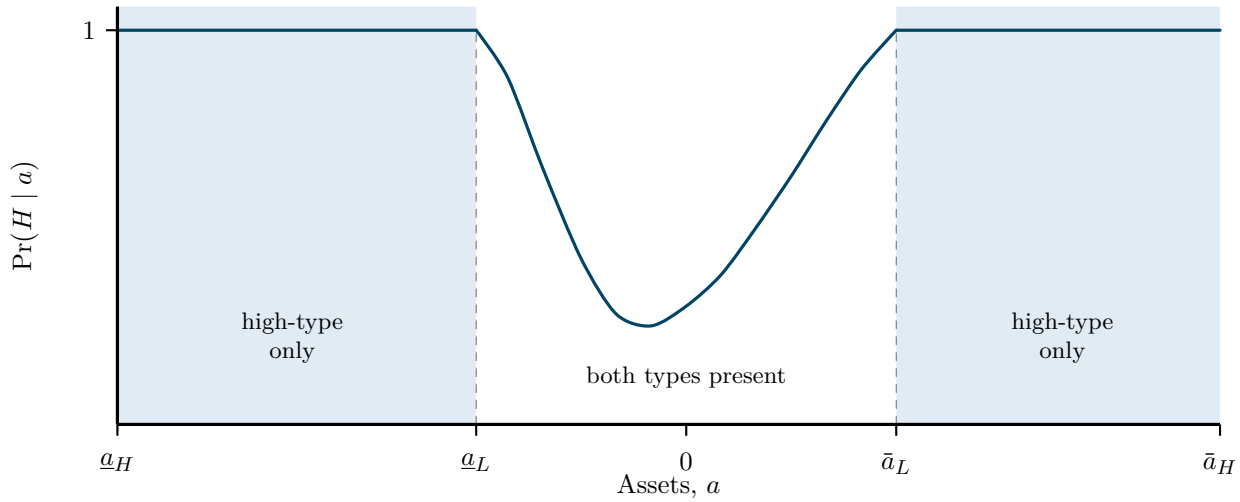
When income is IID, $\underline{z}^P = \overline{z}^P$, so the restriction holds for every $\eta > 1$ and Proposition 1 is recovered.

Proof. Part (i) follows exactly as in the proof of Proposition 1, since neither lemma uses the IID assumption. For part (ii), wealth a_t is chosen in period $t-1$ with knowledge of z_{t-1}, so given z_{t-1}, today's productivity is unrelated to today's wealth. Hence

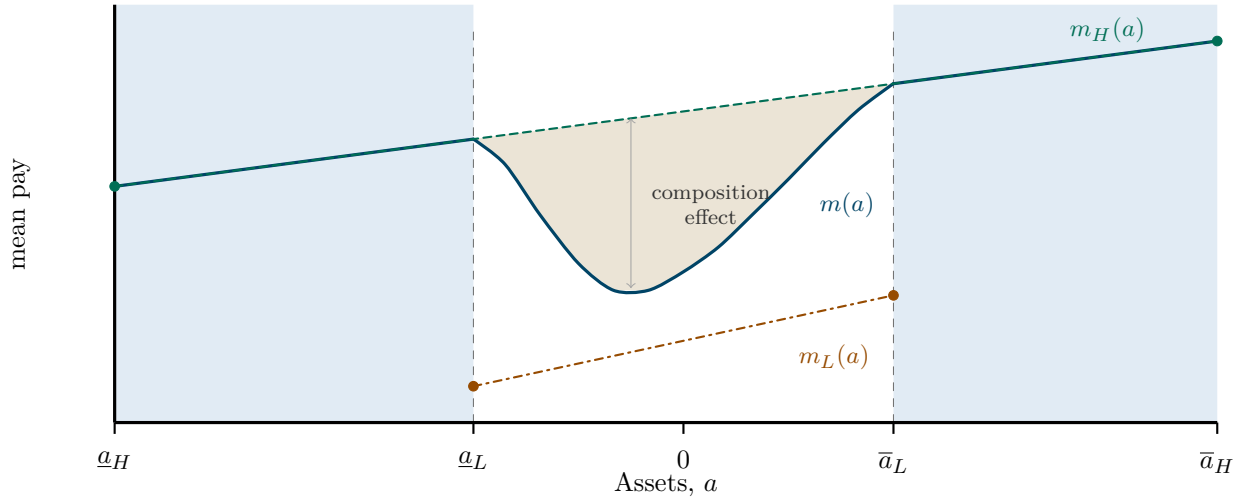
$$m_j(a) = \theta_j \, E\Big[\, E[z_t \mid z_{t-1}] \;\Big|\; a_t = a,\, j \,\Big],$$

which lies between $\theta_j \underline{z}^P$ and $\theta_j \overline{z}^P$. It follows that $m_H(a) - m_L(a) \geq \theta_L \Big(\eta \, \underline{z}^P - \overline{z}^P \Big) > 0$. Wherever both types are present, $m(a) = m_H(a) - p_L(a) \Big[m_H(a) - m_L(a) \Big] < m_H(a)$. ■

The restriction $\eta > \overline{z}^P / \underline{z}^P$ has a simple interpretation: the pay gap across types must dominate the variation in expected pay within each type induced by persistence. When this holds, a high type in the worst income state still expects higher pay next period than a low type in the best state, so pooling types at any wealth level cannot erase the gap. More generally, the argument requires only that $m_H(a) > m_L(a)$ wherever the types overlap, and it is silent on the shape of m_j within each type. Because $m_H(a)$ now varies with wealth, the V-shape is defined relative to the high-type profile rather than in levels: pooled pay coincides with $m_H(a)$ in the tails and lies strictly below it where the types overlap. Figure F.1 illustrates this, contrasting with the IID case shown in Figure 7 of the main text.



(a) Conditional probability of the high type, $\Pr(H \mid a)$



(b) Within- and across-type mean pay

Notes. This figure extends Figure 7 to the case of persistent income shocks. Panel (a) reproduces the conditional probability $\Pr(H \mid a)$ from the IID example, which retains the same shape as in the IID case since the composition mechanism is unchanged. Panel (b) plots within-type mean pay $m_H(a)$ and $m_L(a)$, which now slope upward when persistent shocks induce a positive correlation between current wages and wealth within each type. The composition effect (shaded wedge) nevertheless pulls $m(a)$ sharply below $m_H(a)$ in the overlap region, embedding a V-shape in the overall upward trend. The qualitative pattern therefore does not require flat within-type means.

Figure F.1: Pay–wealth profile with persistent income shocks

G Quantitative Methods and Additional Model Results

G.1 Computational Details

We solve the household problem separately for each permanent type, compute the associated stationary distribution, and aggregate using the population weights of each type.

Earnings process We discretize the permanent component α on 27 nodes with a support spanning three standard deviations of our estimated permanent component standard deviation. We discretize the persistent component z using 21 nodes following Tauchen (1986) and the transitory component ν on 5 nodes. Measurement error is not a state variable. It enters only when we compute moments of observed pay, as an independent multiplicative lognormal error.

Grids The wealth grid has 1,800 nodes on $[\underline{a}(\alpha)$, \$3 million] and contains zero as a node. Forty percent of nodes lie in the borrowing region. They use cosine (Chebyshev) spacing, which concentrates points near both the borrowing limit and zero. The remaining nodes lie in the saving region, with quadratic spacing concentrated near zero. Policies are constructed on a common cash-on-hand grid of 3,500 nodes that is dense on both sides of zero. This grid is augmented with nodes at the kinks of each earnings state's policy function, where $a' = 0$ and $a' = \underline{a}(\alpha)$.

Smoothing the interest-rate wedge The discrete jump from r_L to r_B at zero wealth kinks the budget constraint. This creates a region of cash on hand over which $a' = 0$ is optimal, and it can make the Euler-equation operator locally expansive. We therefore replace equation (7) with the logistic approximation

$$r(a) = r_L + \frac{r_B - r_L}{1 + \exp(a/\varepsilon)}, \qquad \varepsilon = 0.25,$$

that is, a smoothing width of \$250. The Euler equation then uses the effective gross return $1 + r(a') + a'r'(a')$, which equals $1 + r_L$ or $1 + r_B$ wherever $r'(a') = 0$. We find this approximation aids with numerical stability, particularly in estimation where the wedge varies, but that the unsmoothed problem behaves similarly.

Household problem We solve each type's problem with the endogenous grid method (Carroll, 2006). At each iteration we invert the Euler equation on the wealth grid and take the upper envelope over three sets of candidates on the common cash-on-hand grid: interior solutions on monotone segments, the borrowing constraint, and the upper bound of the grid (Iskhakov et al., 2017; Clausen

and Strub, 2020). For non-converging points we switch to value function iteration with Howard improvement steps, warm-started from the final EGM value function. This step is a contraction regardless of the wedge.

Stationary distribution We compute each type's stationary distribution over (a, z, ν) by non-stochastic simulation (Young, 2010). Next-period wealth is assigned to the two adjacent grid nodes using linear-interpolation weights. Combined with the earnings transition matrix, these weights define a sparse transition matrix on the joint state space. Wealth percentiles, pay by wealth percentile, the poverty measures and the targeting rules are all computed exactly from the type-weighted distributions. Given our focus on percentiles of the distribution, we find this has lower overheads than the large number of Monte Carlo agents we would otherwise require.

Estimation We minimize the criterion in two stages. We first evaluate it at 164 Sobol points over $\gamma \in [1, 6]$ and $r_B - r_L \in [0.005, 0.25]$. We then run Nelder–Mead searches from the eight best Sobol points, and retain the global minimum. In the Nelder–Mead step we impose an additional criterion that starting values must be "well-separated". In practice, we find this allows us to explore a wider variety of basins of attraction and identify more local minima.

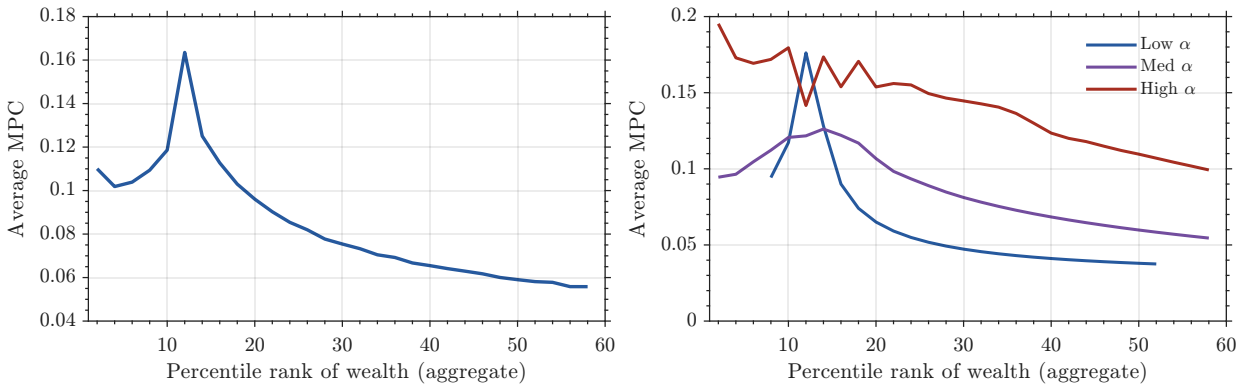
G.2 Additional Numerical Results

G.2.1 Marginal Propensities to Consume

The selection mechanism in our model also has implications for marginal propensities to consume. Figure G.1 plots the MPC out of a small transitory transfer. A large empirical literature documents substantial heterogeneity in MPCs across the wealth distribution (Fagereng, Holm, and Natvik, 2021; Lewis, Melcangi, and Pilossoph, 2019), and a key question is the extent to which this heterogeneity reflects differences in liquid wealth, permanent income, or both (Kaplan, Violante, and Weidner, 2014). Our model highlights a novel channel: endogenous selection through precautionary saving and debt utilization.[15]

In the aggregate, the MPC is itself non-monotonic in wealth. This arises from two distinct forces. The first is compositional: because the mix of worker types varies across the wealth distribution, the average MPC at a given wealth percentile reflects the behavior of economically very different

[15]The overall level of MPCs in the model is broadly consistent with the values typically obtained in one-asset heterogeneous-agent models.



(a) Aggregate Consumption (b) Consumption by Type

Figure G.1: Aggregate MPC by Worker Type

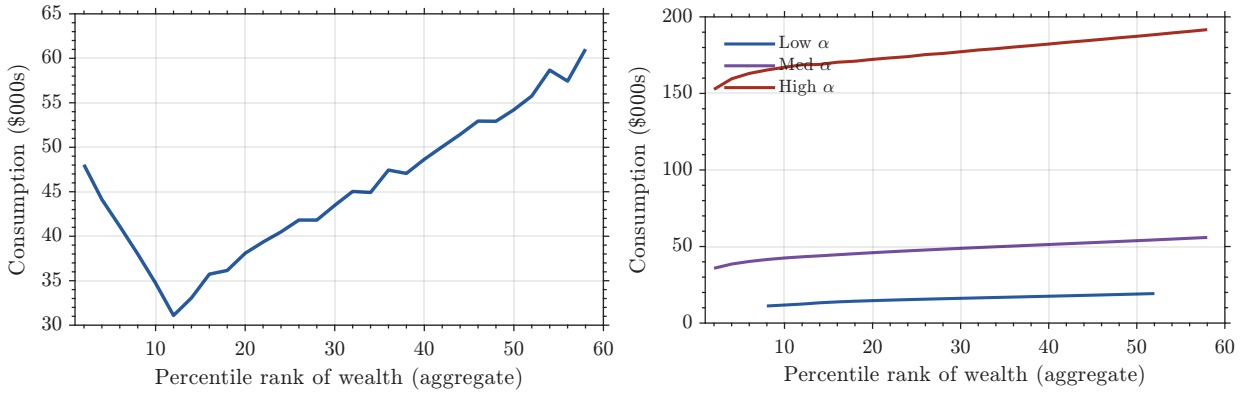
Notes. The left panel plots the average marginal propensity to consume out of a small transitory transfer against the percentile rank of wealth in the simulated stationary distribution. The right panel plots the same relationship separately for each of three types—low, medium and high. In both panels, the average at each wealth percentile integrates over the distribution of persistent labor income realizations conditional on wealth. The discontinuity at zero wealth reflects the wedge between borrowing and lending rates described in equation (7). The model is evaluated at the parameter values reported in Tables 3 and 4.

households. The second is the wedge between borrowing and lending rates. Even within a given type, households crossing into negative wealth face a discrete increase in the cost of intertemporal substitution, generating a local increase in MPCs around zero wealth (see Koşar et al. 2025 and Ahn et al. 2018 for discussions of related mechanisms).

While the borrowing wedge amplifies the non-monotonicity and sharpens the kink around zero wealth, the right panel of Figure G.1 shows that the dominant force is compositional. This observation has important implications for the targeting of fiscal transfers. Wealth rank is an imperfect proxy for household type, and so policies aimed at low-wealth households may reach very different groups of workers depending on where in the wealth distribution they are targeted. In particular, heavily indebted households are often high-type workers borrowing against strong future earnings prospects, whereas households clustered around zero wealth are disproportionately lower-type workers with substantially lower expected lifetime resources.

G.2.2 Consumption

Figure G.2 shows that the same logic extends to consumption. Within each worker type, consumption is increasing in wealth, reflecting both greater resources and weaker precautionary-saving motives. But the aggregate consumption profile is non-monotonic because the composition of households changes across the wealth distribution. As with pay and lifetime utility, the aggregate relationship



(a) Aggregate Consumption

(b) Consumption by Type

Figure G.2: Aggregate Consumption Function by Worker Type

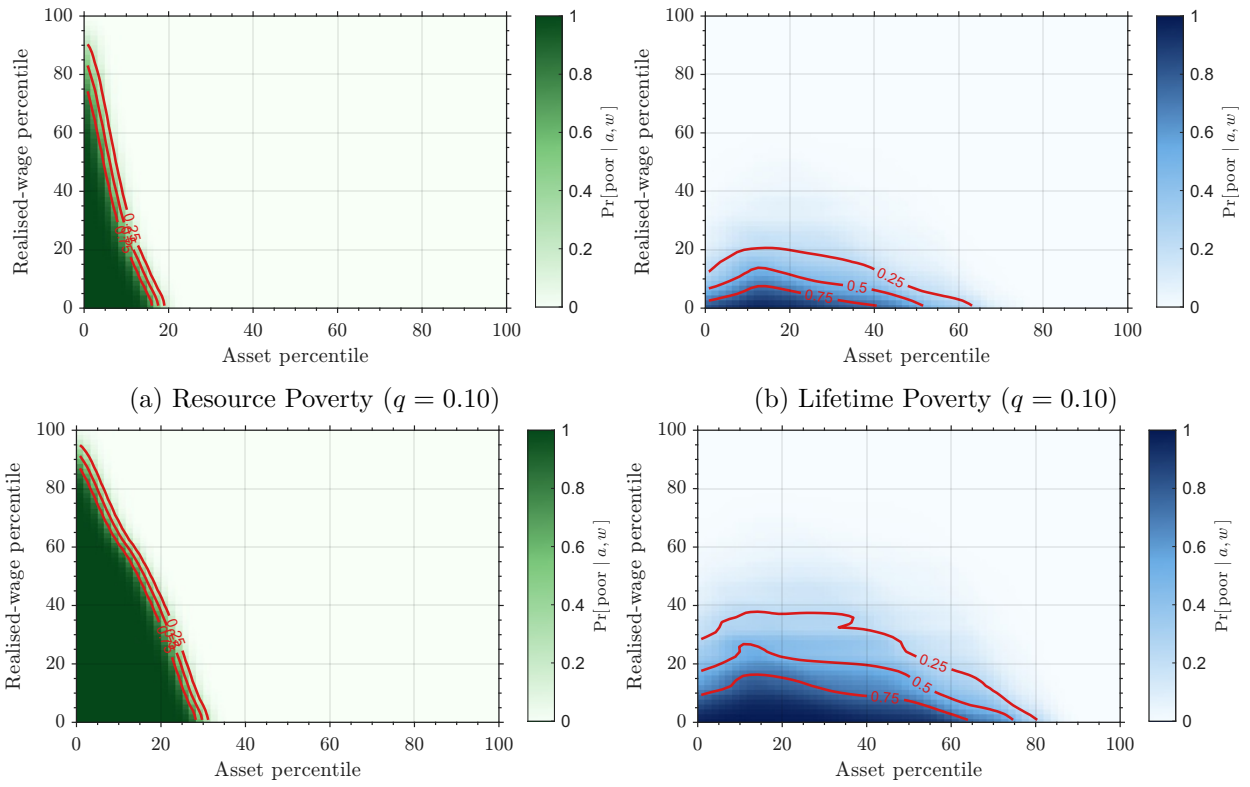
Notes. The left panel plots average consumption against the percentile rank of wealth in the simulated stationary distribution. The right panel plots the same relationship separately for each of three types—low, medium and high. In both panels, the average at each wealth percentile integrates over the distribution of persistent labor income realizations conditional on wealth. The model is evaluated at the parameter values reported in Tables 3 and 4.

is driven not by within-type behavior, but by the endogenous selection of worker types across the wealth distribution.

Appendix E.10 shows that consumption is also non-monotonic in wealth in the data. The empirical and model profiles have different magnitudes and their quantitative features do not coincide exactly. However, we note also that the empirical definition of consumption differs conceptually from the notion of consumption in the model: our empirical measure of consumption excludes certain important categories (such as durables), while consumption in the model is the entire gap between net income and savings. We therefore note simply that the qualitative relationship between consumption and wealth in the model corresponds with the data.

G.3 Alternative Poverty Thresholds

In this subsection, we repeat the analysis of resource and lifetime poverty over the distributions of wealth and pay from Section 6 for different thresholds. In particular, in Figure G.3 we report those who are in the bottom 10% and bottom 20% of the distributions of each notion of poverty. The overall picture is very similar to that for the bottom 5% reported in Section 6.



(a) Resource Poverty ($q = 0.10$)

(b) Lifetime Poverty ($q = 0.10$)

(c) Resource Poverty ($q = 0.20$)

(d) Lifetime Poverty ($q = 0.20$)

Figure G.3: Additional Thresholds Alternative Poverty Metrics

Notes. Each panel plots Pr[poor | a, w] from equation (15) within cells defined by percentile ranks of wealth and realized pay in the stationary distribution of the calibrated model, with $q = 0.10$ and $q = 0.20$; red lines are iso-poverty contours. Panels (a) and (c) use cash on hand, Panels (b) and (d) use lifetime utility. Parameters as in Tables 3 and 4.

References

Abowd, John M., Francis Kramarz, and David N. Margolis. 1999. "High Wage Workers and High Wage Firms." *Econometrica* .

Ahn, SeHyoun, Greg Kaplan, Benjamin Moll, Thomas Winberry, and Christian Wolf. 2018. "When inequality matters for macro and macro matters for inequality." *NBER macroeconomics annual* 32 (1):1–75.

Aiyagari, S. Rao. 1994. "Uninsured Idiosyncratic Risk and Aggregate Saving." *The Quarterly Journal of Economics* Vol. 109, No. 3.:659–684.

Blundell, Richard, Luigi Pistaferri, and Itay Saporta-Eksten. 2016. "Consumption Inequality and Family Labor Supply." *The American Economic Review* 106 (2):387–435.

Bonhomme, Stephane, Kerstin Holzheu, Thibaut Lamadon, Elena Manresa, Magne Mogstad, and Bradley Setzler. 2023. "How Much Should We Trust Estimates of Firm Effects and Worker Sorting?" *Journal of Labor Economics* 41 (2):291–322.

Carroll, Christopher D. 2006. "The Method of Endogenous Gridpoints for Solving Dynamic Stochastic Optimization Problems." *Economics Letters* 91 (3):312–320.

Clausen, Andrew and Carlo Strub. 2020. "Reverse Calculus and Nested Optimization." *Journal of Economic Theory* 187:105019.

Fagereng, Andreas, Martin B Holm, and Gisle J Natvik. 2021. "MPC heterogeneity and household balance sheets." *American Economic Journal: Macroeconomics* 13 (4):1–54.

Guvenen, Fatih, Fatih Karahan, Serdar Ozkan, and Jae Song. 2021. "What do data on millions of US workers reveal about lifecycle earnings dynamics?" *Econometrica* 89 (5):2303–2339.

Huggett, Mark. 1993. "The risk-free rate in heterogeneous-agent incomplete-insurance economies." *Journal of economic Dynamics and Control* 17 (5-6):953–969.

Iskhakov, Fedor, Thomas H. Jørgensen, John Rust, and Bertel Schjerning. 2017. "The Endogenous Grid Method for Discrete-Continuous Dynamic Choice Models with (or without) Taste Shocks." *Quantitative Economics* 8 (2):317–365.

Kaplan, Greg, Giovanni L Violante, and Justin Weidner. 2014. "The wealthy hand-to-mouth." Tech. rep., National Bureau of Economic Research.

Koşar, Gizem, Davide Melcangi, Laura Pilossoph, and David G Wiczer. 2025. "Stimulus through insurance: The marginal propensity to repay debt." Tech. rep., National Bureau of Economic Research.

Lewis, Daniel J, Davide Melcangi, and Laura Pilossoph. 2019. "Latent heterogeneity in the marginal propensity to consume." *FRB of New York Staff Report* (902).

Ma, Qingyin, John Stachurski, and Alexis Akira Toda. 2020. "The income fluctuation problem and the evolution of wealth." *Journal of Economic Theory* 187:105003.

Ma, Qingyin and Alexis Akira Toda. 2022. "Asymptotic Linearity of Consumption Functions and Computational Efficiency." *Journal of Mathematical Economics* 98:102562.

Schechtman, Jack and Vera L. S. Escudero. 1977. "Some Results on "An Income Fluctuation Problem"." *Journal of Economic Theory* 16 (2):151–166.

Tauchen, George. 1986. "Finite State Markov-Chain Approximations to Univariate and Vector Autoregressions." *Economics Letters* 20 (2):177–181.

Young, Eric R. 2010. "Solving the Incomplete Markets Model with Aggregate Uncertainty Using the Krusell–Smith Algorithm and Non-Stochastic Simulations." *Journal of Economic Dynamics and Control* 34 (1):36–41.